EUROPEAN MiCROSOFT FABRiC Community Conference

VIENNA 15-18 SEPTEMBER 2025

JOIN THE CONVERSATION

#FABCONEUROPE25

Mastering the Essentials of Data Factory

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Principal Product Manager
Microsoft

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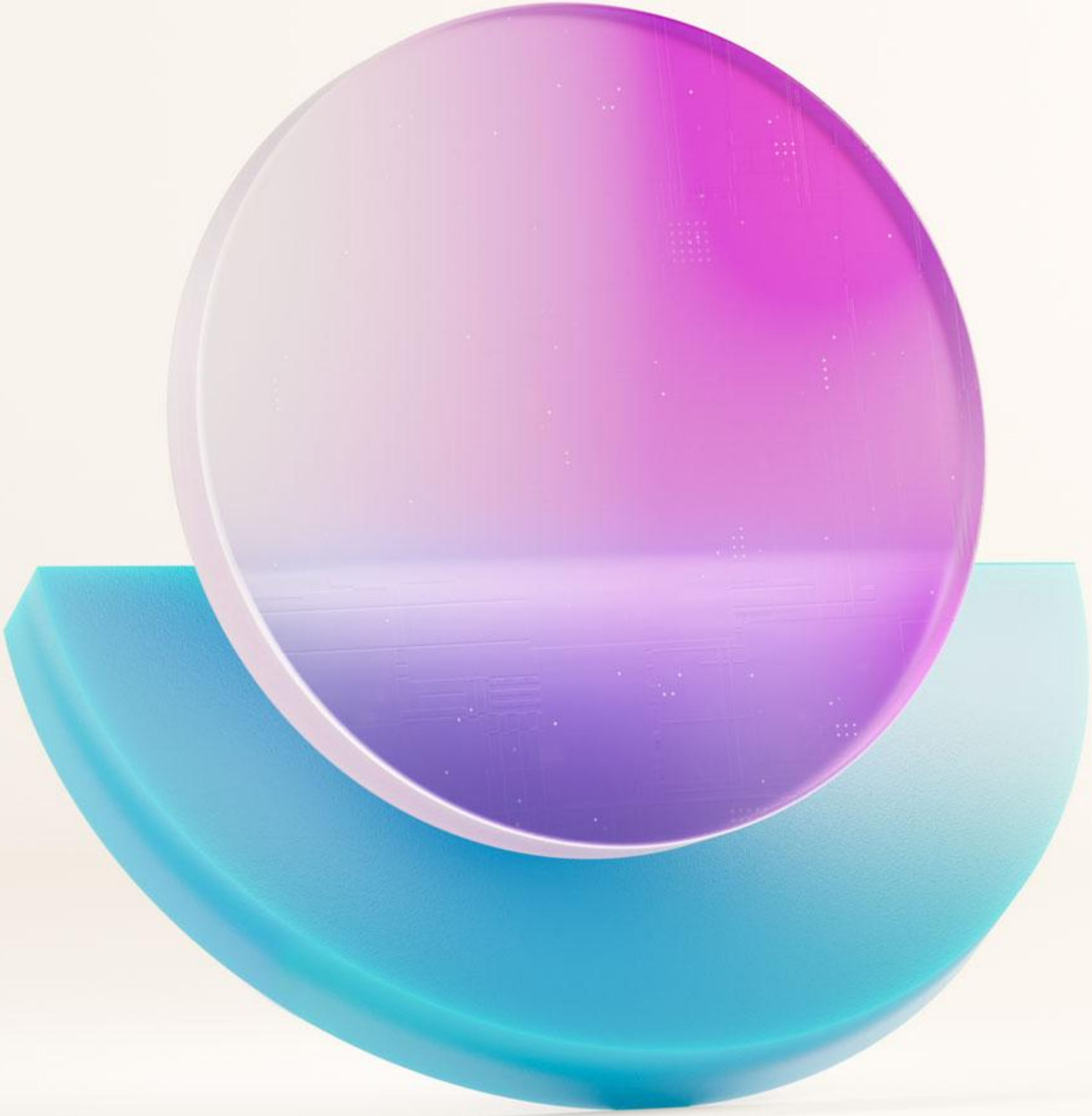
Microsoft

Agenda (times are approximate and will be fluid with the event)

Morning		
9:00 AM – 9:15 AM		Introduction & setup
9:15 AM – 10:30 AM	Eran, Wilson, Mark	Microsoft Fabric and Data Factory overview
10:30 AM – 11:00 AM		⏳ Morning Break
11:00 AM – 1:00 PM	Eran, Wilson, Mark	Orchestrating data movement
1:00 PM – 2:00 PM		⏳ Break for lunch
Afternoon		
2:00 PM – 3:15 PM	Eran, Wilson, Mark	Transforming and copying data
3:15 PM – 3:45 PM		⏳ Afternoon Break
3:45 PM – 5:00 PM	Eran, Wilson, Mark	Transforming and copying data

Microsoft Fabric
Community Conference



Let's get some prep work out of the way

Workshop content

Slide content

Designed to provide overviews and highlight key concepts

+

Demo videos

Pre-recorded insights into how these concepts and features function within Fabric Data Factory that are out of the scope for a hands-on lab in this workshop

+

Lab experience: Hands-on demos

Experience Microsoft Fabric **for free** with our demo environment and guided tutorials on Fabric Data Factory's key concepts and features

Everyone learns differently

Visual

Reads diagrams

Takes detailed notes

Studies notes on slides

Great sense of direction

Auditory

Participates vocally

Enjoys story-telling

Engages in open discussions

Understands changes in tone

Kinesthetic

Likes to explore

Learns by doing

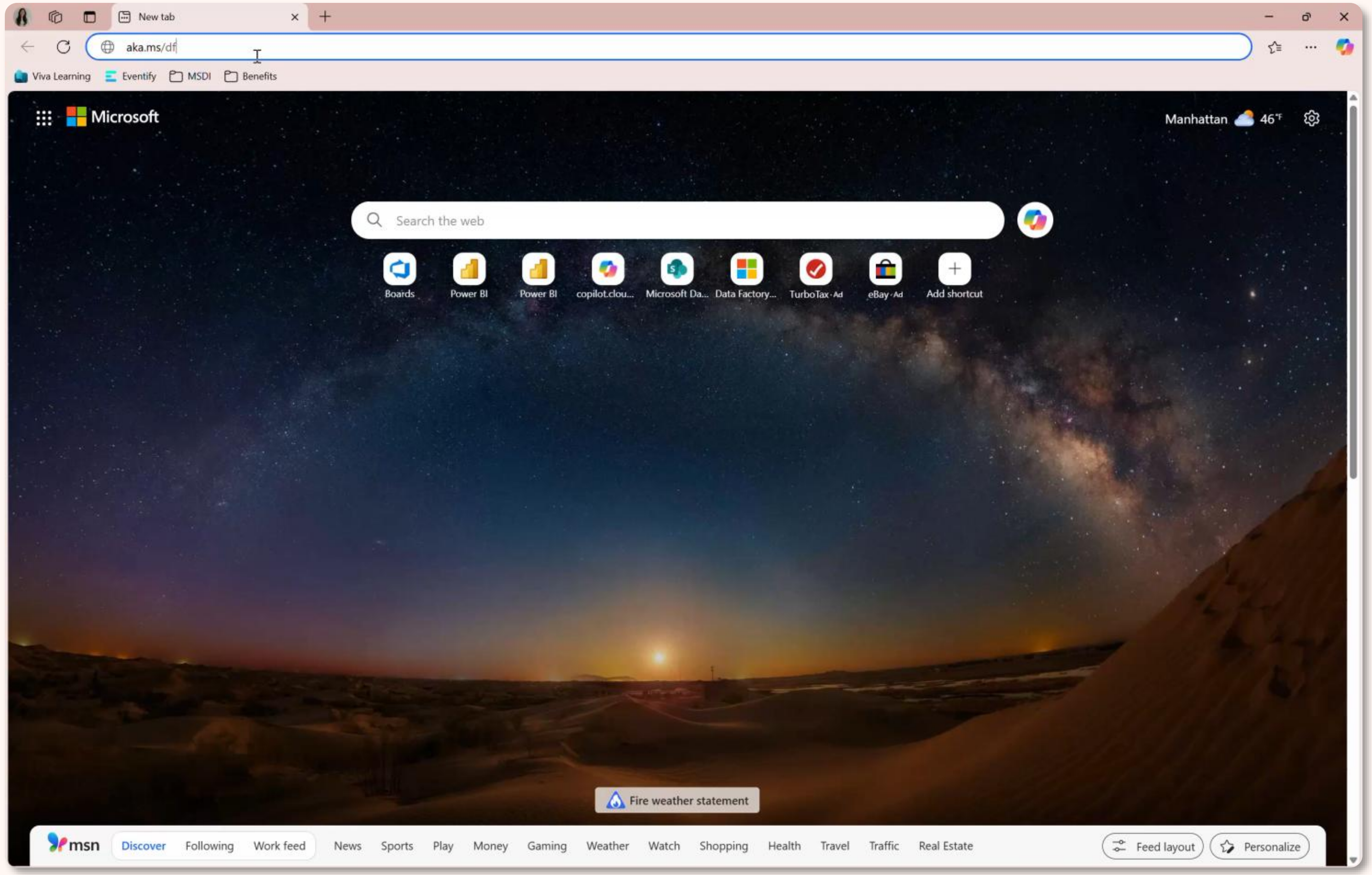
Gets joy from building

Enjoys the “clicks” and “keys”

Today's workshop material + How to access lab instructions

All materials are available at:
https://aka.ms/dfiad

Start the lab with:
Getting started



Logging in with your free Fabric Demo User Profile

Open a new incognito browser:

Ctrl + Shift + N

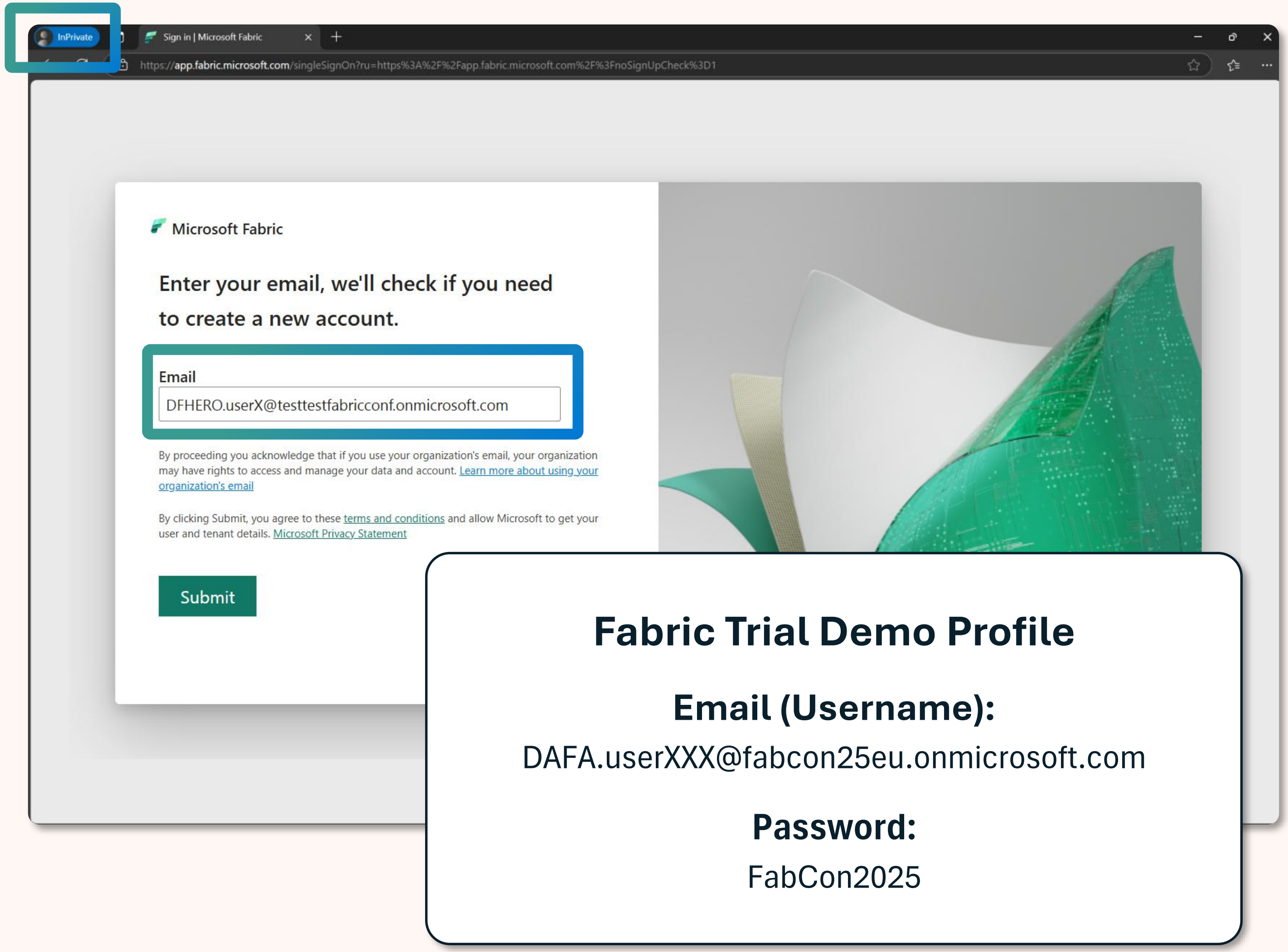
Login with demo user profile on:

https://fabric.microsoft.com

*Do not use multi-factor authentication

Username: on your paper handout

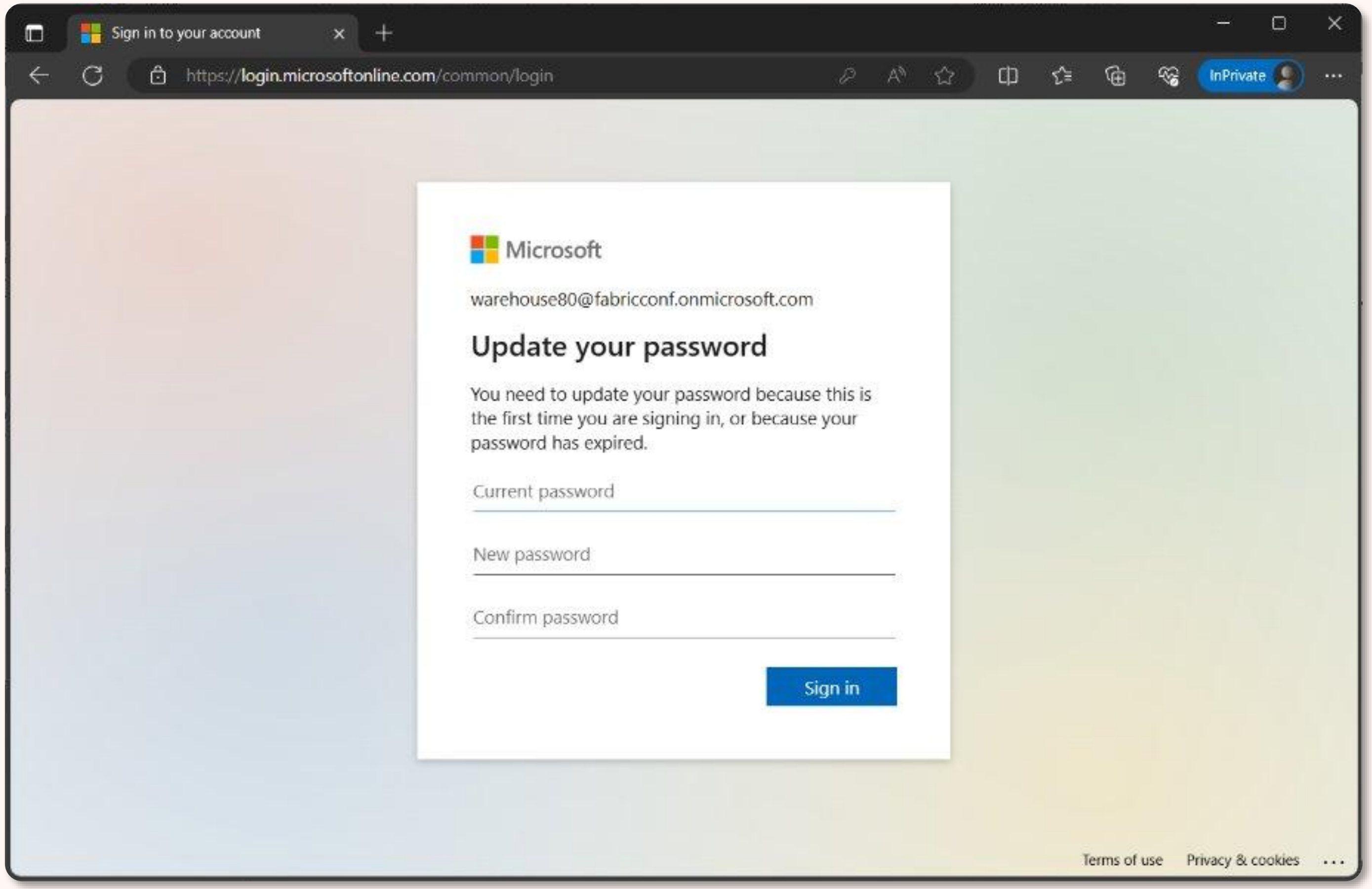
Password: FabCon2025



Logging in with your free Fabric Demo User Profile

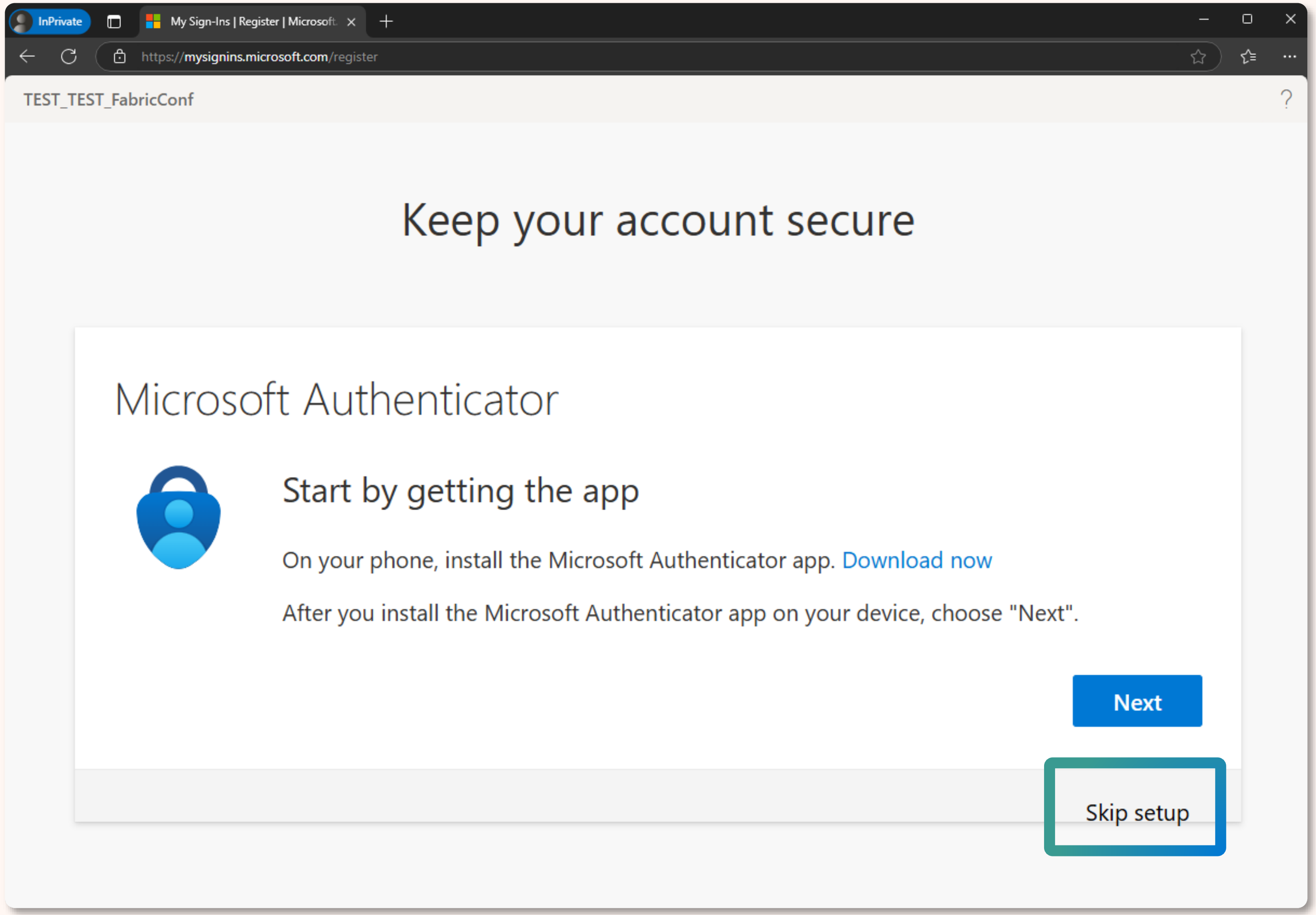
The first time you login, you will be prompted to **change the password**

*make sure you remember this new password



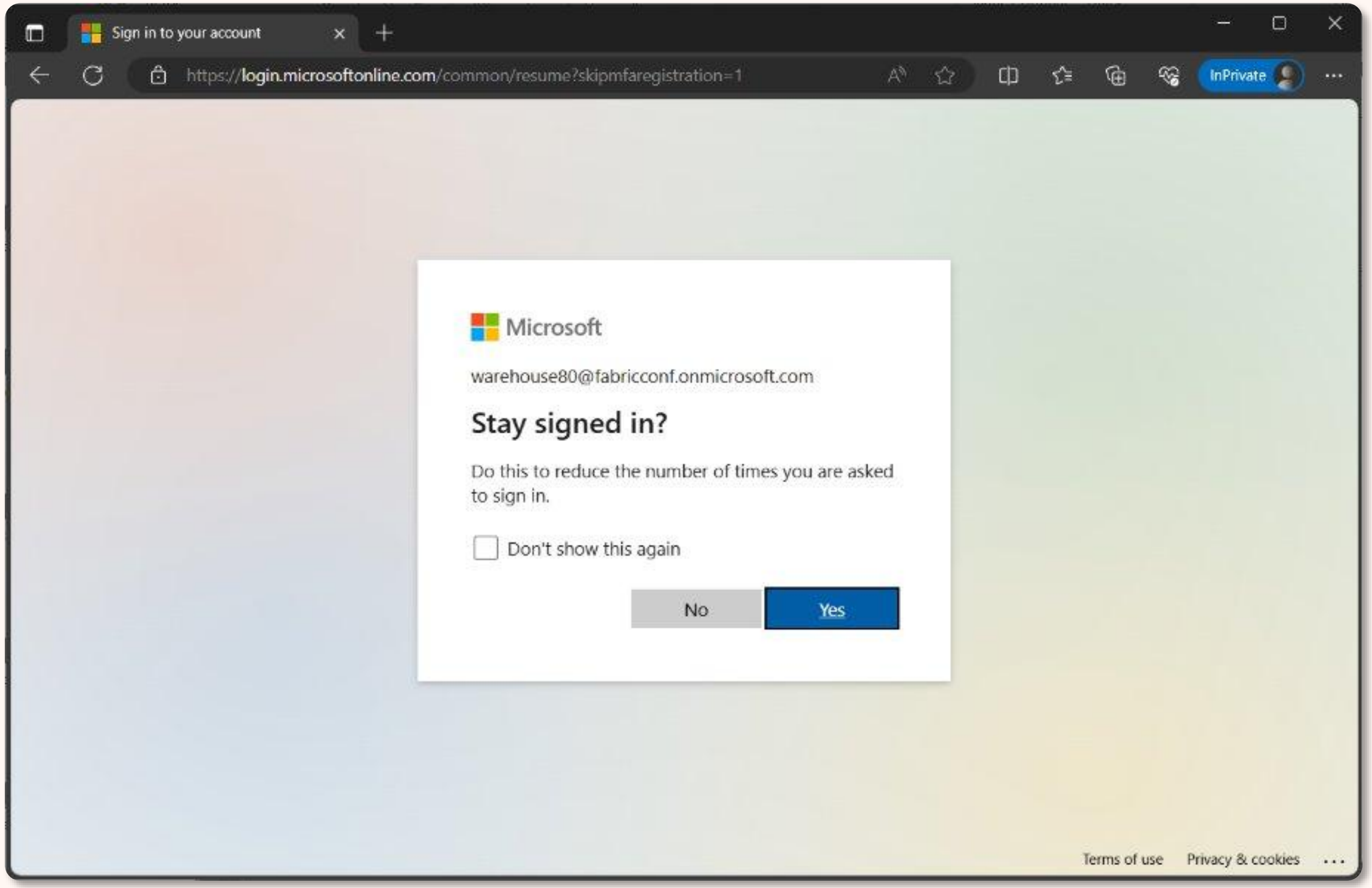
Logging in with your free Fabric Demo User Profile

Do *not* use multi-factor authentication by making sure to click **"Skip setup"**



Logging in with your free Fabric Demo User Profile

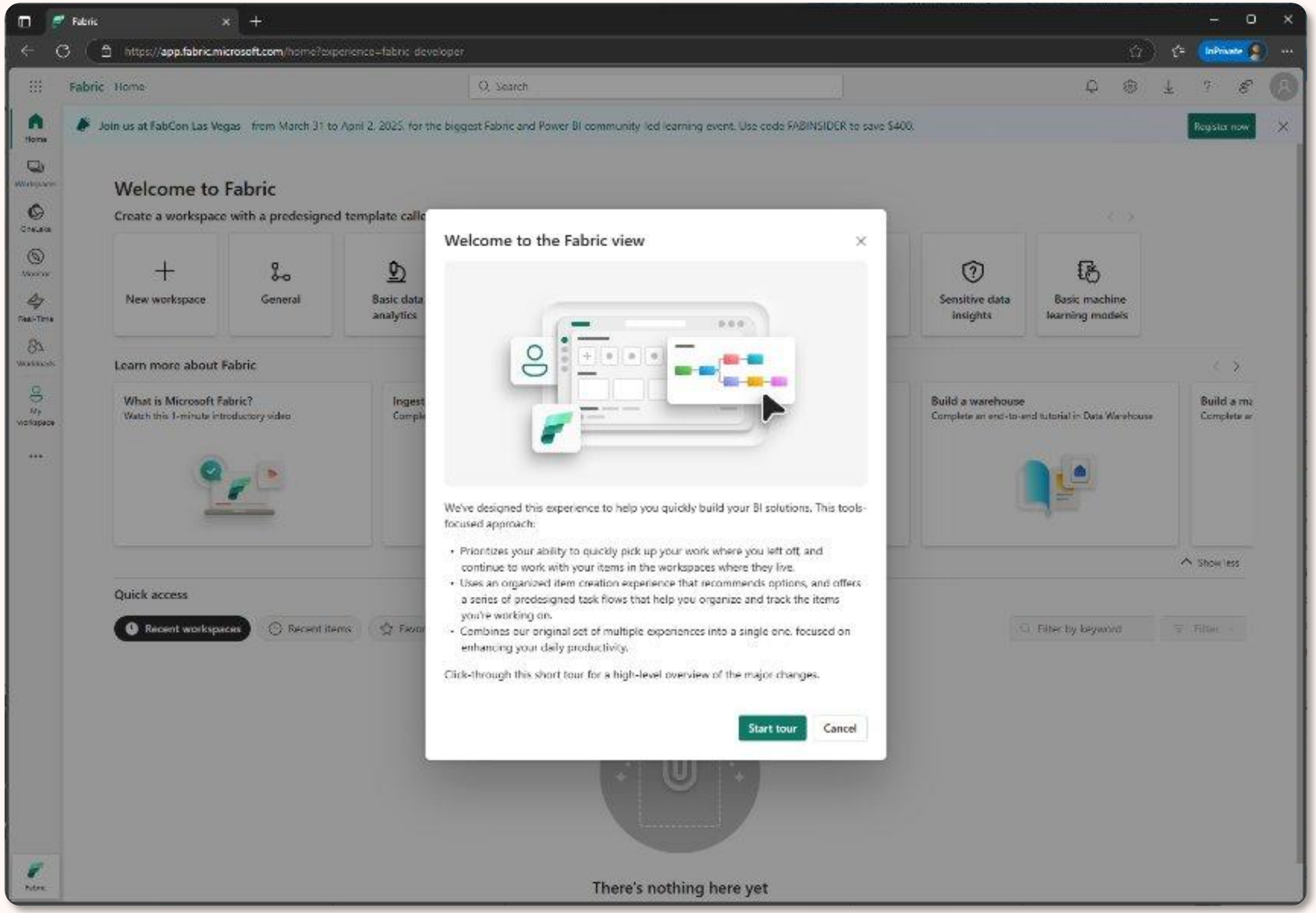
You can select **either option** you prefer for staying signed options



Logging in with your free Fabric Demo User Profile

You are now ready to work with Fabric!

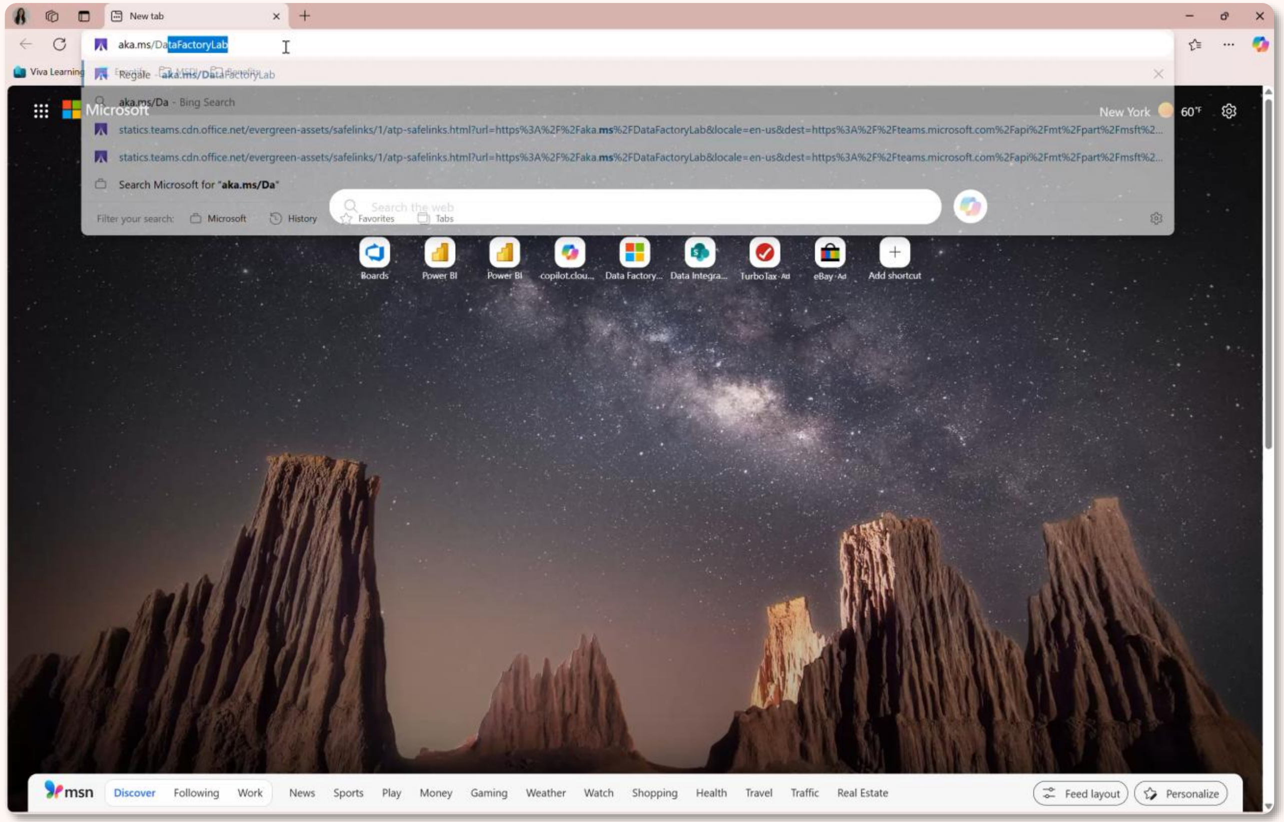
*make sure to **hold onto your handout with your Fabric email (username)**



Troubleshooting with the lab

Something not working on Fabric?

Raise your hand, and we'll have one **our proctors help to troubleshoot** your issues!



Click-by-click lab experience:

https://aka.ms/DataFactoryLab

Make sure the following links are bookmarked

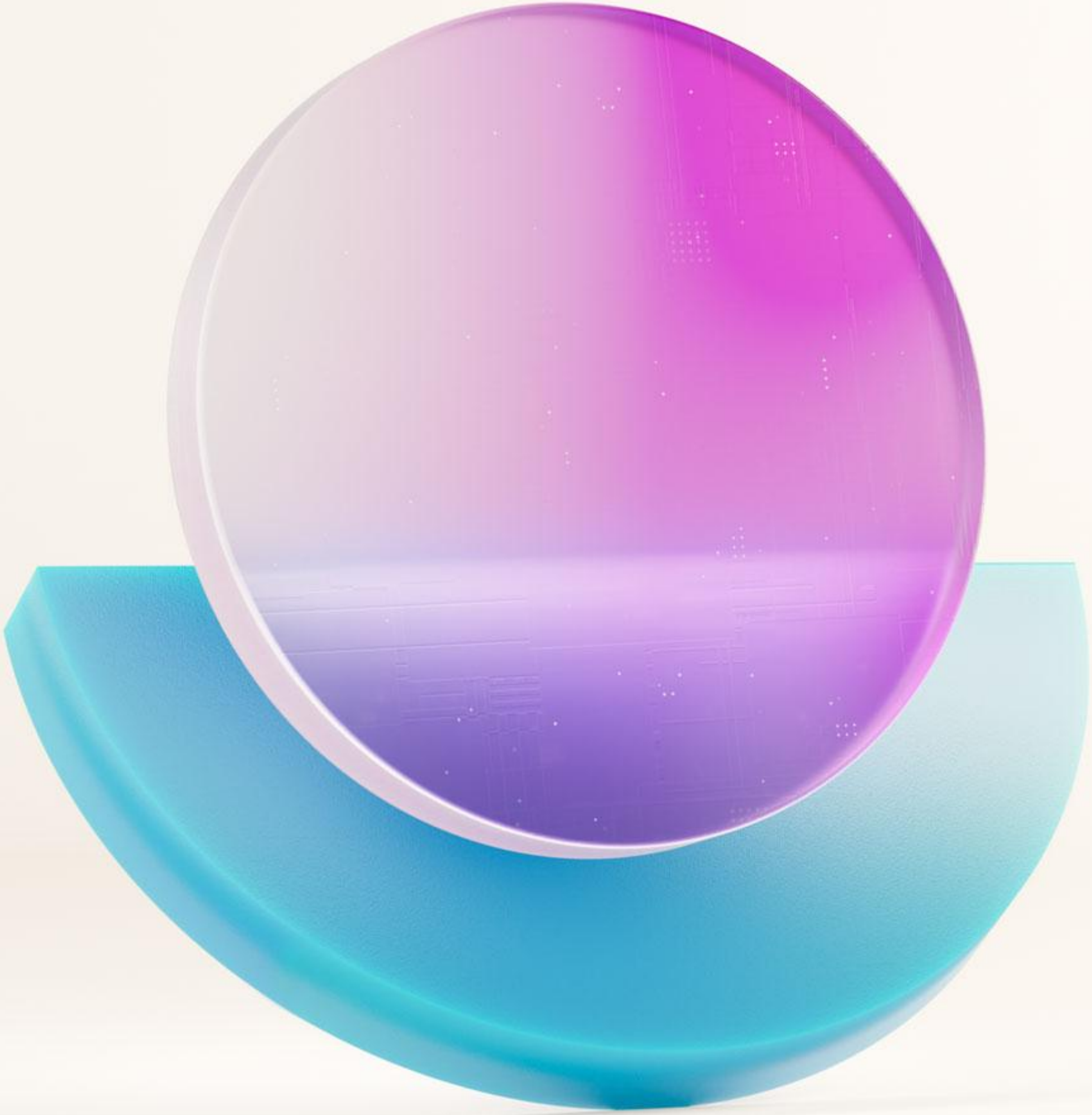
Lab instructions:

https://aka.ms/dfiad

Click-by-click lab:

https://aka.ms/DataFactoryLab

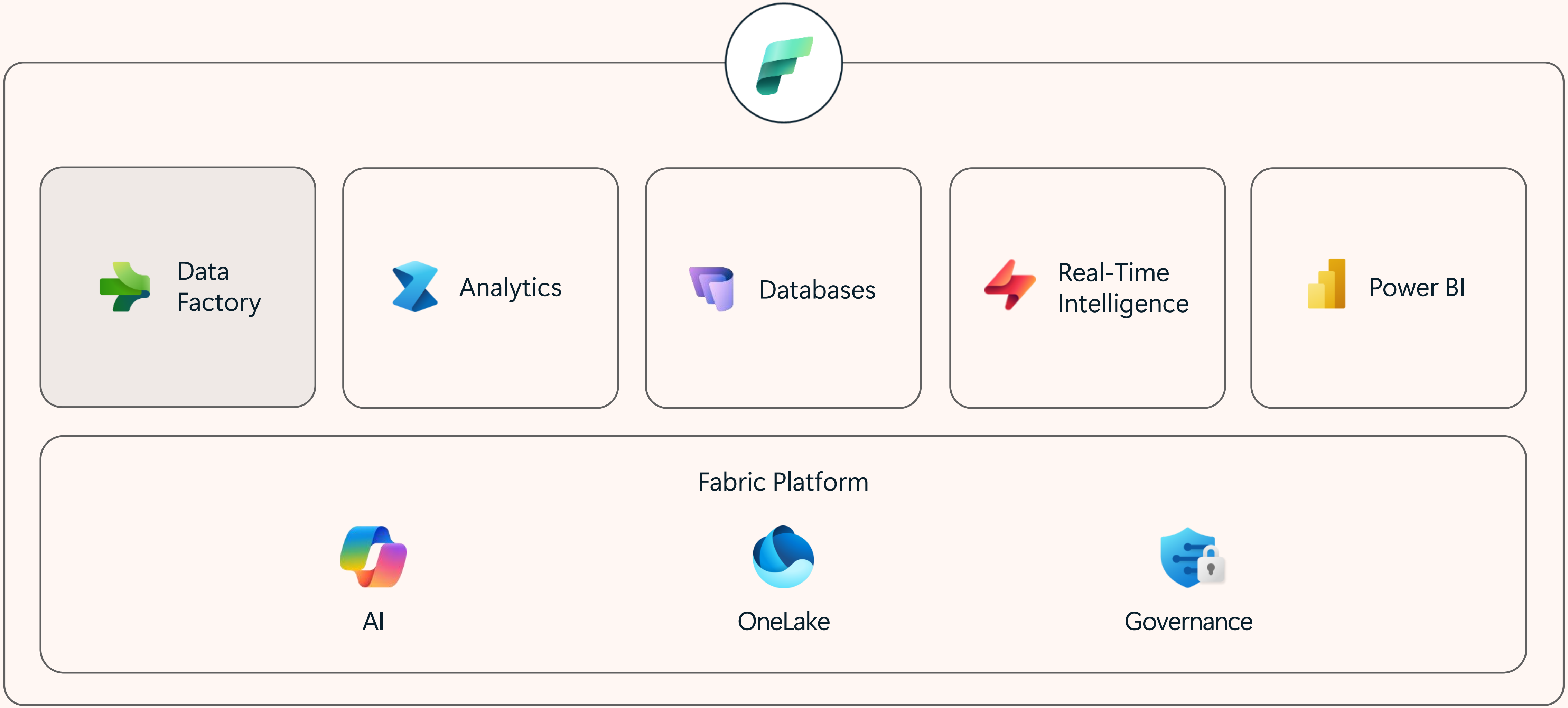
Microsoft Fabric
Community Conference



An Introduction to Microsoft Fabric

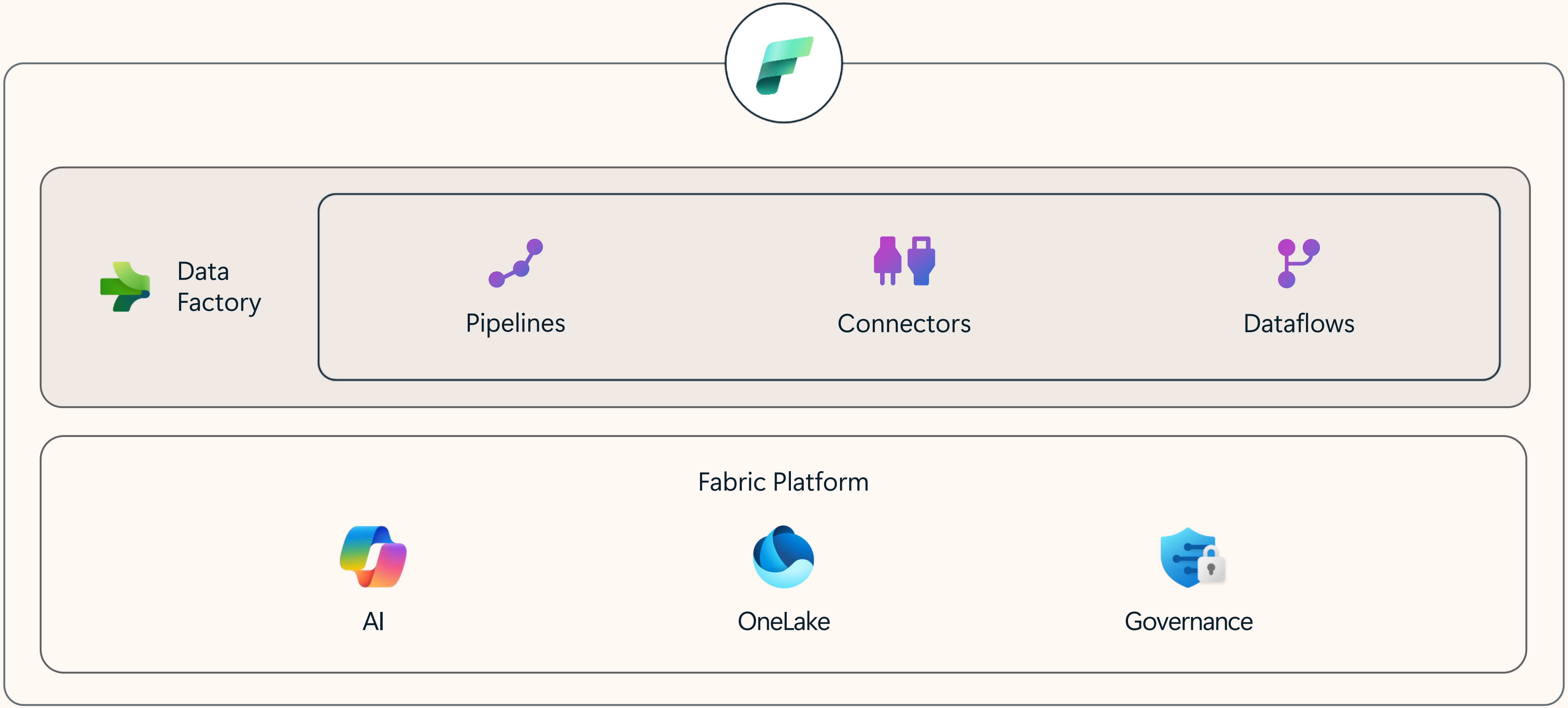
Microsoft Fabric

The unified data platform for AI transformation



Microsoft Fabric

The unified data platform for AI transformation



Microsoft Fabric

The unified data platform for AI transformation

Professional Data Integration ⟷ Citizen Data Integration

Microsoft Azure

Enterprise data integration in Azure

- Azure Data Factory
- Synapse Link
- Purview Integrated

+

Self-service data prep with Power Query

- Excel
- Power BI
- Office 365
- Dynamics
- Power Platform

=

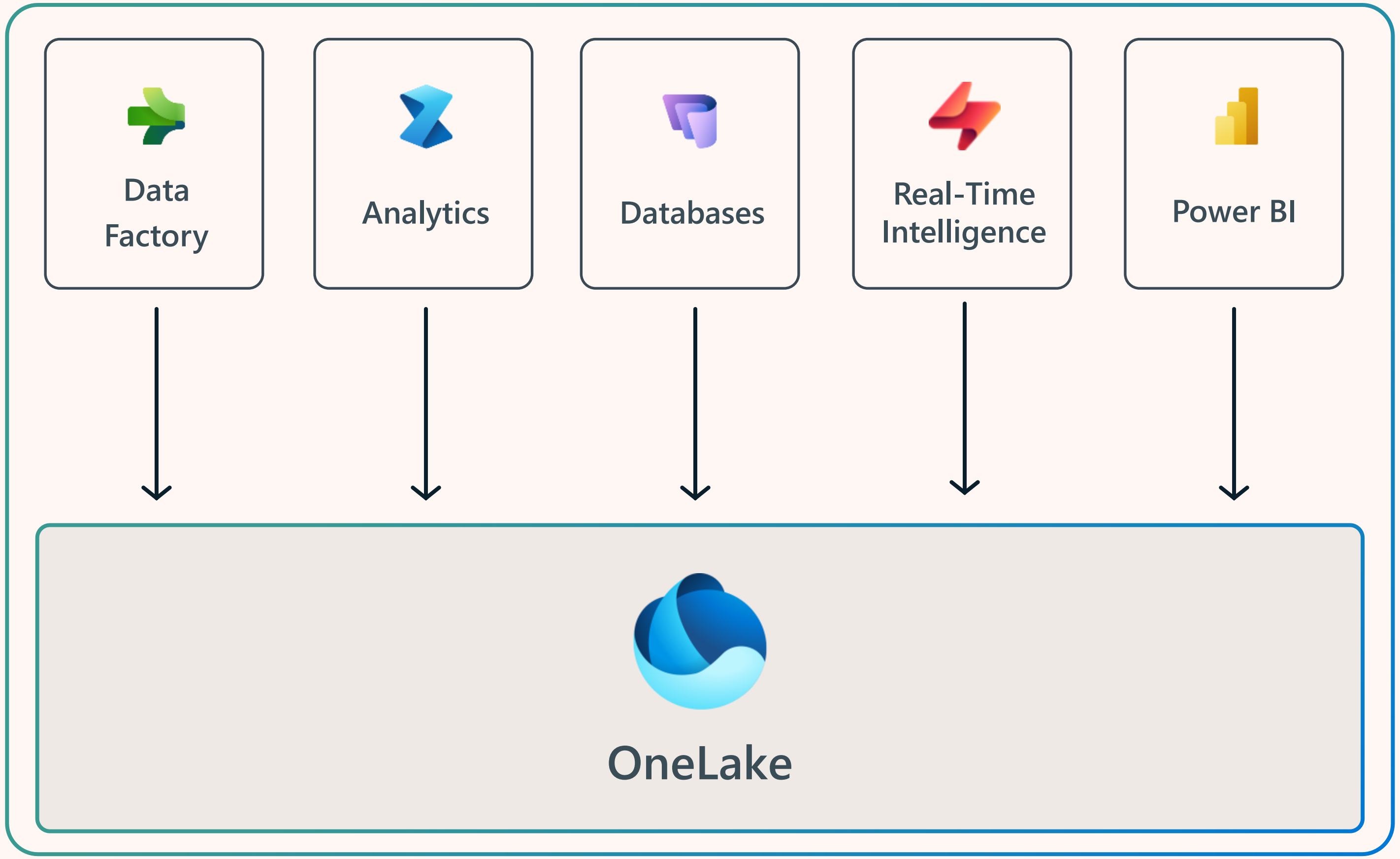
Microsoft Fabric

Data Factory in Microsoft Fabric

- Integrated SaaS platform
- No-code & code tools
- Copilot & AI-powered
- Open Source

OneLake for All Data

The OneDrive for Data



A **single SaaS lake** for the whole organization

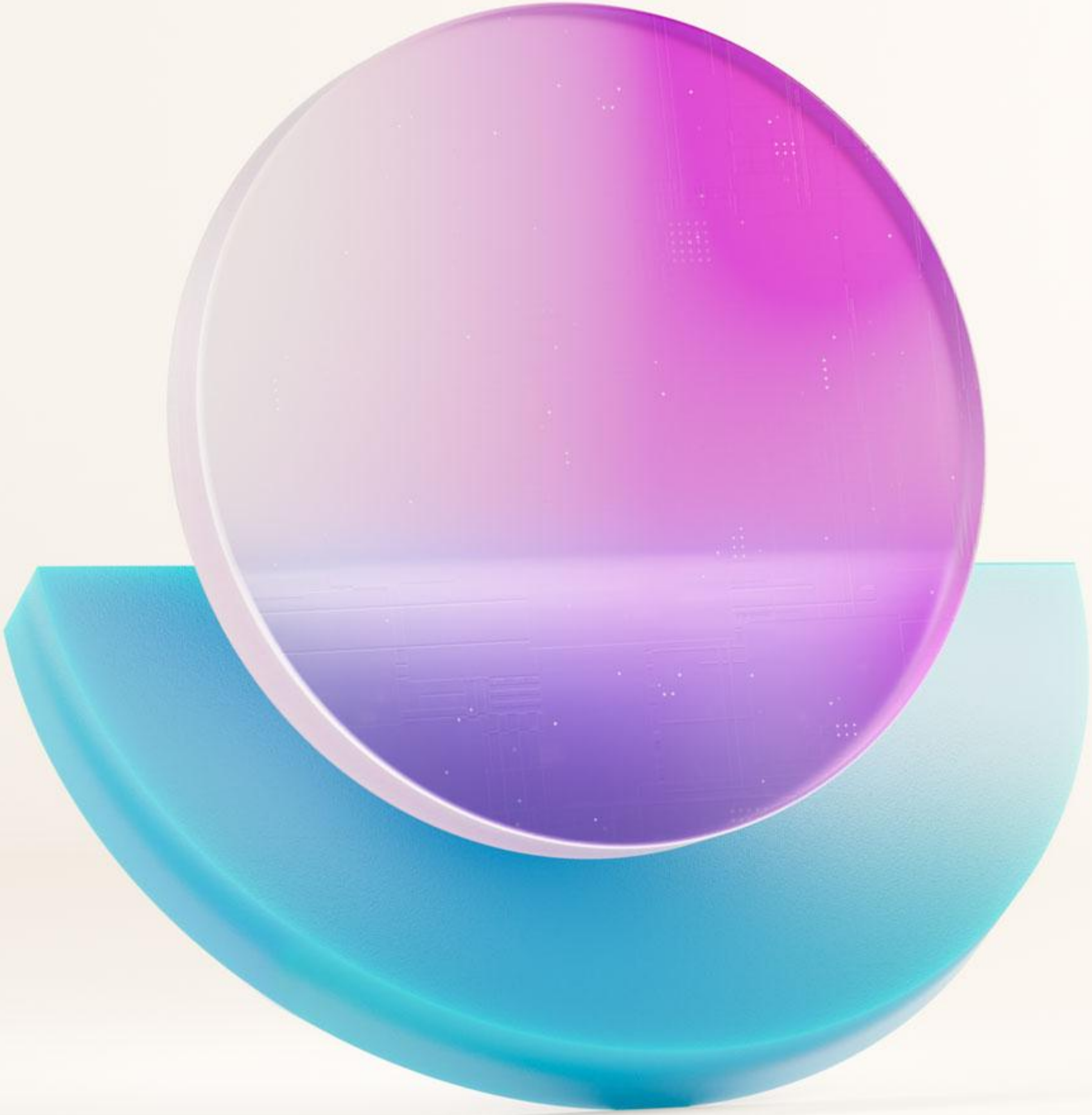
Provisioned **automatically** with the tenant

All workloads automatically store their data in the OneLake workspace folders

All the data is organized in an intuitive **hierarchical namespace**

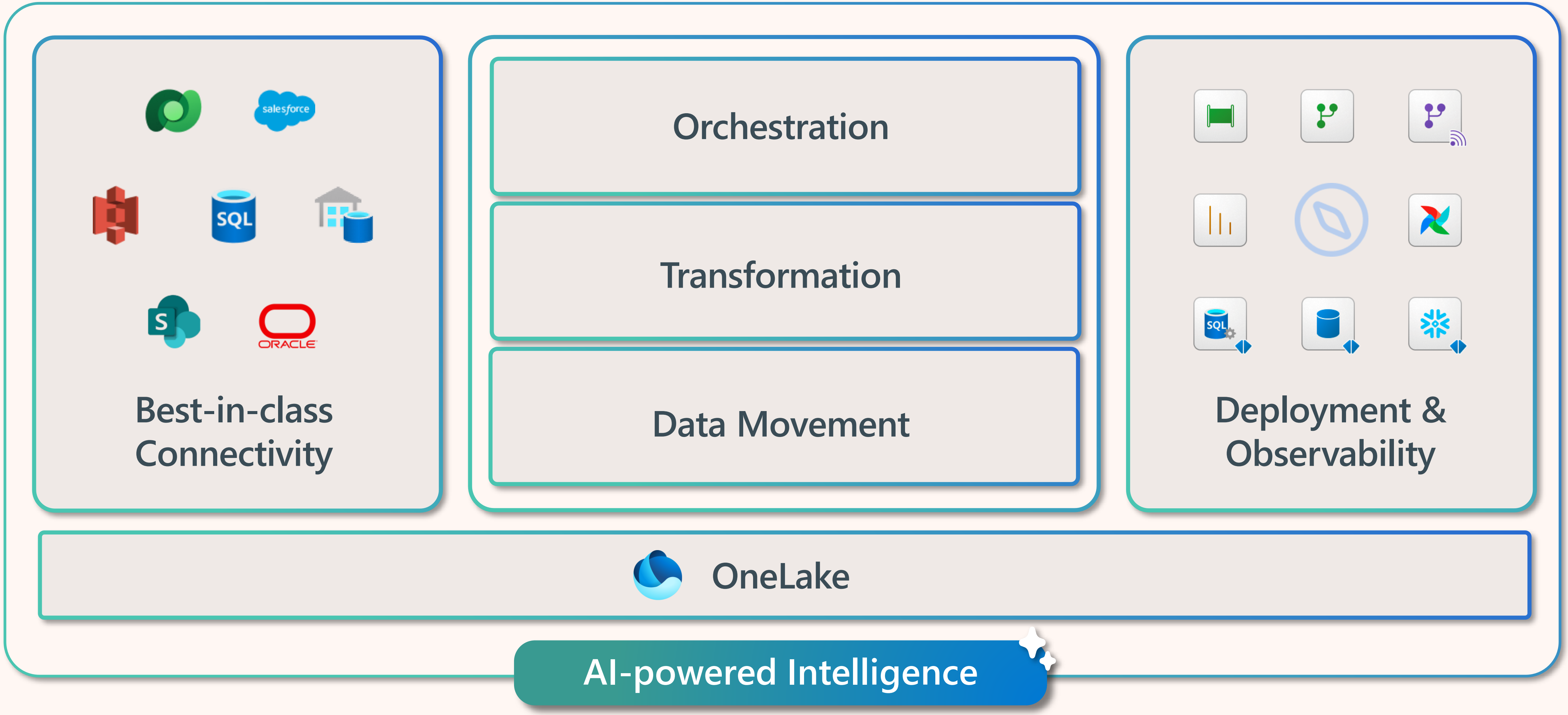
The data in OneLake is automatically indexed for **discovery, Sensitivity labels, lineage, PII scans, sharing, governance and compliance**

Microsoft Fabric
Community Conference



An Introduction to Fabric Data Factory

Data Factory
in Microsoft Fabric



Data Factory
in Microsoft Fabric

Items in Microsoft Fabric

Data pipeline

Ingest data at scale and schedule data workflows.

Copy Job

Copy data easily with settings for full, incremental, or event-based.

Dataflow *Gen2*

Prep, clean, and transform data.

Apache Airflow Job

Simplify creation and management of Apache Airflow environments.

Mirrored Database

Easily replicate data from an existing source into OneLake.

Azure Data Factory

Mount ADF into Fabric to monitor all pipelines in one platform.

Data pipelines
Ingest and orchestrate activities at scale

No code
Data pipelines

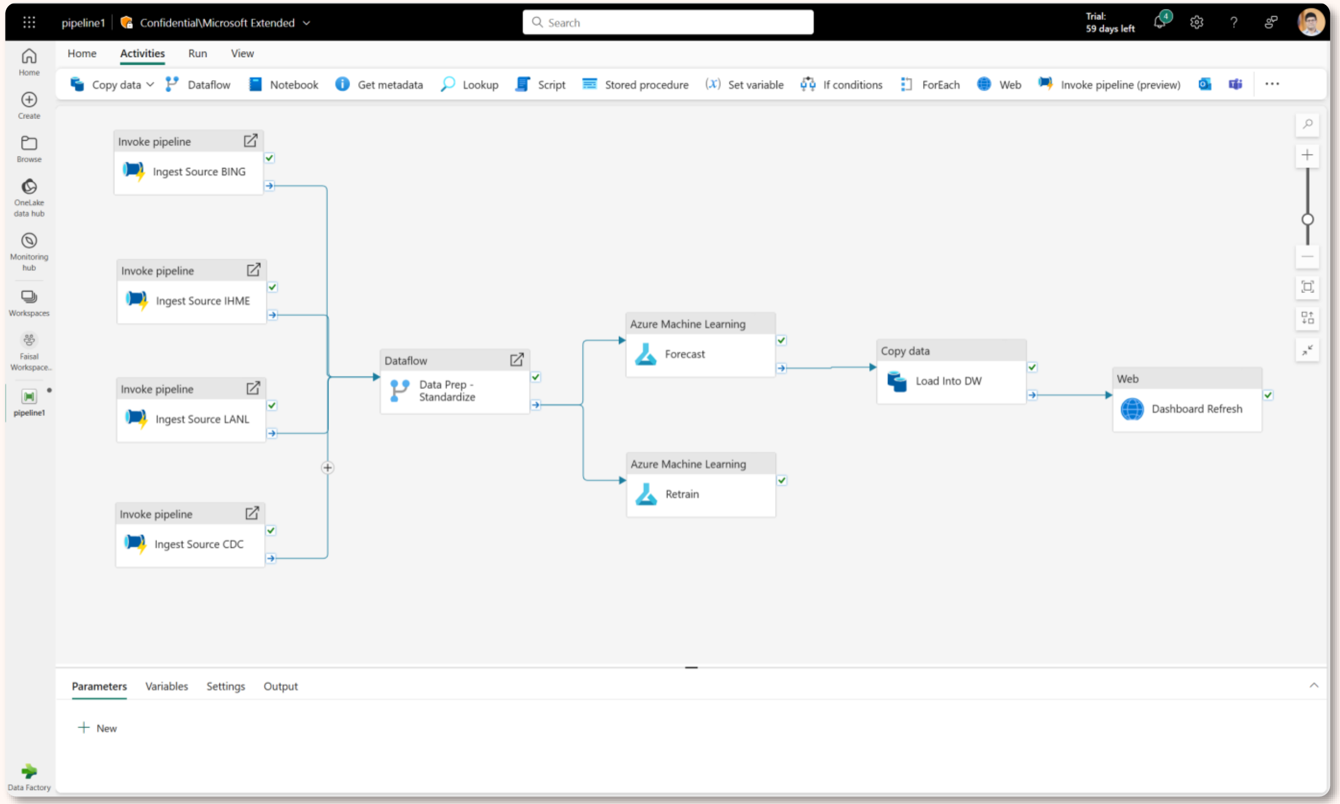
Intuitive, **cloud-based** user experience

Orchestrate data movement, transformation, cleansing, control flow

Triggered and **scheduled** execution models

User Data Functions for specialized functions

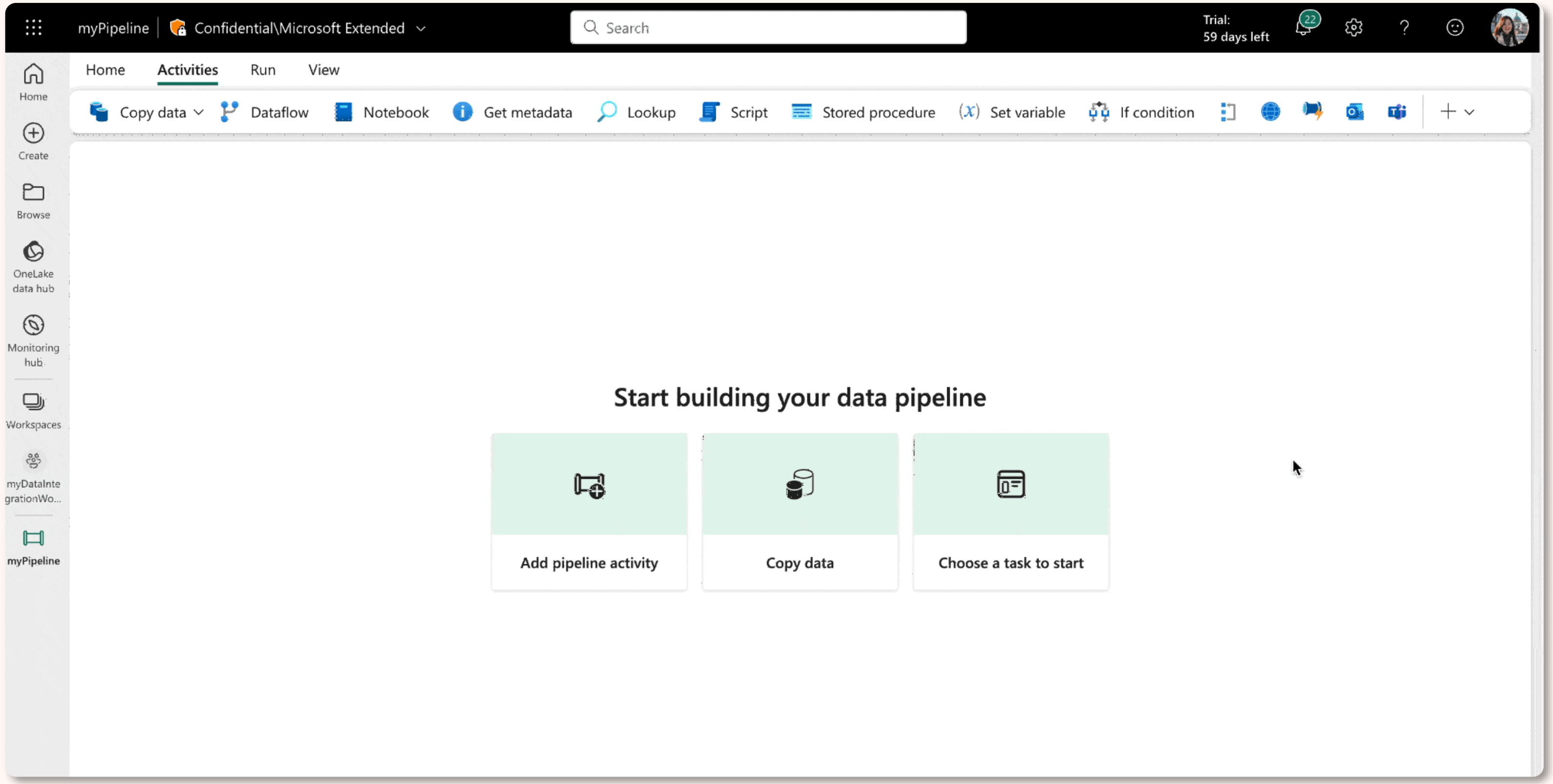
AI Powered with **Copilots**



Data pipelines

Ingest and orchestrate activities at scale

Familiar authoring canvas experience

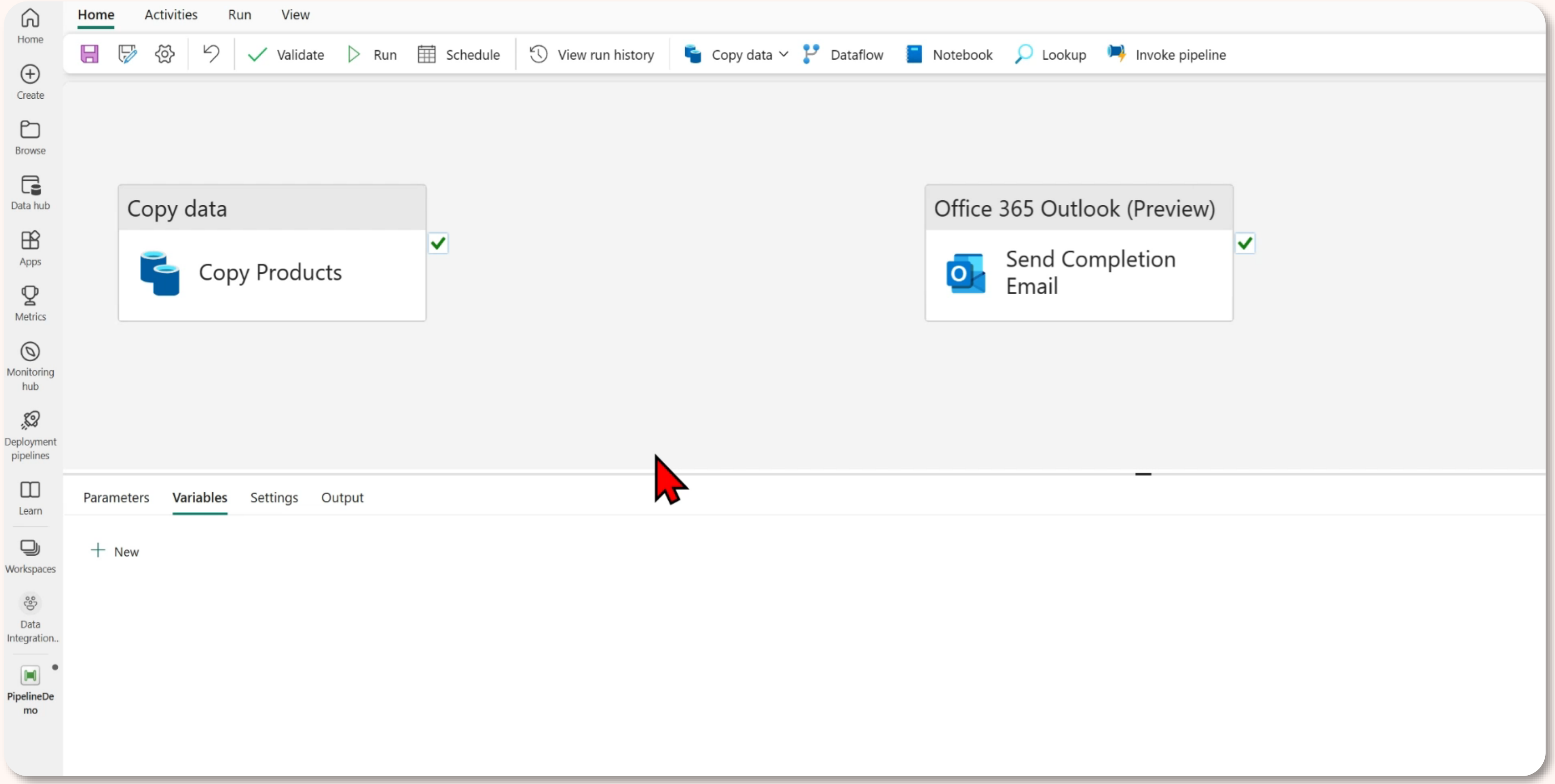


OneLake

Data pipelines

Ingest and orchestrate activities at scale

Empower every person to integrate data



OneLake

Data pipelines

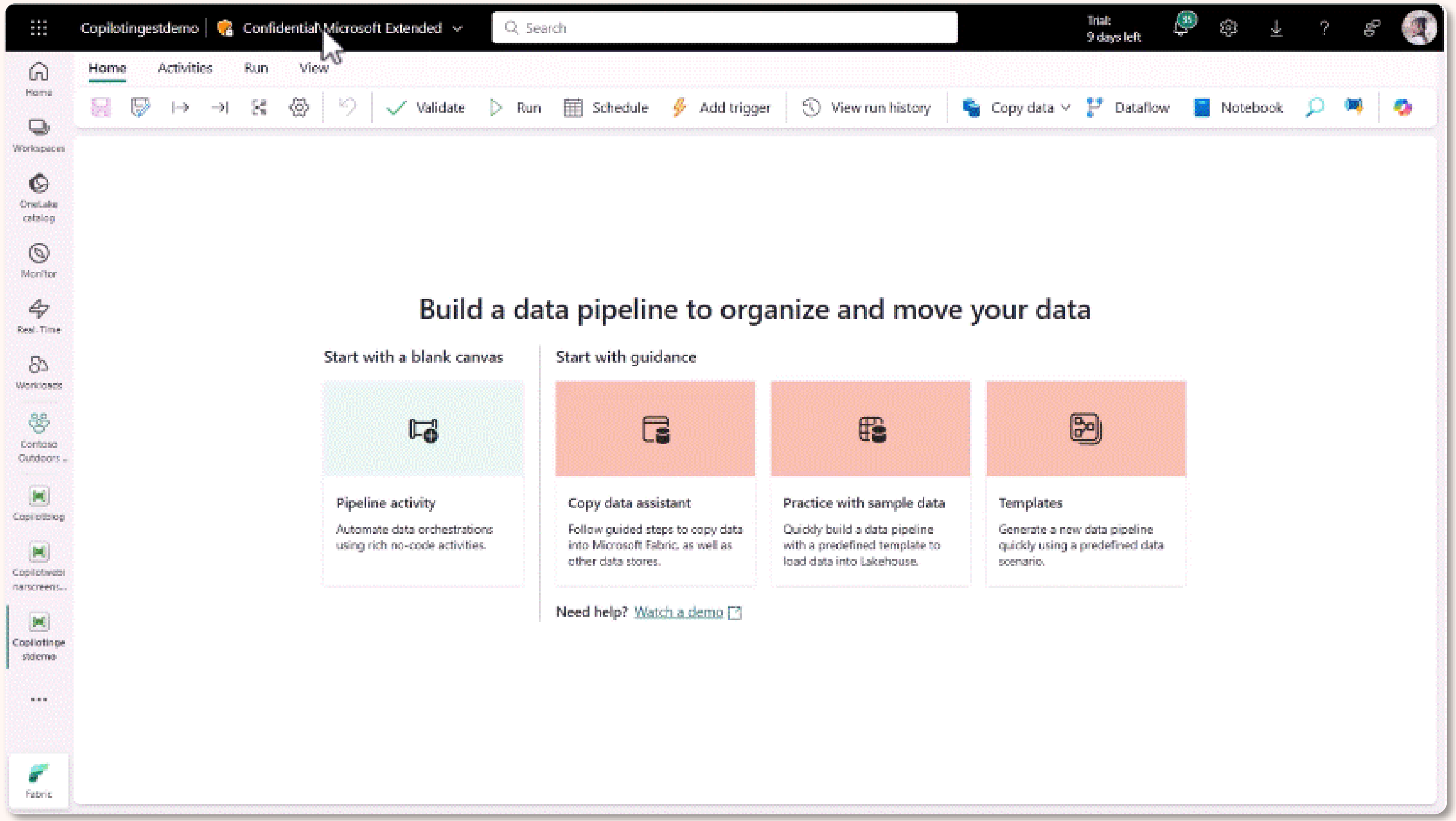
Ingest and orchestrate activities at scale

AI-powered Intelligence

Pipeline generation

Getting from **text to action**

Enabling you to be productive during development of data pipelines



Copy job

Integrate data from built-in connectors

Generally Available

Copy Job item

Streamline copying data into and across OneLake

Setup automatic **incremental refresh**

Built-in support for **copying data between workspaces**

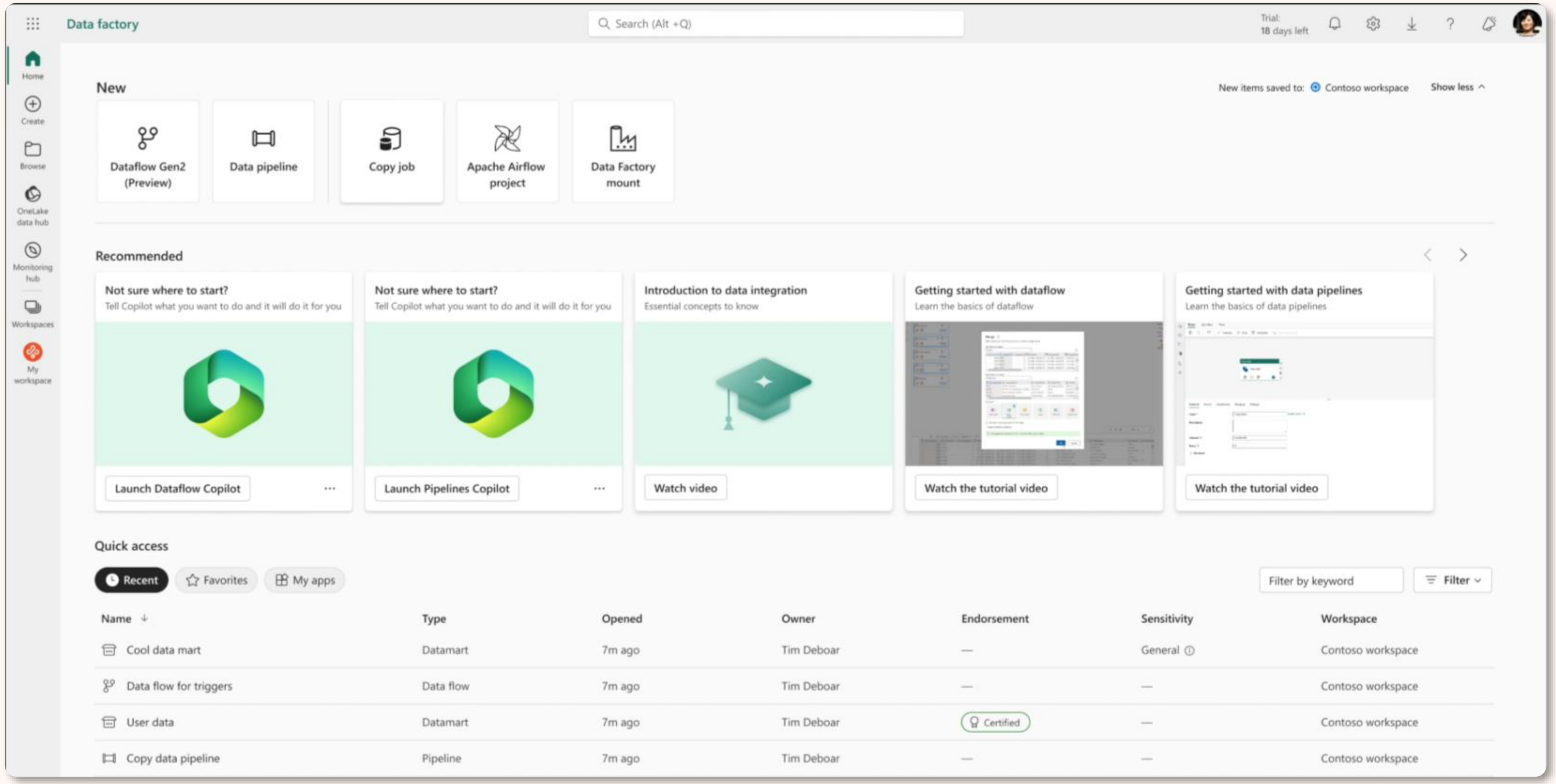
Copy job

Integrate data from built-in connectors

Generally Available

Supports both batch and near real-time incremental copy* (CDC)

(*Incremental copy is still in Public Preview)



OneLake

Dataflows Gen2

Ingest and transform data at scale

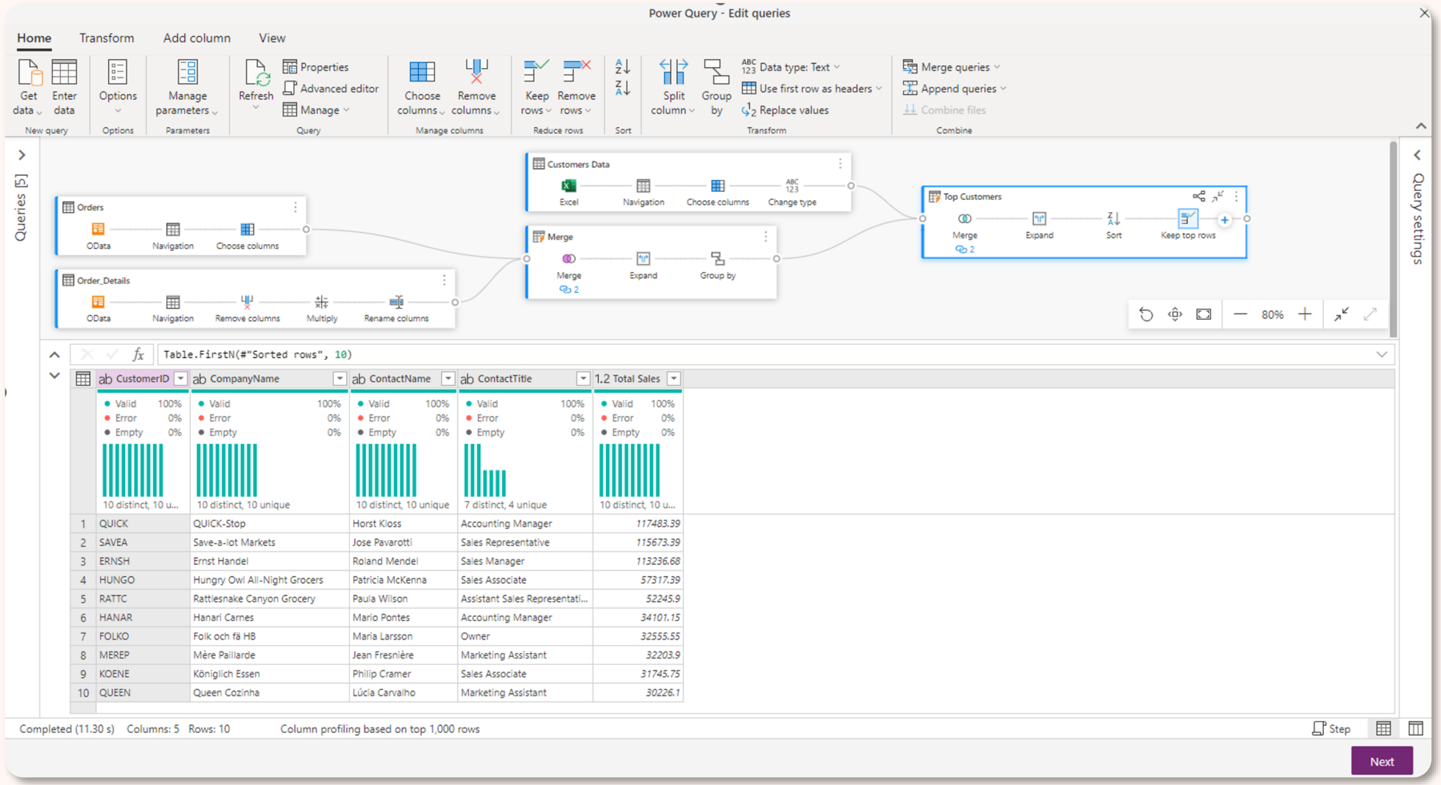
Dataflows Gen2: Power Query with enterprise scale

Based on **Power Query**, used by millions

File, relational, multi-dimensional, SaaS data experiences

Scalable transformations via Fabric compute

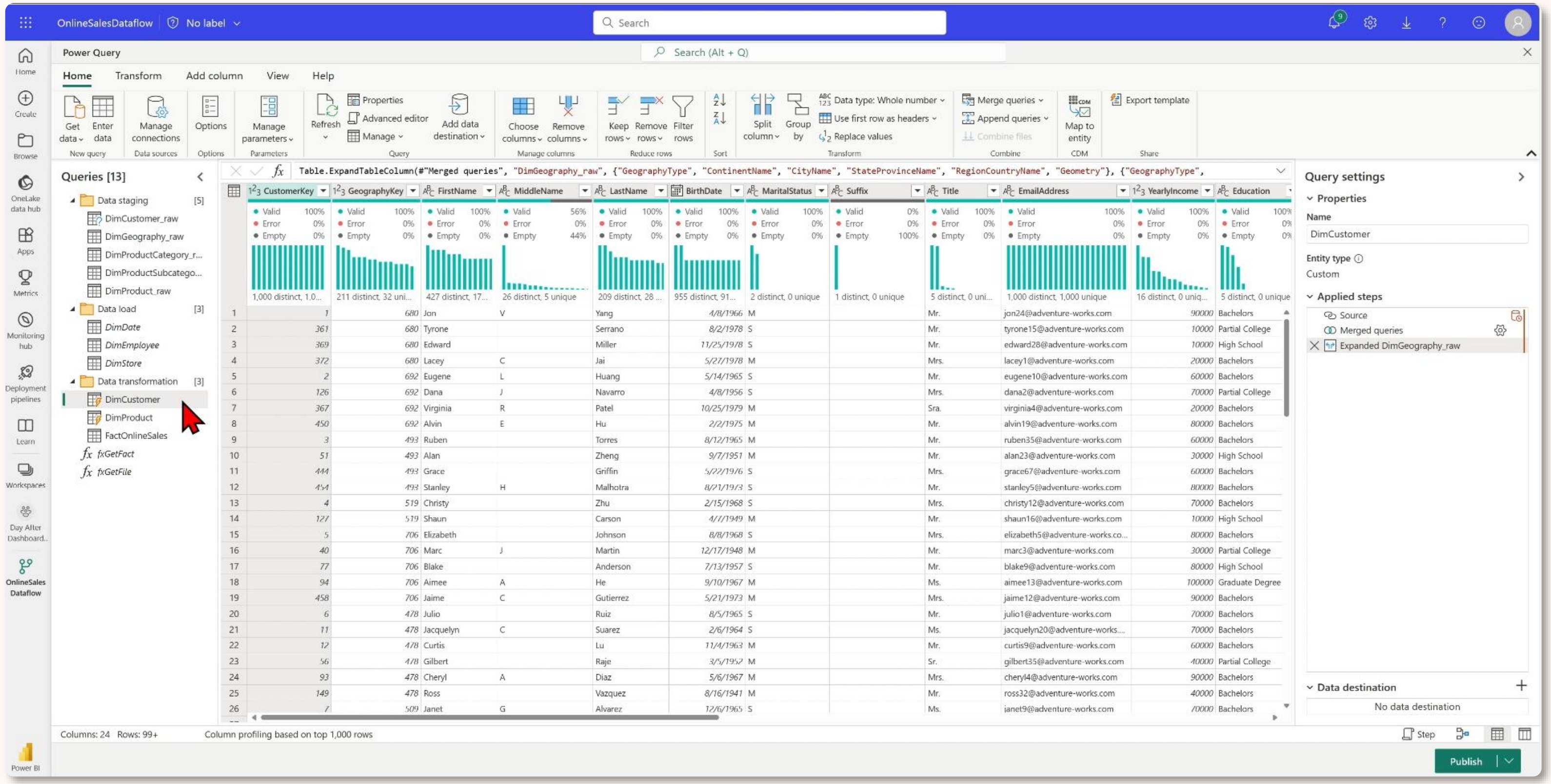
AI Powered with **Copilots**



Dataflows Gen2

Ingest and transform data at scale

Enterprise-scale data ingestion and transformation



OneLake

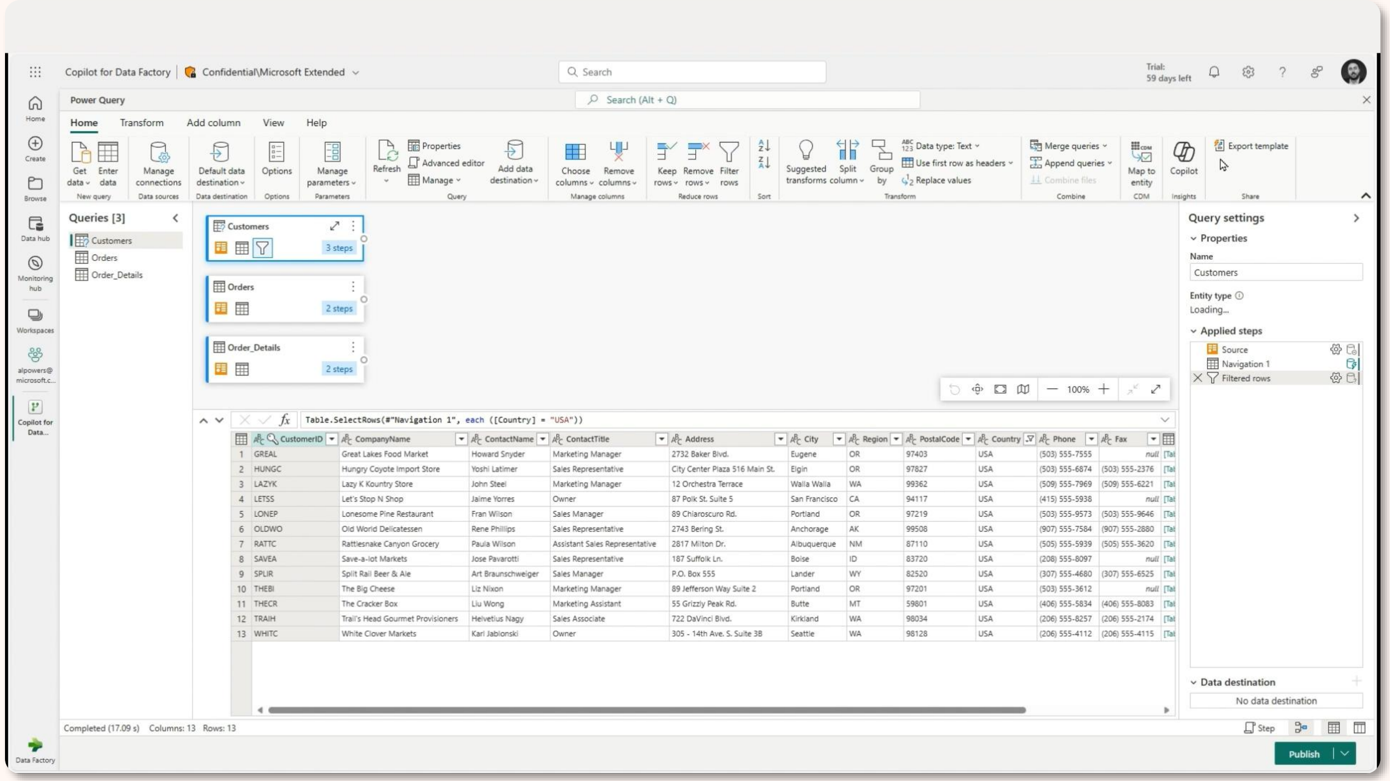
Dataflows Gen2
Ingest and transform data at scale

AI-powered Intelligence

Transformation using Dataflows Gen2

Transform your data quickly using **Copilot** in Dataflows Gen2

Intelligently create new queries as per your requirements



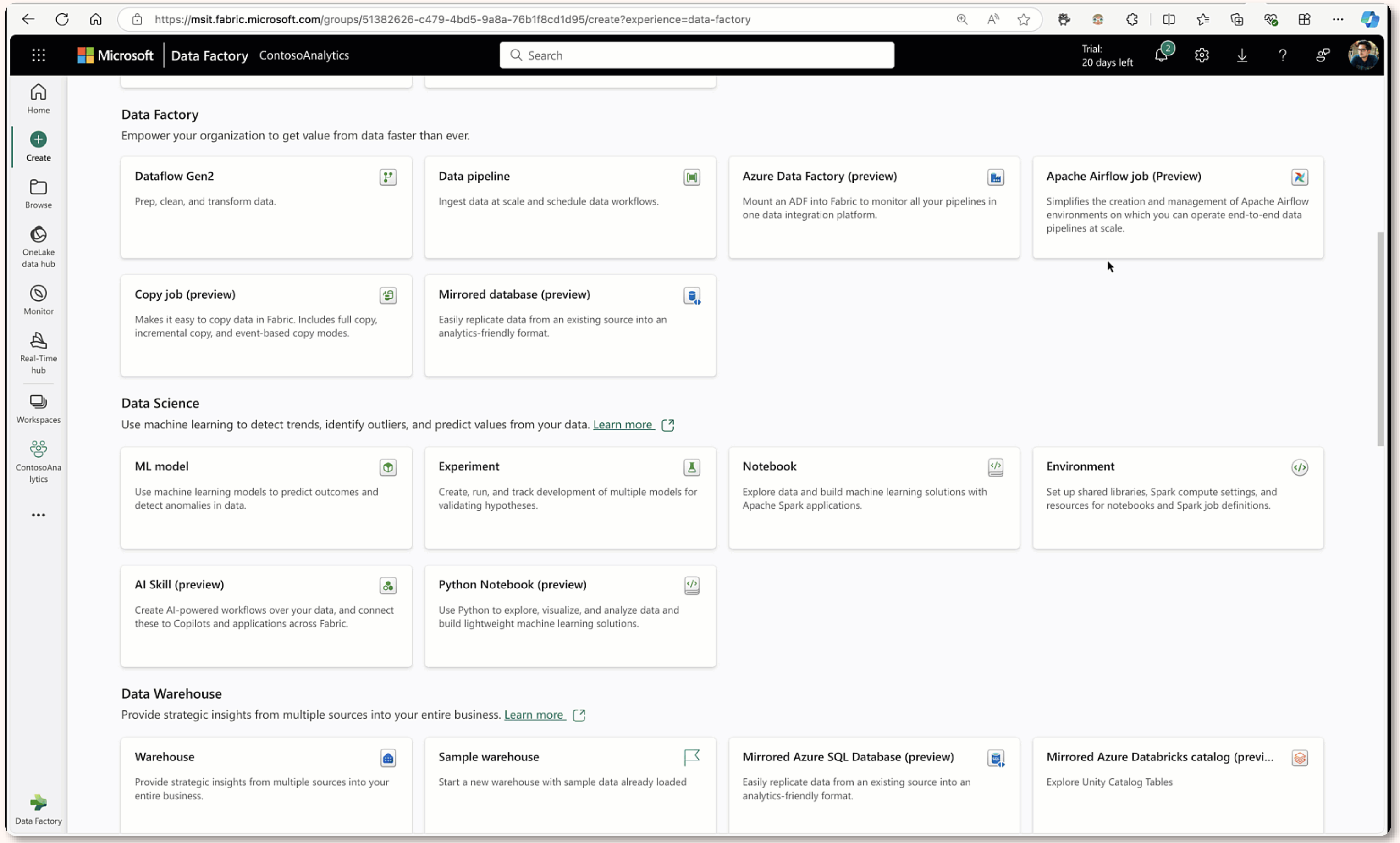
Apache Airflow Job

Data Factory’s workflow orchestration manager

Native support for Apache Airflow

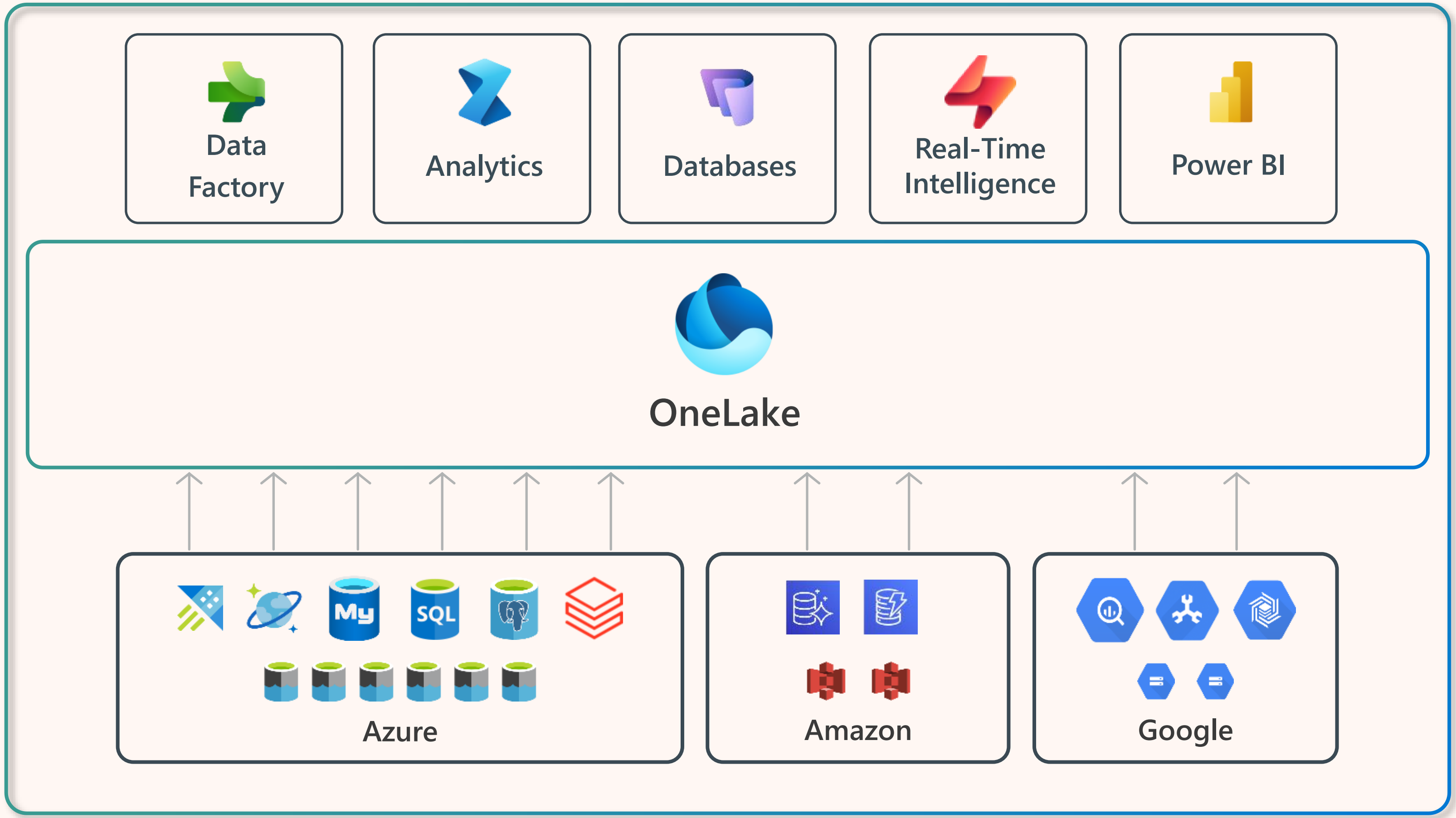
Apache Airflow job - Pro code, Python based data orchestration based on **Open-Source Apache Airflow project**

Serverless model with instant provisioning and **automatic scaling**



Mirrored Database

Bring data from various systems together



Frictionless linking of **external databases, with full replicas** created with a couple of clicks

Available for both **multi-cloud and on-premises** databases

Real time updates of the replicas using the CDC feeds of the database

Data is stored in **Delta Parquet** tables, with all Fabric services instantly available

Azure Data Factory
Mount your ADF items to Microsoft Fabric

Upgrade Pathways for ADF & Synapse customers to Fabric

Data Factory in Fabric

Data pipelines in Fabric feature parity with ADF

1

Available today

ADF Connectors in Fabric

Connectors to Lakehouse and Data Warehouse (DW)

2

Available today

Mounting ADF in Fabric

Azure Data Factory works as-is experience in Fabric

3

Available today

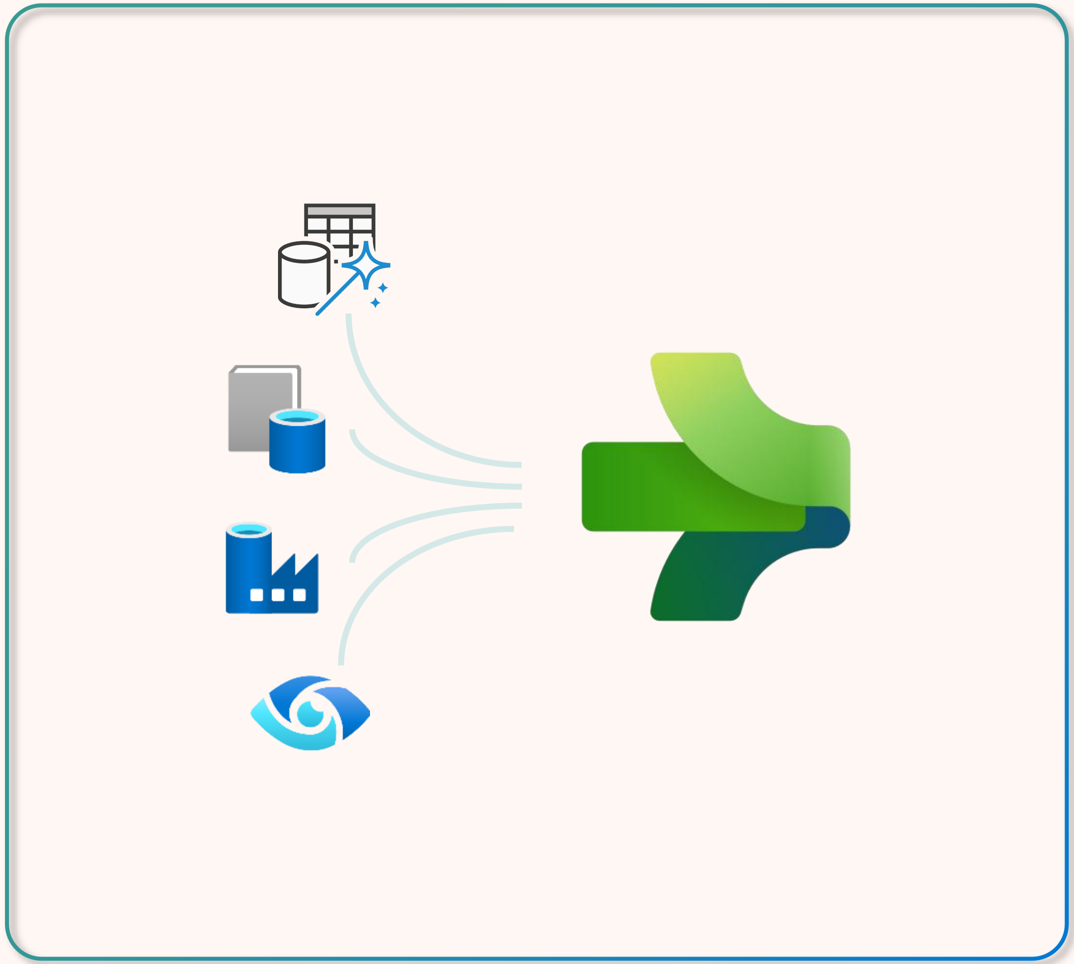
Upgrade to Data Factory in Fabric

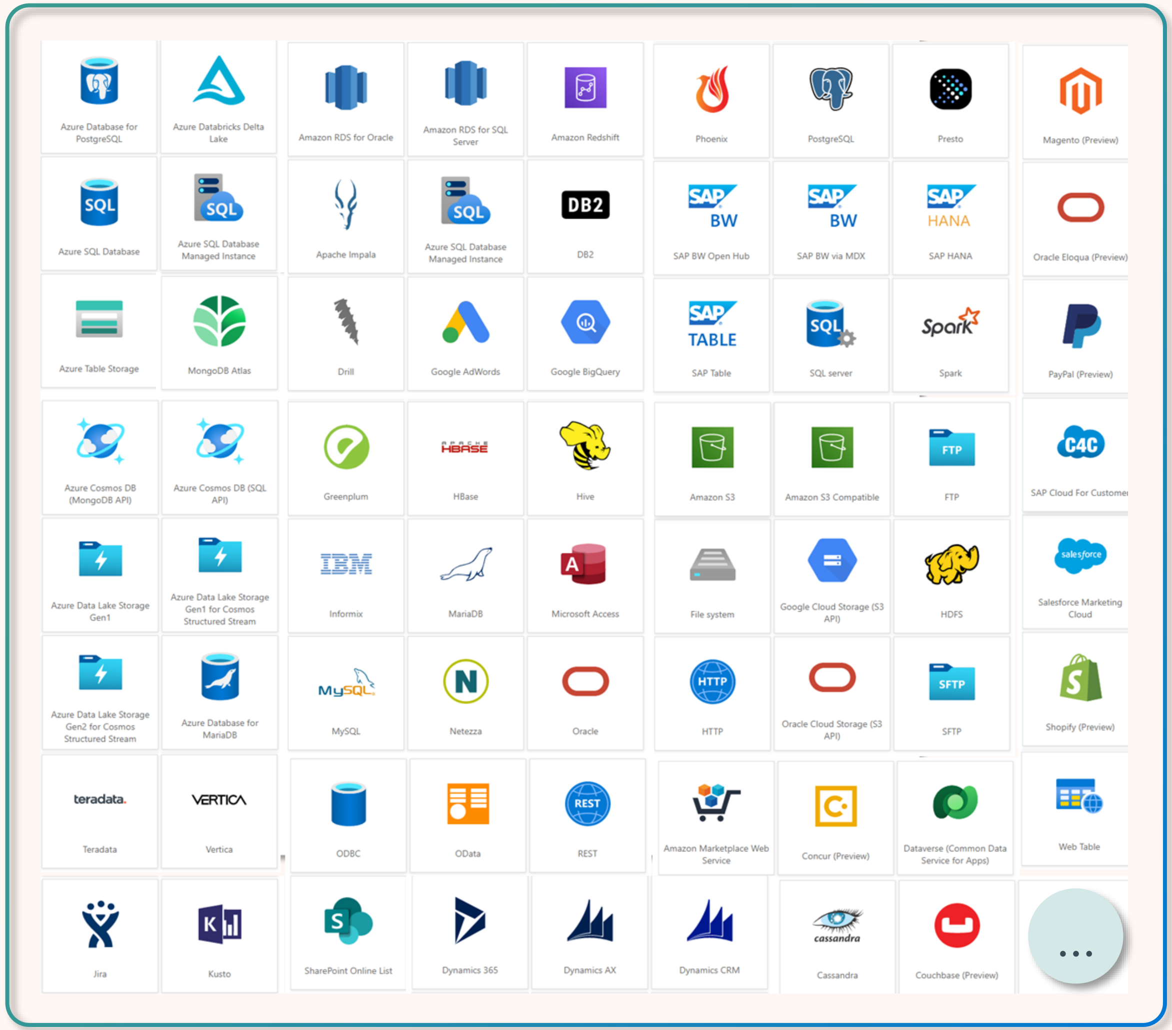
Upgrade your mounted ADF pipelines to native Fabric pipelines

4

Unifying data in OneLake

Seamlessly connect to more than 170+ data sources





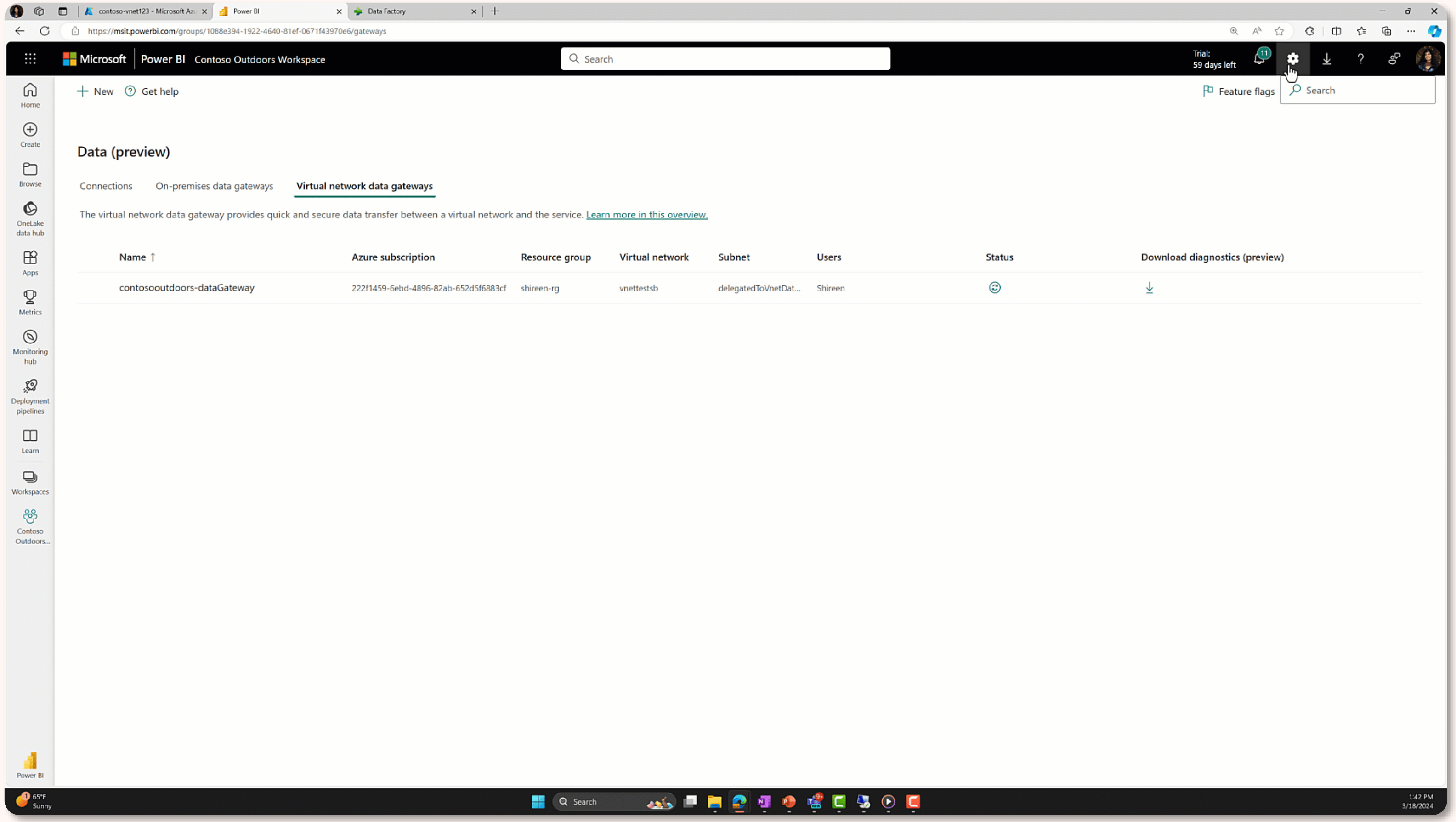
Virtual Network & On Premises Data Gateways

Secure connectivity to your Azure and private data services

VNet & On Prem Gateways

VNet Gateways enable network security compliant access to secured resources

On Prem Gateways enable line of sight to on-prem data sources



CI/CD & Monitoring Hub in Microsoft Fabric

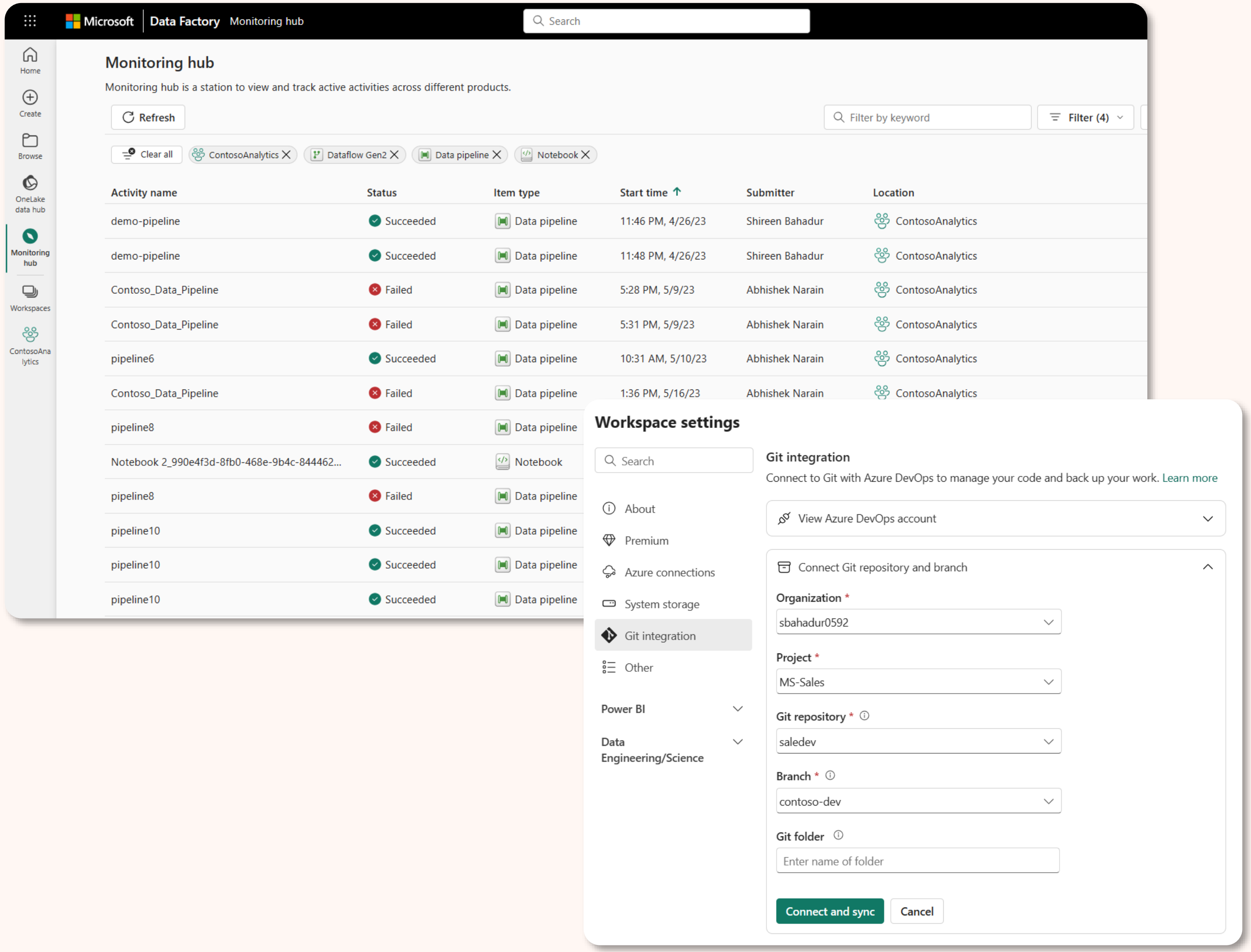
Rich monitoring capabilities and DataOps

Connect to Git with **Azure DevOps** or **GitHub** (cloud) repositories effortlessly

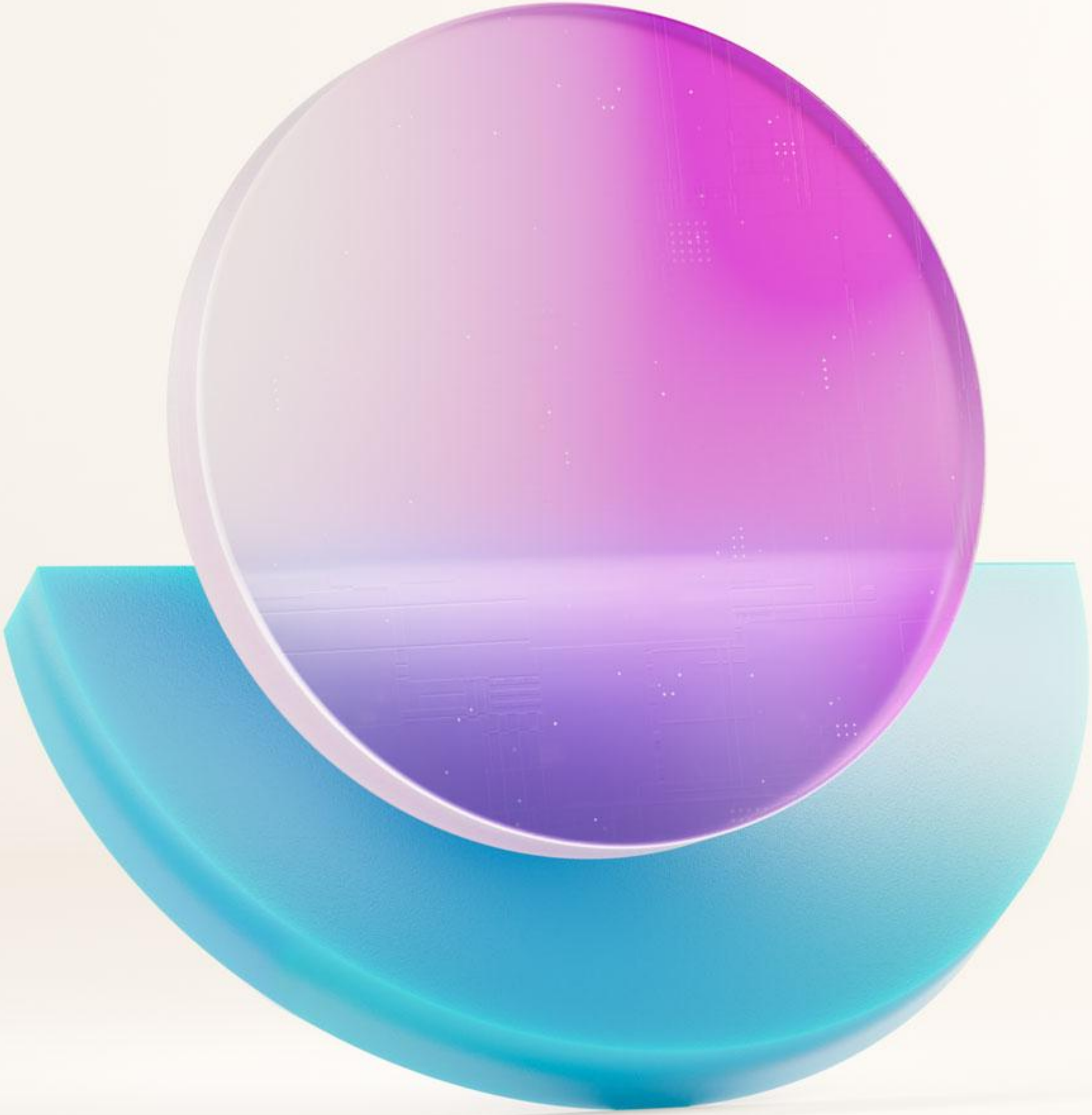
Utilize a code-free UI and API-based deployment for streamlined processes.

Diff and merge Fabric items with ease

View and track all your activities across all workspaces in one place



Microsoft Fabric
Community Conference



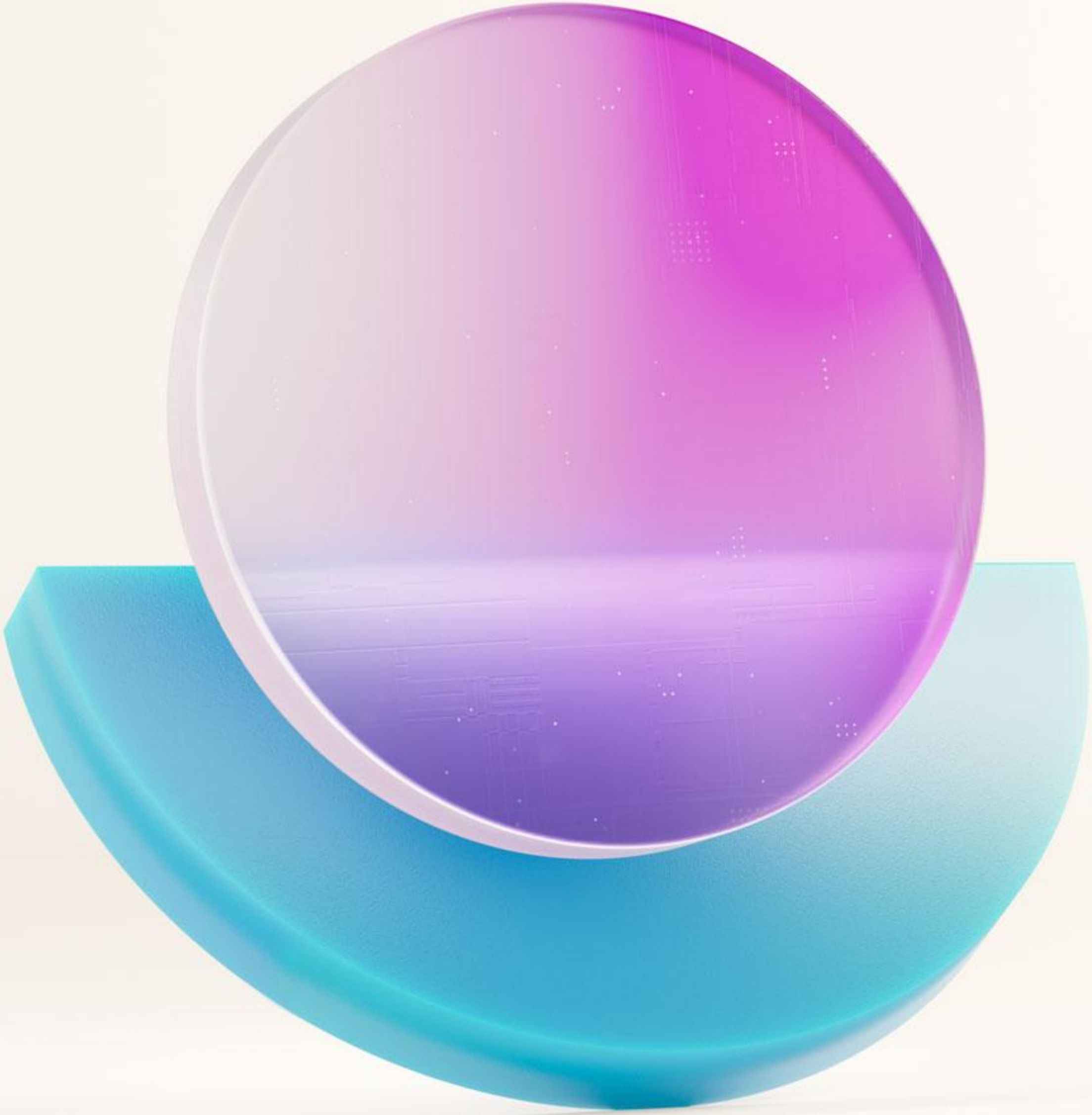
Questions?

Let’s take a short break…

Be back @ 11:00

Influence our product roadmap
and ensure Fabric meets your real-life needs
https://aka.ms/fabricideas

Microsoft Fabric
Community Conference



Unify your data with Lakehouses

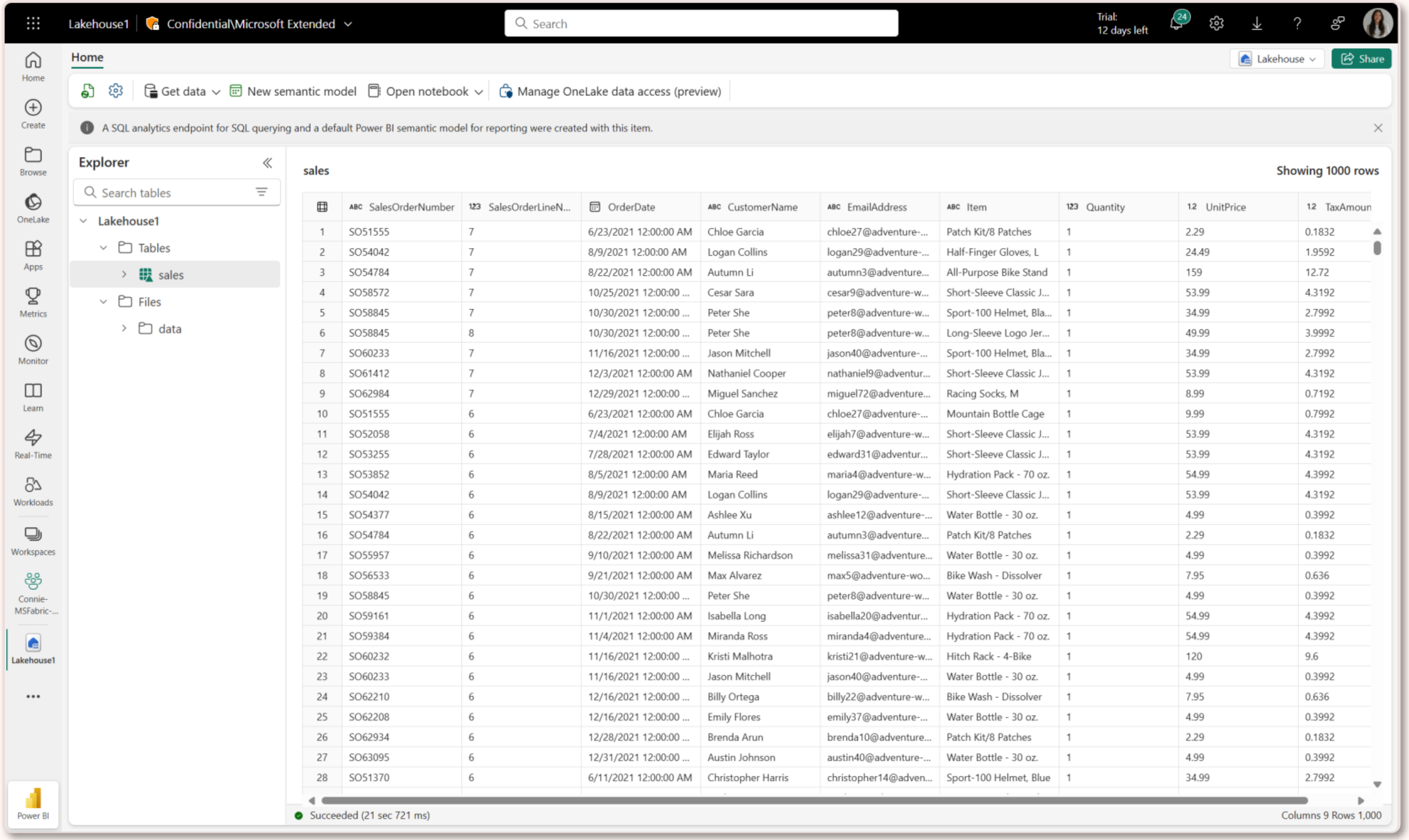
Unified data management with a Lakehouse
in Microsoft Fabric

Store, **manage** and **analyze** all your data in a single location and easily share across the entire enterprise

Flexible and scalable solution that enables organizations to handle large data volumes of all types and sizes

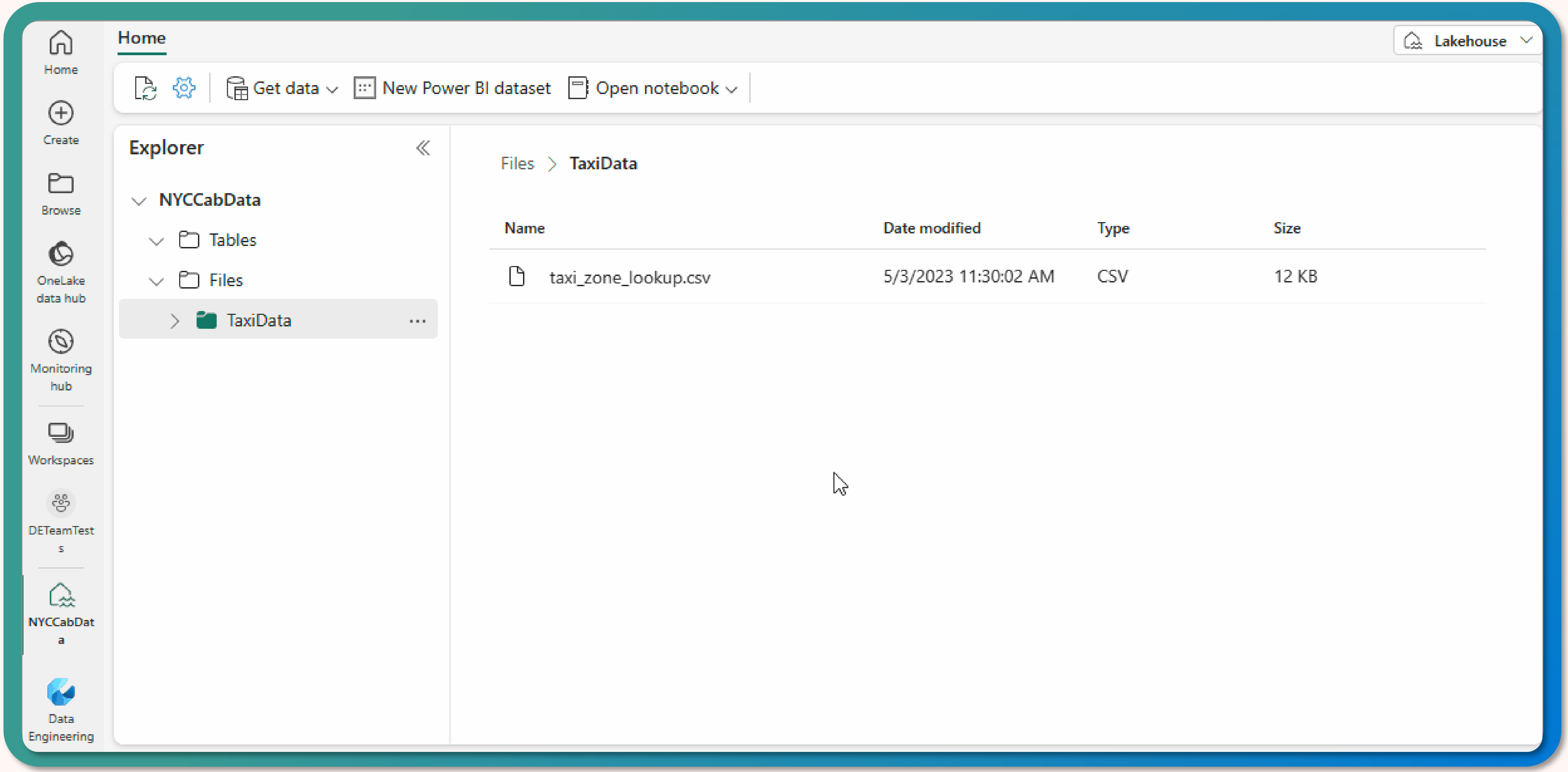
Built-in SQL endpoint unlocks Data Warehouse capabilities on top of your Lakehouse with no data movement

Address the challenges of traditional Data Lakes by adding a **Delta Lake storage** layer directly on top of the cloud Data Lake



	SalesOrderNumber	SalesOrderLineN...	OrderDate	CustomerName	EmailAddress	Item	Quantity	UnitPrice	TaxAmoun
1	SO51555	7	6/23/2021 12:00:00 AM	Chloe Garcia	chloe27@adventure-...	Patch Kit/8 Patches	1	2.29	0.1832
2	SO54042	7	8/9/2021 12:00:00 AM	Logan Collins	logan29@adventure-...	Half-Finger Gloves, L	1	24.49	1.9592
3	SO54784	7	8/22/2021 12:00:00 AM	Autumn Li	autumn3@adventure...	All-Purpose Bike Stand	1	159	12.72
4	SO58572	7	10/25/2021 12:00:00 ...	Cesar Sara	cesar9@adventure-w...	Short-Sleeve Classic J...	1	53.99	4.3192
5	SO58845	7	10/30/2021 12:00:00 ...	Peter She	peter8@adventure-w...	Sport-100 Helmet, Bla...	1	34.99	2.7992
6	SO58845	8	10/30/2021 12:00:00 ...	Peter She	peter8@adventure-w...	Long-Sleeve Logo Jer...	1	49.99	3.9992
7	SO60233	7	11/16/2021 12:00:00 ...	Jason Mitchell	jason40@adventure-...	Sport-100 Helmet, Bla...	1	34.99	2.7992
8	SO61412	7	12/3/2021 12:00:00 AM	Nathaniel Cooper	nathaniel9@adventur...	Short-Sleeve Classic J...	1	53.99	4.3192
9	SO62984	7	12/29/2021 12:00:00 ...	Miguel Sanchez	miguel72@adventure...	Racing Socks, M	1	8.99	0.7192
10	SO51555	6	6/23/2021 12:00:00 AM	Chloe Garcia	chloe27@adventure-...	Mountain Bottle Cage	1	9.99	0.7992
11	SO52058	6	7/4/2021 12:00:00 AM	Elijah Ross	elijah7@adventure-w...	Short-Sleeve Classic J...	1	53.99	4.3192
12	SO53255	6	7/28/2021 12:00:00 AM	Edward Taylor	edward31@adventur...	Short-Sleeve Classic J...	1	53.99	4.3192
13	SO53852	6	8/5/2021 12:00:00 AM	Maria Reed	maria4@adventure-w...	Hydration Pack - 70 oz.	1	54.99	4.3992
14	SO54042	6	8/9/2021 12:00:00 AM	Logan Collins	logan29@adventure-...	Short-Sleeve Classic J...	1	53.99	4.3192
15	SO54377	6	8/15/2021 12:00:00 AM	Ashlee Xu	ashlee12@adventure-...	Water Bottle - 30 oz.	1	4.99	0.3992
16	SO54784	6	8/22/2021 12:00:00 AM	Autumn Li	autumn3@adventure...	Patch Kit/8 Patches	1	2.29	0.1832
17	SO55957	6	9/10/2021 12:00:00 AM	Melissa Richardson	melissa31@adventure...	Water Bottle - 30 oz.	1	4.99	0.3992
18	SO56533	6	9/21/2021 12:00:00 AM	Max Alvarez	max5@adventure-wo...	Bike Wash - Dissolver	1	7.95	0.636
19	SO58845	6	10/30/2021 12:00:00 ...	Peter She	peter8@adventure-w...	Water Bottle - 30 oz.	1	4.99	0.3992
20	SO59161	6	11/1/2021 12:00:00 AM	Isabella Long	isabella20@adventur...	Hydration Pack - 70 oz.	1	54.99	4.3992
21	SO59384	6	11/4/2021 12:00:00 AM	Miranda Ross	miranda4@adventure...	Hydration Pack - 70 oz.	1	54.99	4.3992
22	SO60232	6	11/16/2021 12:00:00 ...	Kristi Malhotra	kristi21@adventure-w...	Hitch Rack - 4-Bike	1	120	9.6
23	SO60233	6	11/16/2021 12:00:00 ...	Jason Mitchell	jason40@adventure-...	Water Bottle - 30 oz.	1	4.99	0.3992
24	SO62210	6	12/16/2021 12:00:00 ...	Billy Ortega	billy22@adventure-w...	Bike Wash - Dissolver	1	7.95	0.636
25	SO62208	6	12/16/2021 12:00:00 ...	Emily Flores	emily37@adventure-...	Water Bottle - 30 oz.	1	4.99	0.3992
26	SO62934	6	12/28/2021 12:00:00 ...	Brenda Arun	brenda10@adventure...	Patch Kit/8 Patches	1	2.29	0.1832
27	SO63095	6	12/31/2021 12:00:00 ...	Austin Johnson	austin40@adventure-...	Water Bottle - 30 oz.	1	4.99	0.3992
28	SO51370	6	6/11/2021 12:00:00 AM	Christopher Harris	christopher14@adven...	Sport-100 Helmet, Blue	1	34.99	2.7992

Automatic discovery of Delta Lake tables
in a Lakehouse

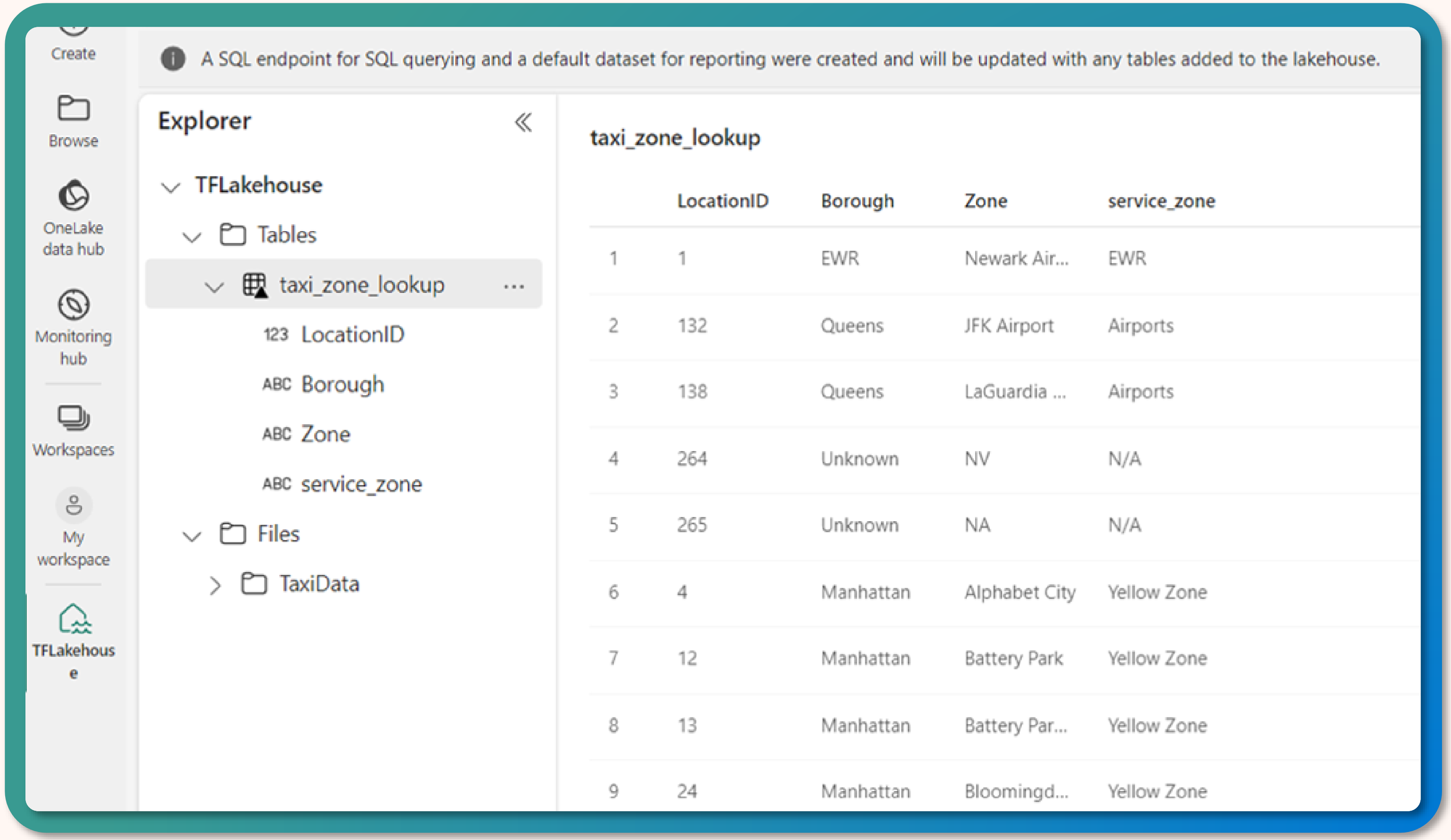


Automatic discovery of Delta Lake tables
in a Lakehouse

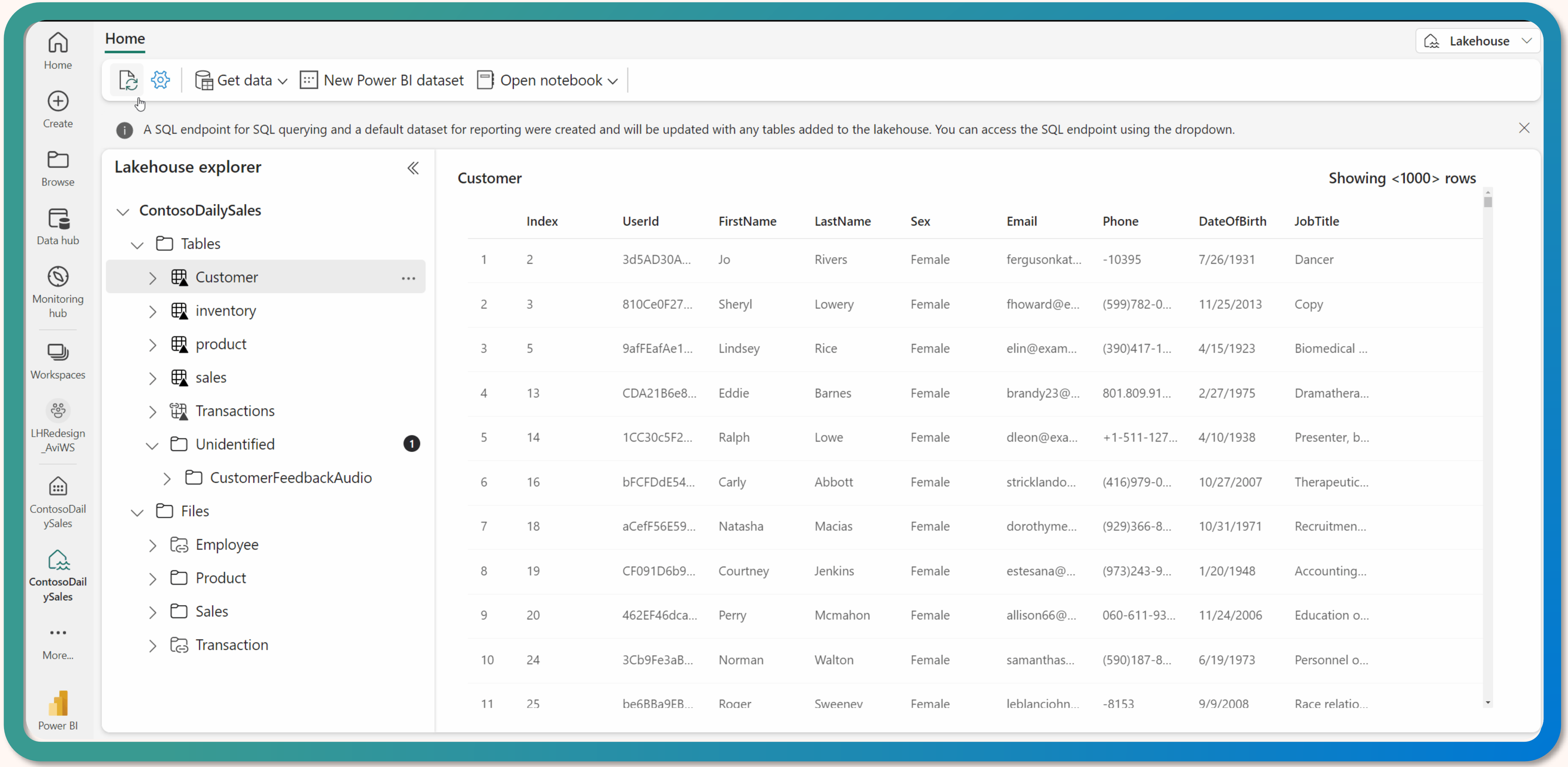
The Lakehouse explorer provides a **tree-like view** of the objects in the Microsoft Fabric Lakehouse item.

It has a key capability of discovering and displaying tables that are described in the metadata repository and in OneLake storage.

Delta Lake - Achieve seamless data access across all compute engines in Microsoft Fabric



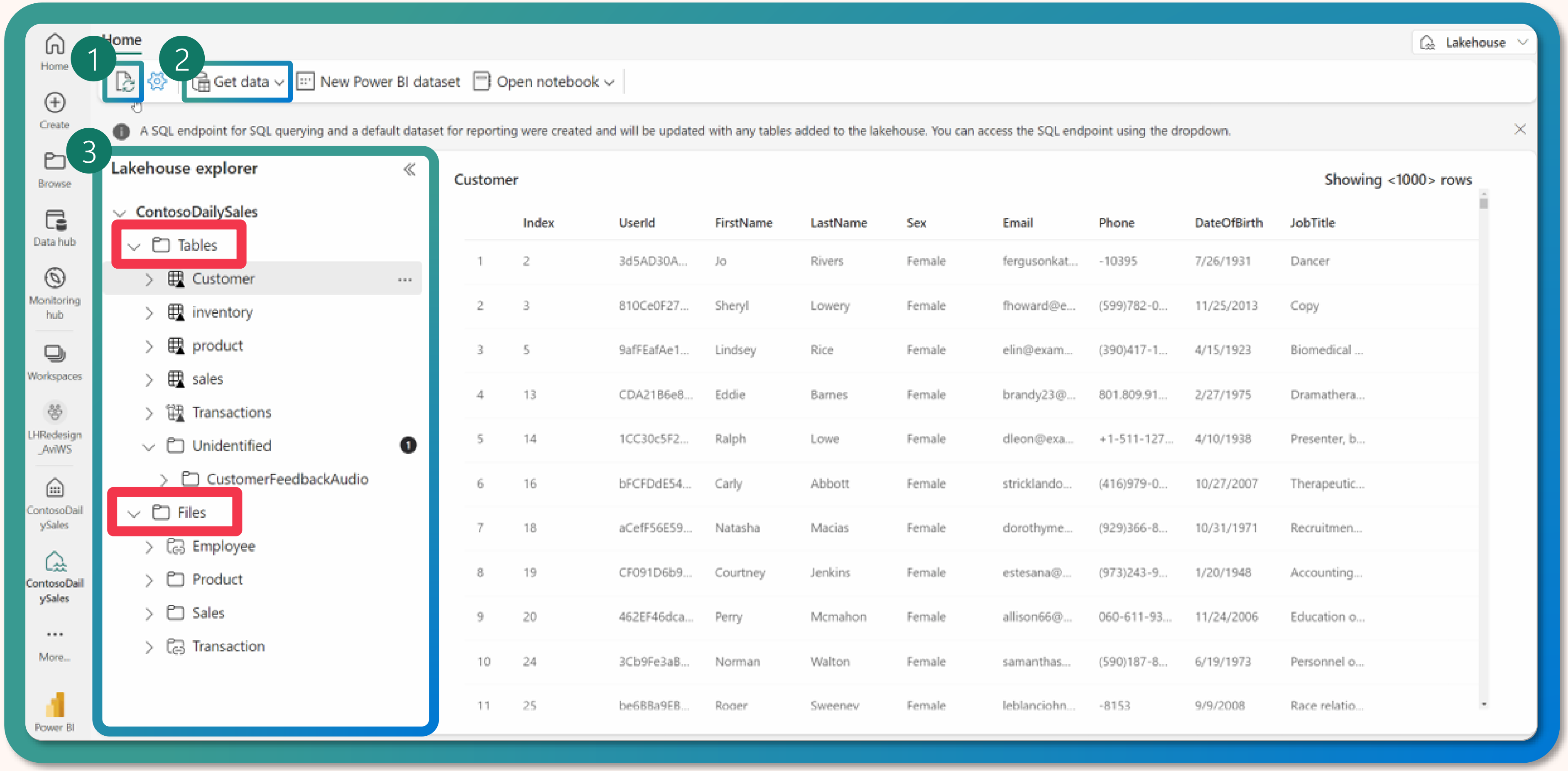
Navigating the Fabric Lakehouse explorer
in Microsoft Fabric



Navigating the Fabric Lakehouse explorer
in Microsoft Fabric

Navigation

1. Refresh Lakehouse metadata
2. Get data
3. Lakehouse explorer for tables and files

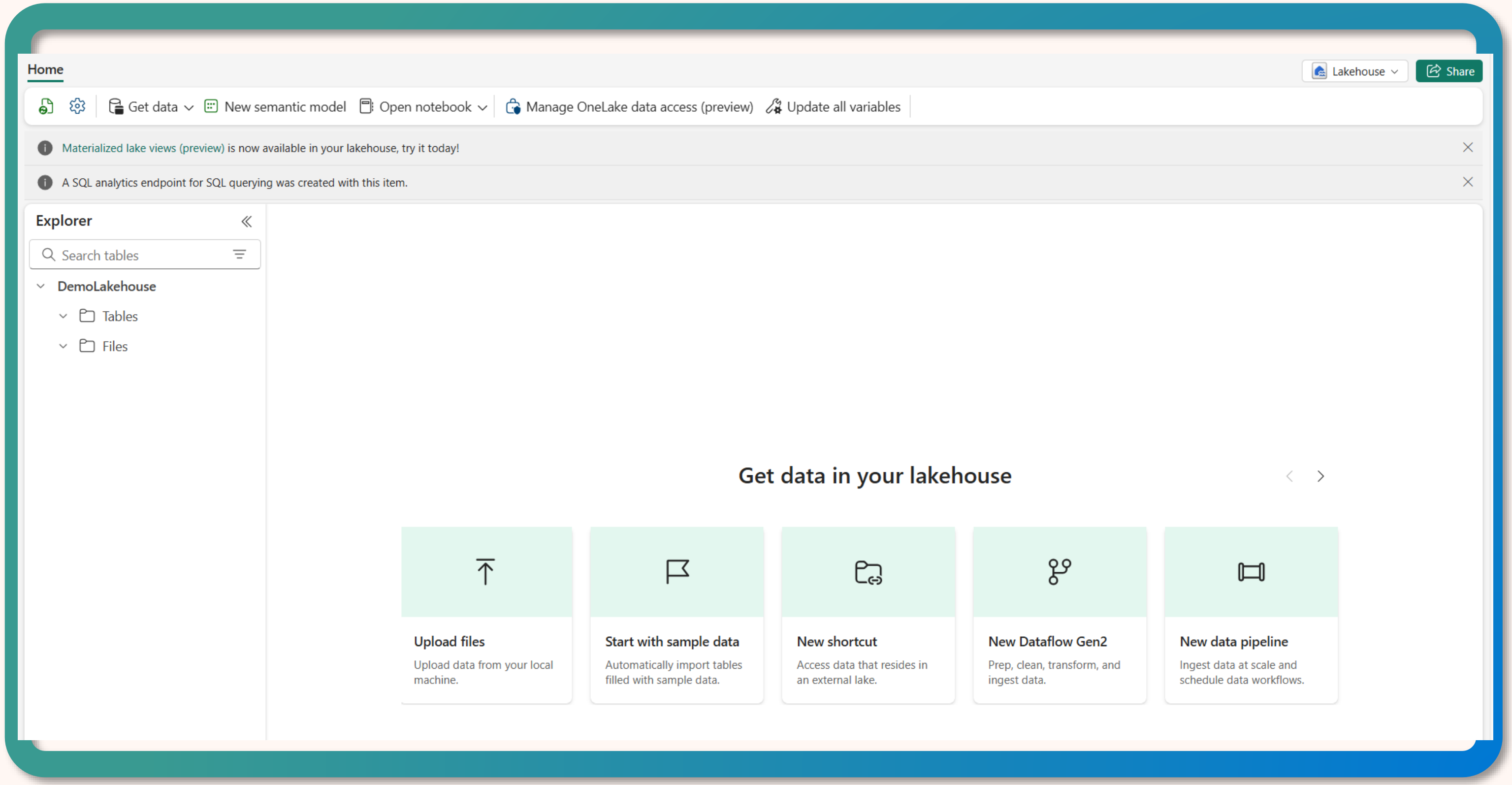


Lakehouse SQL endpoint
in Microsoft Fabric

The Lakehouse creates a serving layer by automatically generating a SQL endpoint

You can **only read data** from delta tables using SQL endpoint

Save functions, views, and set SQL object-level security



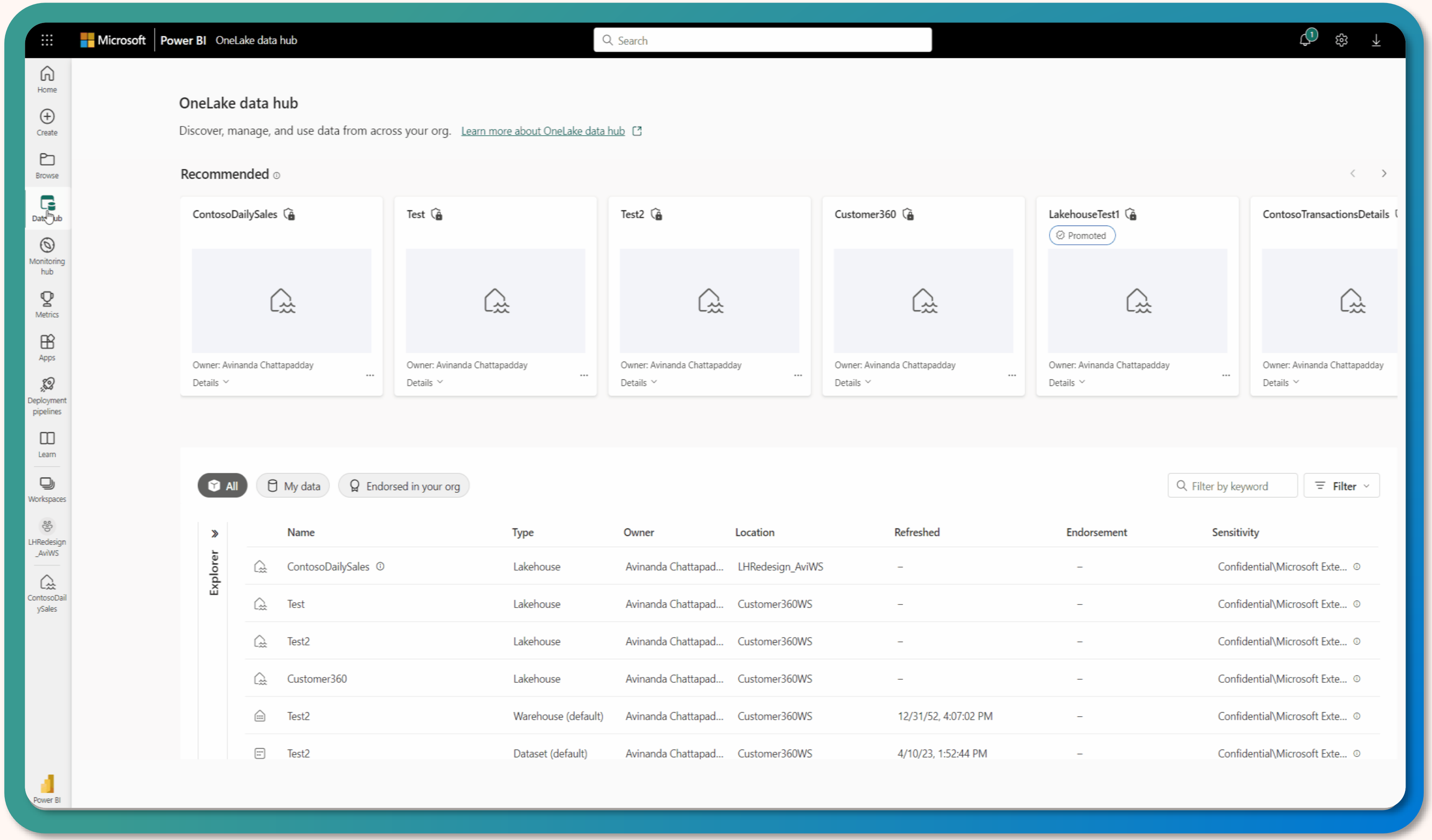
Share your Lakehouse with consumers in Microsoft Fabric

Grant access **without adding users** to your workspace

Use **SQL Security** or custom permissions for SQL Endpoint access

Discover accessible Lakehouses in the OneLake catalog

Access Lakehouse data via Spark, SQL Endpoint, and Power BI semantic models



Open and accessible endpoints in Microsoft Fabric

Lakehouse

SQL analytics endpoint / Azure Blob File System (ABFS) / URL endpoint

Warehouse

SQL endpoint / OneLake availability

Eventhouse

Cluster address (Kusto) / OneLake availability

Semantic model

Extensible Markup Language for Analysis (XMLA) endpoint address / OneLake availability

GraphQL

URL endpoint

Delta table and column name validation and rules
in Microsoft Fabric

Tables

Table names can only contain **alphanumeric** characters and **underscores**

Text files

Text files without column headers are replaced with **standard col# notation** as the table column names

Columns

Column names are **validated during the load action** and the action fails if column names are invalid

Interacting with a Lakehouse
in Microsoft Fabric

OneLake file explorer

Explore data in the Lakehouse using Windows File Explorer

Notebooks

Use the Notebook to write code to read, transform and write directly to Lakehouse as tables and/or folders

Data pipelines

Use pipeline copy tool to pull data from other sources and land into the Lakehouse

Copy job

Copy data easily with settings for full, incremental, or event-based.

Dataflows Gen2

Use Dataflows Gen2 to ingest and prepare the data

Apache Spark job definitions

Develop robust applications and orchestrate the execution of compiled Spark jobs in Java, Scala, and Python

And more...

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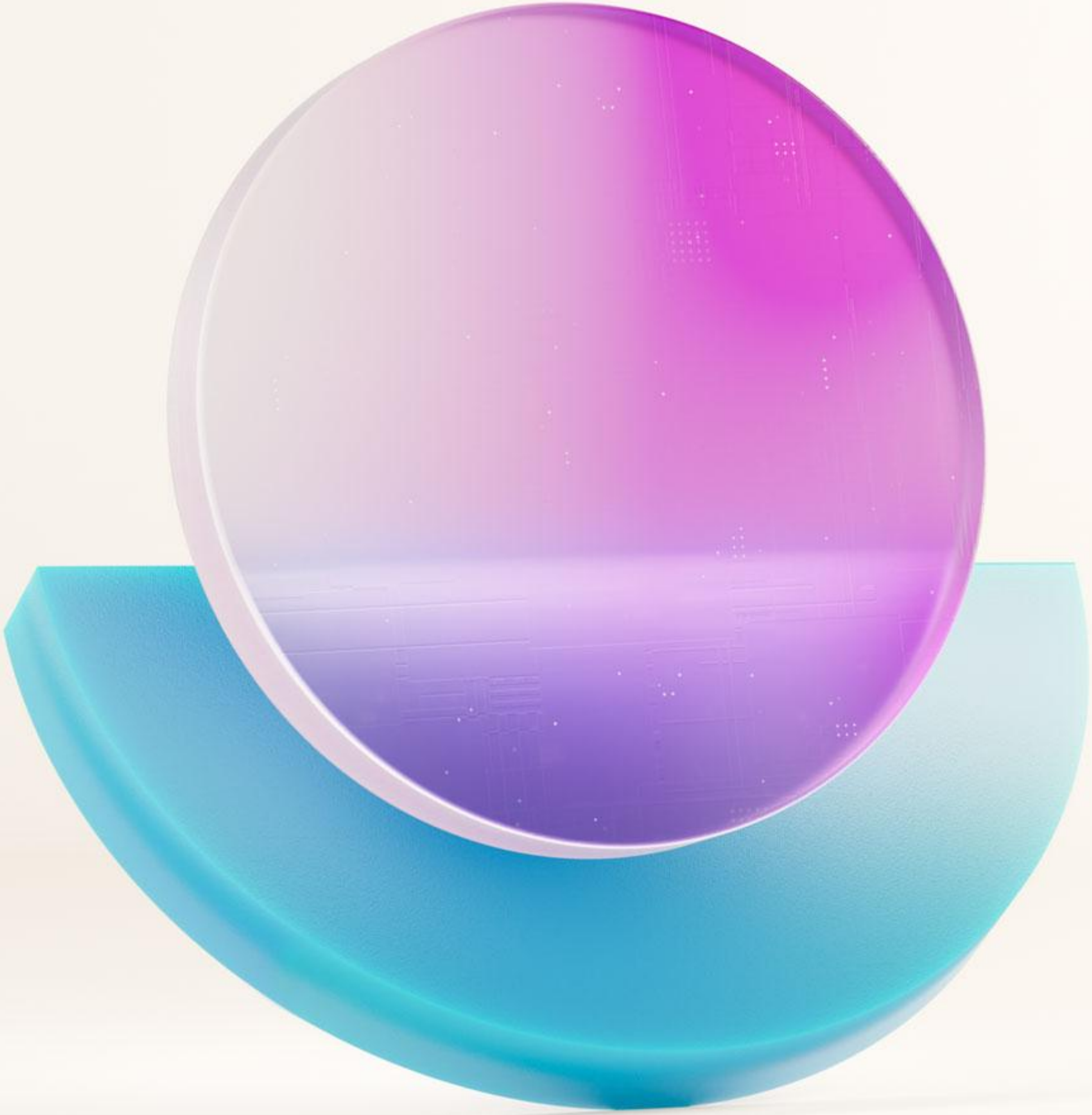
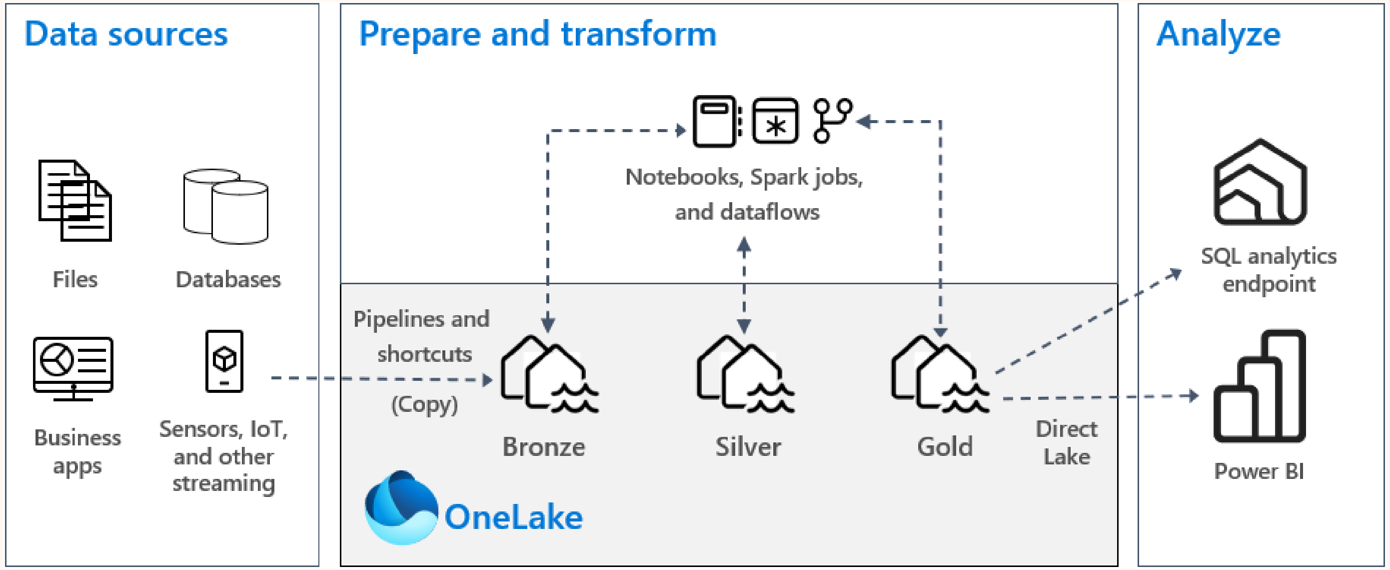


Create Medallion Architecture in a Lakehouse

Medallion Lakehouse architecture

Medallion architecture has three layers that represent different data quality levels: **bronze (raw data), silver (validated data), and gold (enriched data)**

As data moves through these layers, it undergoes validations and transformations to ensure it meets the **ACID properties** (Atomicity, Consistency, Isolation, and Durability) and is optimized for analytics



Medallion architecture

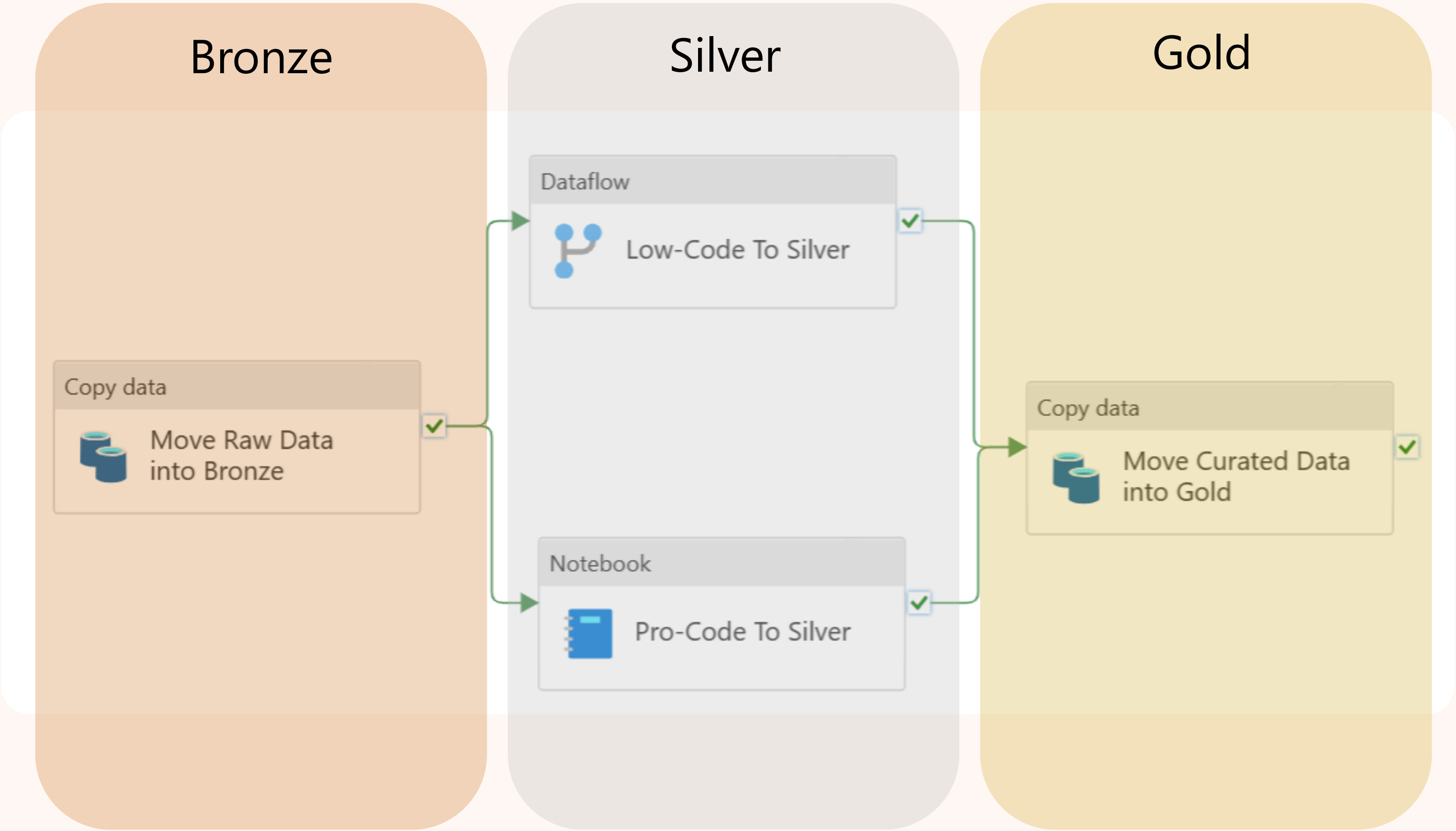
Build a single source of truth for enterprise data products with a multi-layered approach

Distinct layers (or zones)

Bronze (raw zone): This layer stores source data in its original format, which is typically append-only and immutable.

Silver (enriched zone): Data from the bronze layer is cleansed, standardized, and structured as tables. It may also be integrated with other data to provide a comprehensive view of business entities like customers and products.

Gold (curated zone): Data from the silver layer is refined to meet specific business and analytics requirements. Tables in this layer usually follow a star schema design, optimizing them for performance and usability.



Microsoft

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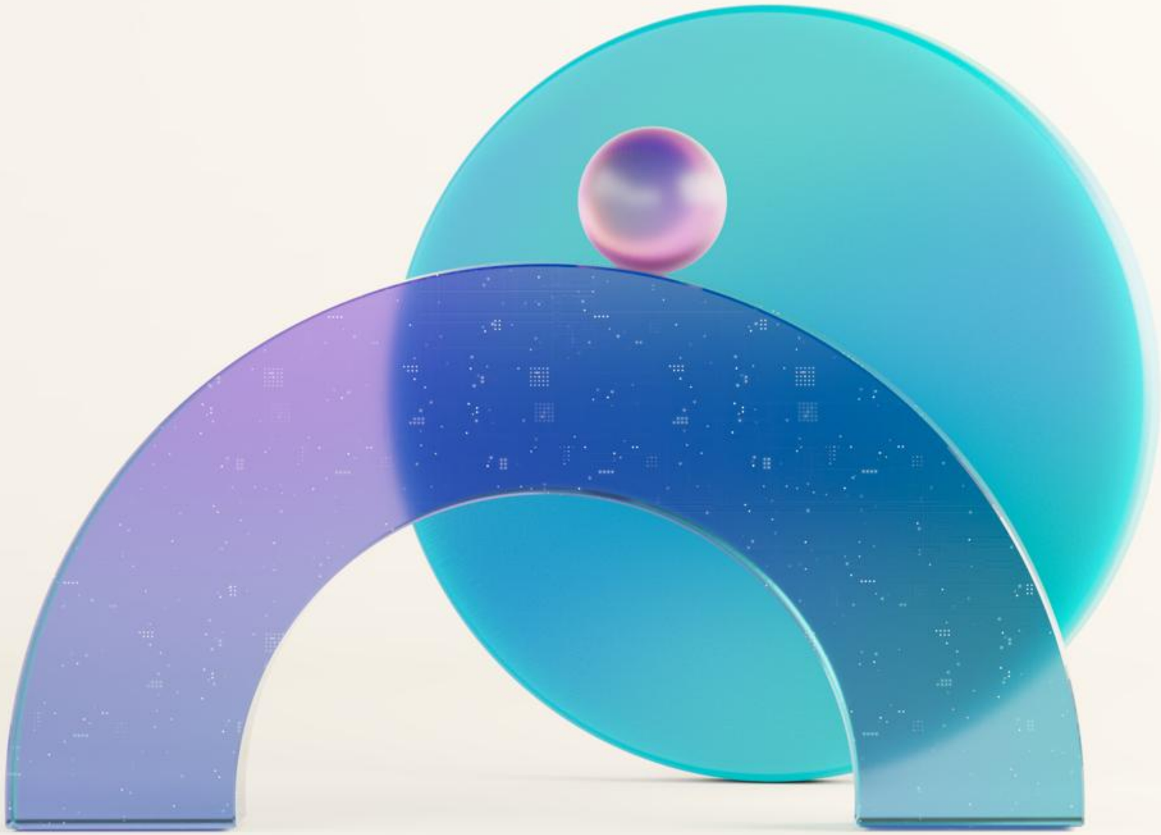

Lab Demo: Let’s Get Started

- Create a **Medallion task flow** with a Lakehouse + Data pipeline
- Use a **Copy activity** in Data pipeline
- Explore **Data pipeline’s authoring canvas**

Reminder of how to access lab instructions

All materials are available at:

https://aka.ms/dfiad

Lab Instructions:

Getting started

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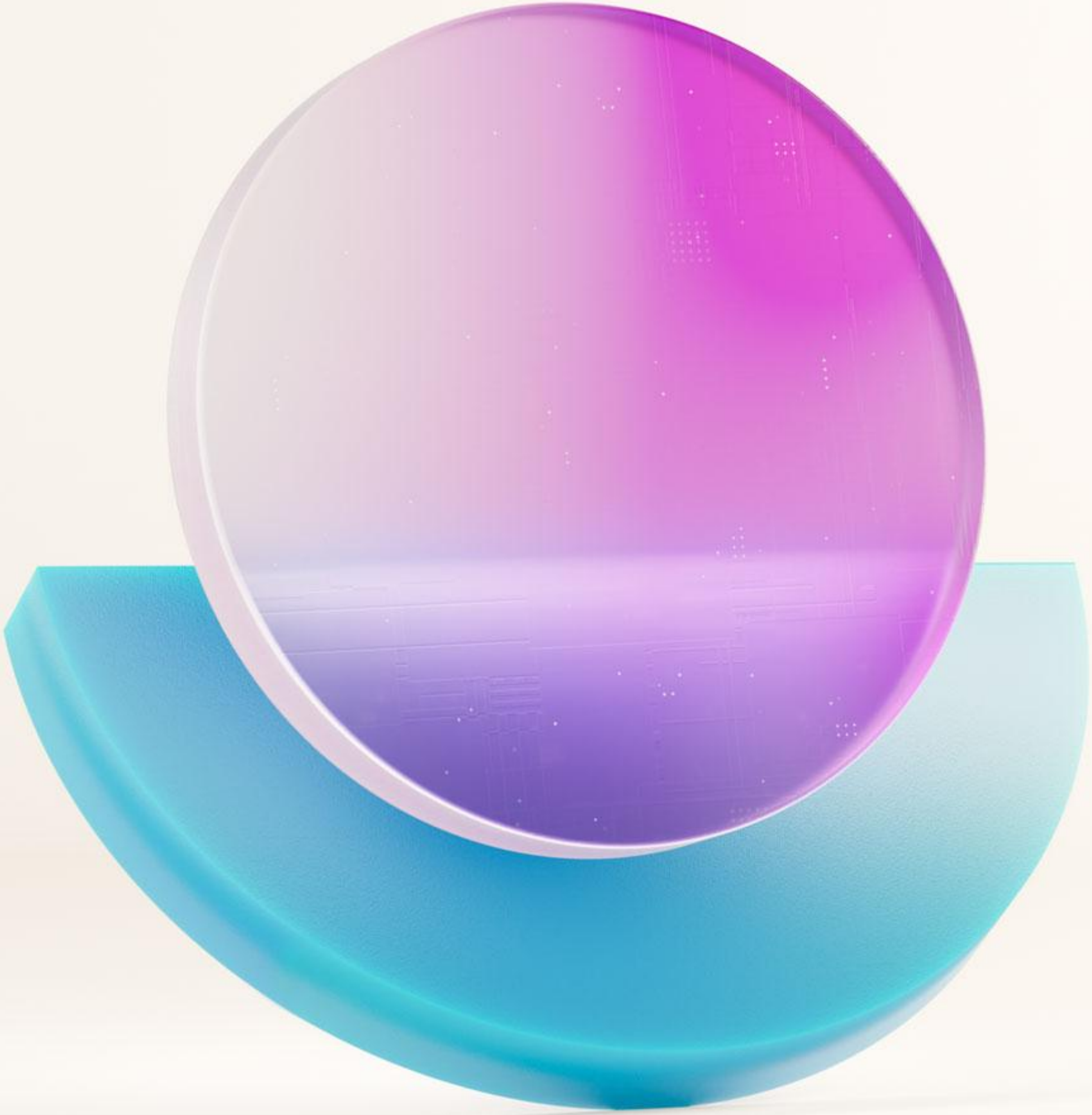


Getting data into a Lakehouse

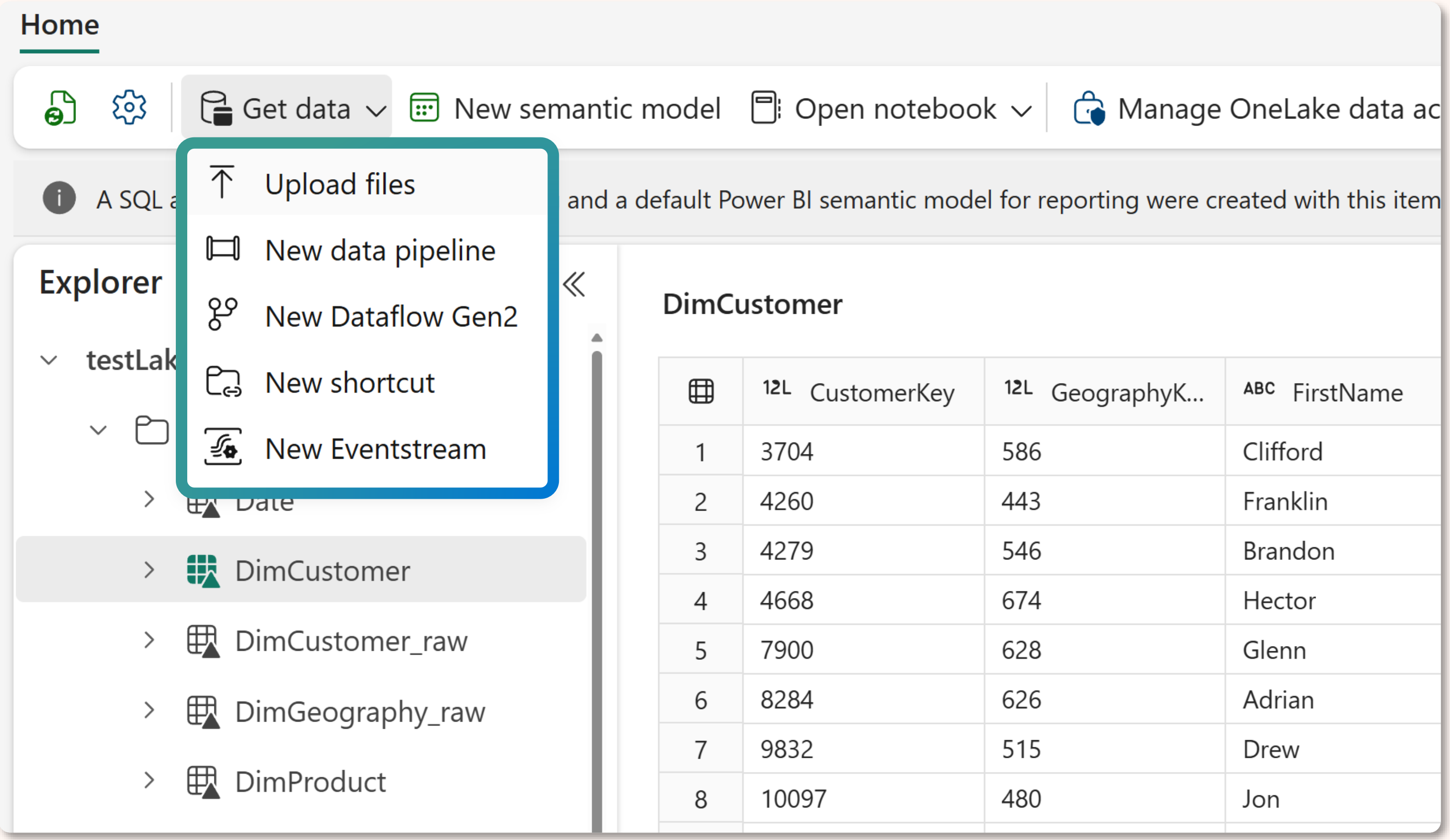
Getting data into a Fabric Lakehouse

File upload from local computer or OneLake file explorer

Run a Copy job, Data pipeline, Dataflow Gen2 or Mirrored database to **ingest data**

Ingest and transform data with a Dataflow Gen2 or Eventstream

Apache Spark libraries in **Notebook code**

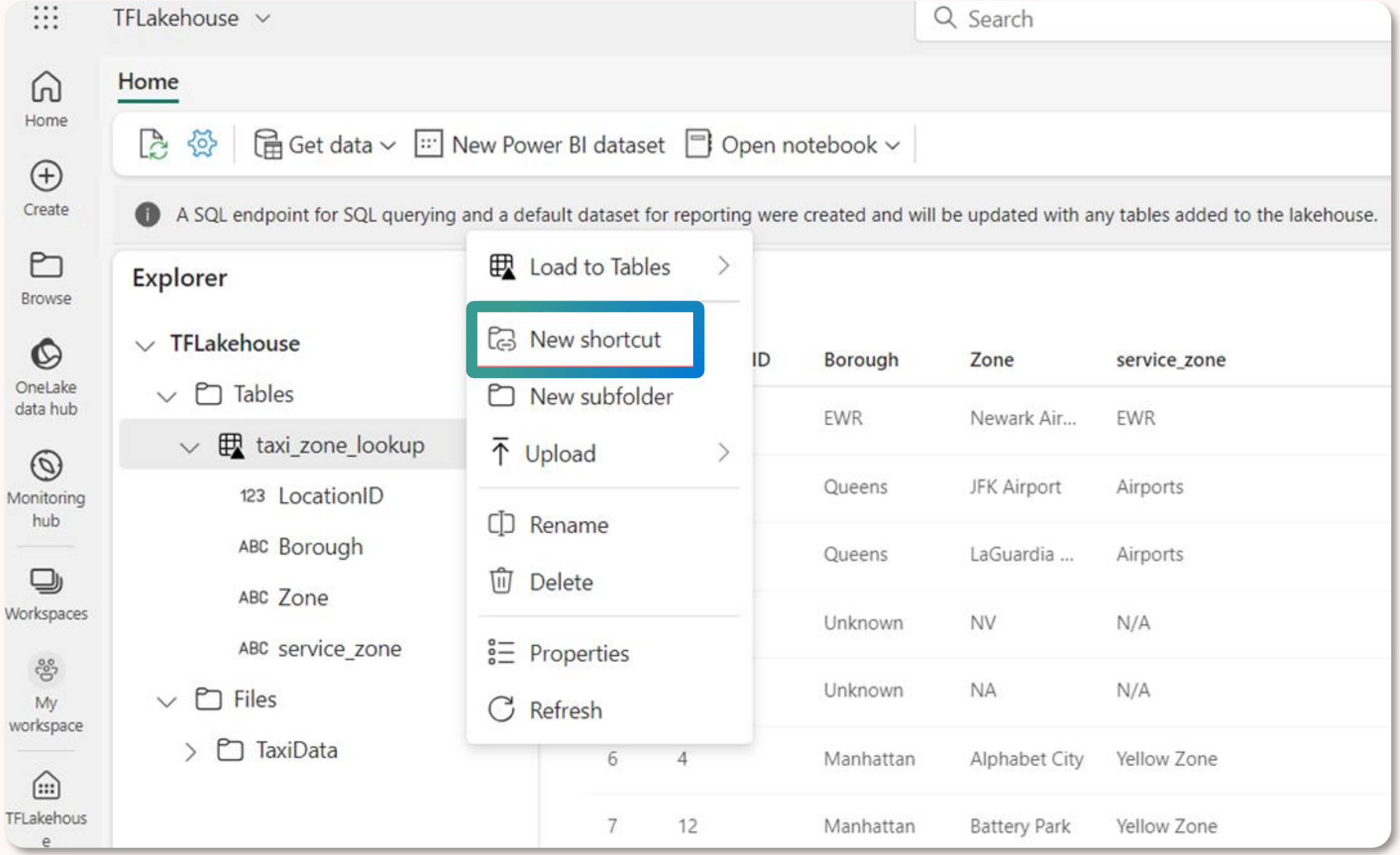


Shortcuts in a Fabric Lakehouse

Shortcuts in a Lakehouse allow users to reference data without copying it from **Tables** or **Files**

Shortcuts unify data from different Lakehouses, workspaces, or external storage, such as:

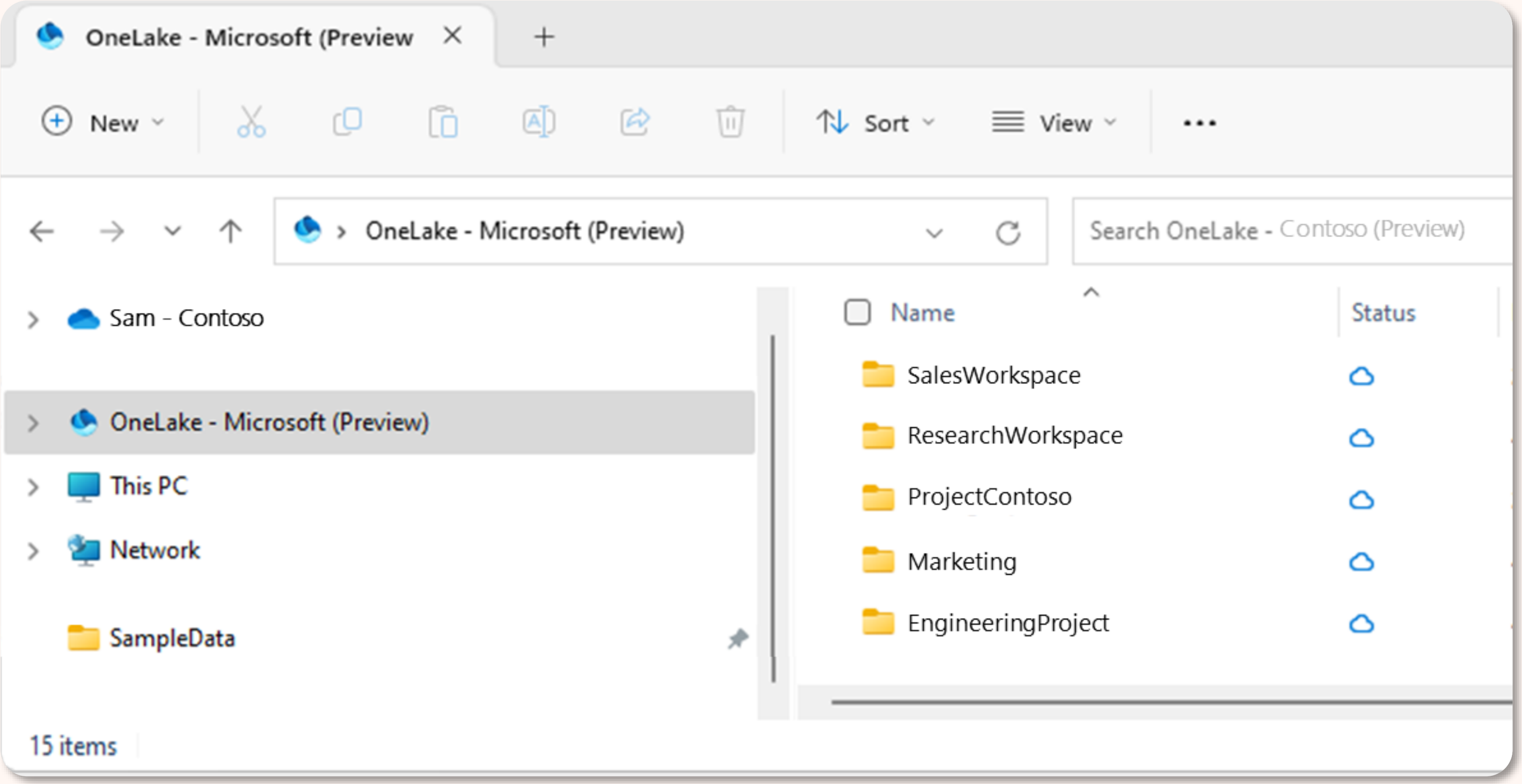
- ADLS Gen2
- AWS S3
- Dataverse



OneLake file explorer

Seamlessly integrates OneLake with Windows File Explorer, **automatically syncing** all OneLake items that you have access to in file explorer

Syncing doesn't download the data, it creates placeholders. You must double-click on a file to download the data locally



Check your knowledge
Data ingestion scenarios

Connecting to existing SQL Server and **copying** data into Delta table on the Lakehouse

Uploading files from your computer

Copying and **merging** multiple tables from other Lakehouses into a new Delta table

Referencing data without copying it from other internal Lakehouses or external sources

Considerations when loading data

Use case	Recommendation
Small file upload from local machine	Use OneLake file explorer / Upload file
Data ingestion only	Use Copy job or Data pipeline
Orchestrated data ingestion and transformation	Use a Data pipeline and Dataflow Gen2
Complex data transformations and orchestration	Use Data pipeline and Notebook

Data ingestion decision guide

	Copy job	Data pipeline	Dataflow Gen2	Spark
Use case	Data ingestion	Data ingestion, lightweight transformation	Data ingestion, data transformation, data wrangling, data profiling	Data ingestion, data transformation, data processing, data profiling
Primary developer persona	Data integrator	Data engineer, data integrator	Data engineer, data integrator, business analyst	Data engineer, data scientist, data developer
Primary developer skill set	ETL	ETL, SQL, JSON	ETL, M, SQL	Spark (Scala, Python, Spark SQL, R)
Code written	No code	No code, low code	No code, low code	Code
Data volume	Low to high	Low to high	Low to high	Low to high
Development interface	Assistant	Assistant, canvas	Power Query	Notebook, Spark job definition
Sources	8+ connectors	30+ connectors	150+ connectors	Hundreds of Spark libraries
Destinations	10+ connectors	18+ connectors	Lakehouse, Warehouse, Azure SQL database, Azure Data explorer	Hundreds of Spark libraries
Transformation complexity	Low: Lightweight – column mapping, table mapping	Low: lightweight - type conversion, column mapping, merge/split files, flatten hierarchy	Low to high: 300+ transformation functions	Low to high: support for native Spark and open-source libraries

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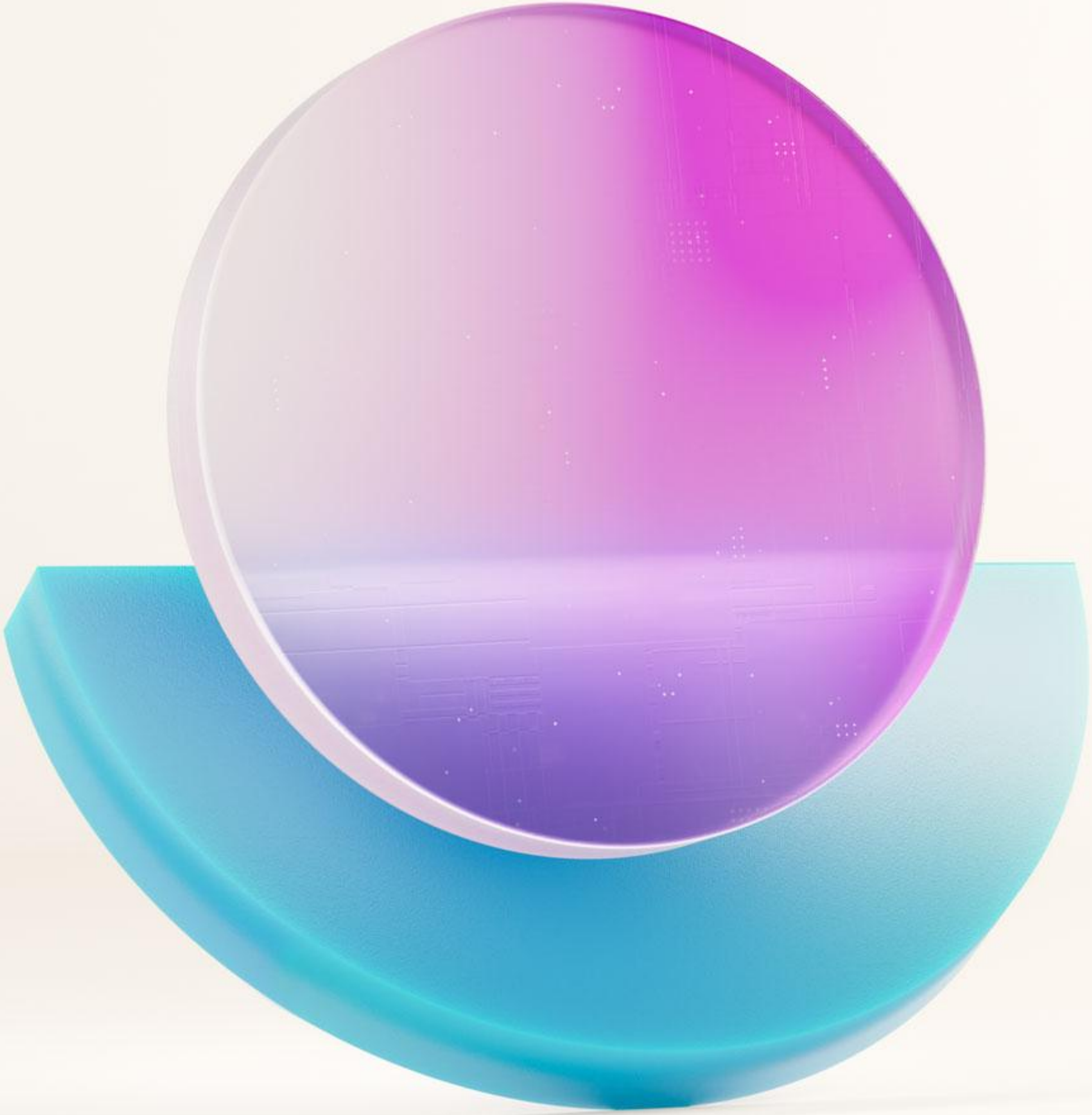

Overview: Data engineering in Fabric

Data engineering in Fabric

Empowers data engineers to transform data at scale and build a Lakehouse architecture

Build a Lakehouse for all your organizational data

Spark runtime with great out of the box performance and robust admin controls

Delightful authoring experience in your tools of choice

Completely integrated into the Fabric foundation

Data engineering in Fabric

Empowers data engineers to transform data at scale and build a Lakehouse architecture

Lakehouse

Store and manage structured and unstructured data in a single location

Apache Spark job definition

Submit batch/streaming job to Spark cluster, apply different transformation logic to the data

Notebook

Create and share documents that contain live code, equations, visualizations, and narrative text

Data pipeline

Collect, process, and transform data from its raw form to a format that you can use for analysis and decision-making

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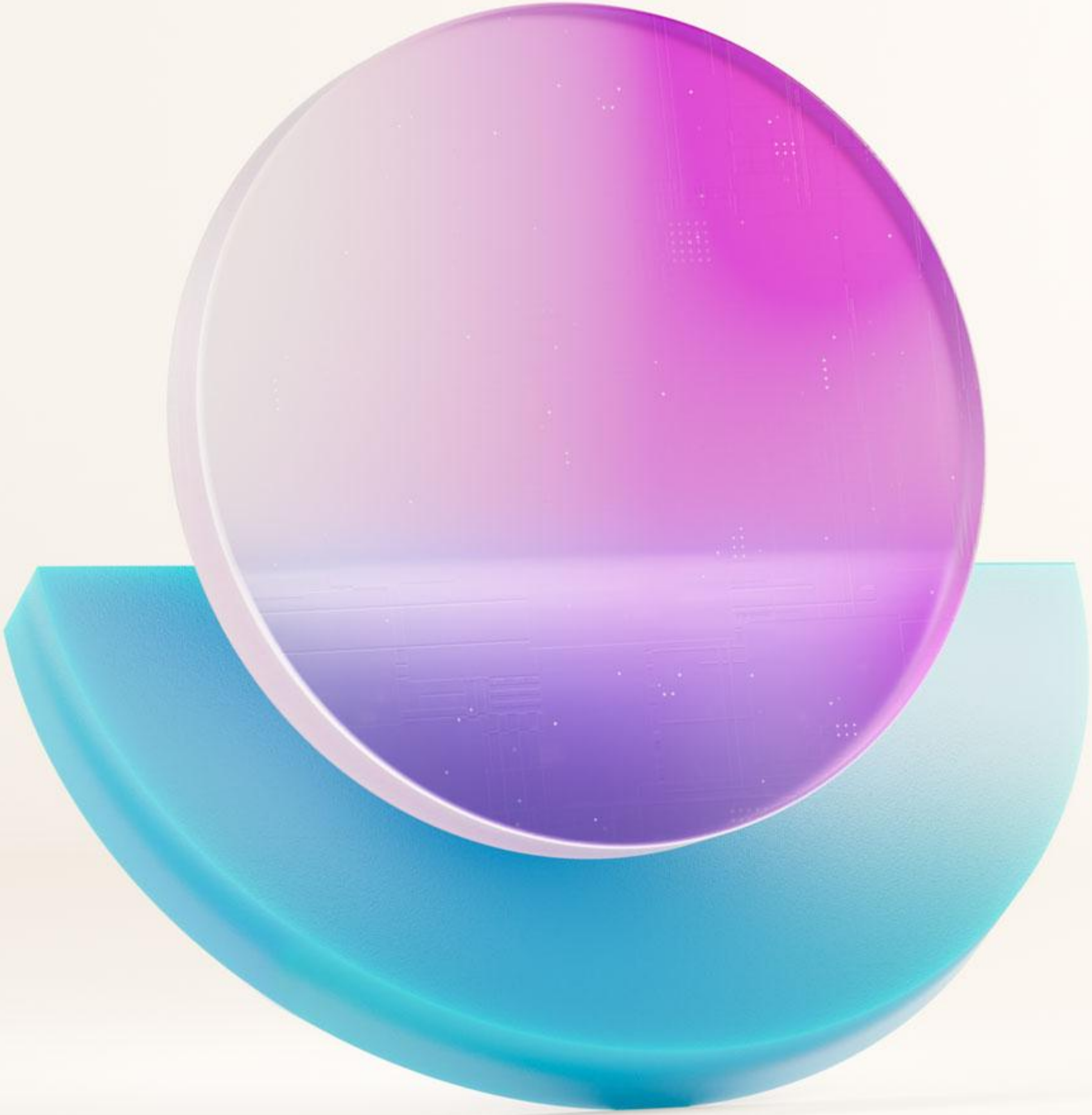


Ingesting and orchestrating data with Data pipelines

Data pipelines
in Fabric Data Factory

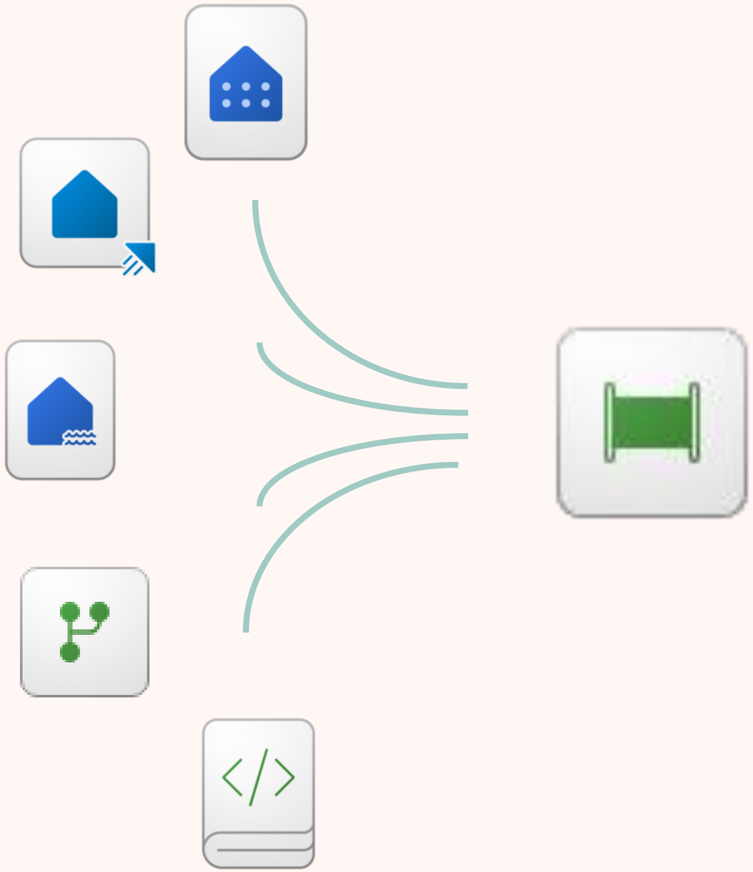
Seamlessly connect and ingest data into Fabric using a no-code interface

Evolution of **Azure Data Factory** pipelines

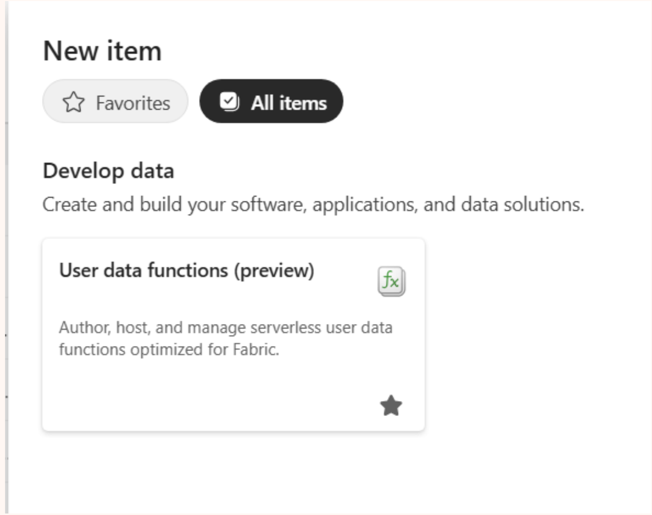
Rich library of **activities**

Fast copy of binary files using Az-Copy

Jumpstart with **Copy Assistant**



User Data Functions in the Functions activity
in Data pipelines

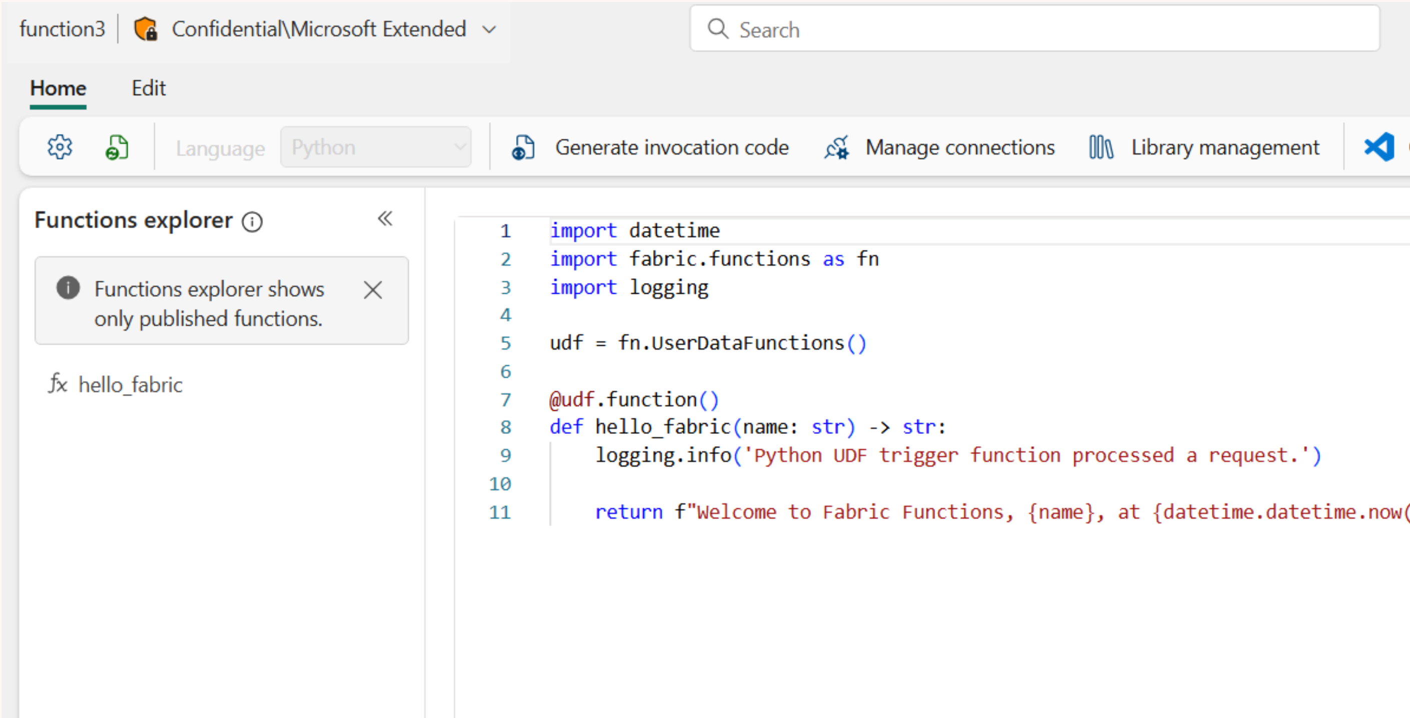


Public Preview

Run specific functions within the pipeline

Fabric User Data Functions allows for custom functions designed for **event-driven scenarios**

Azure functions are serverless functions that can be triggered by various events



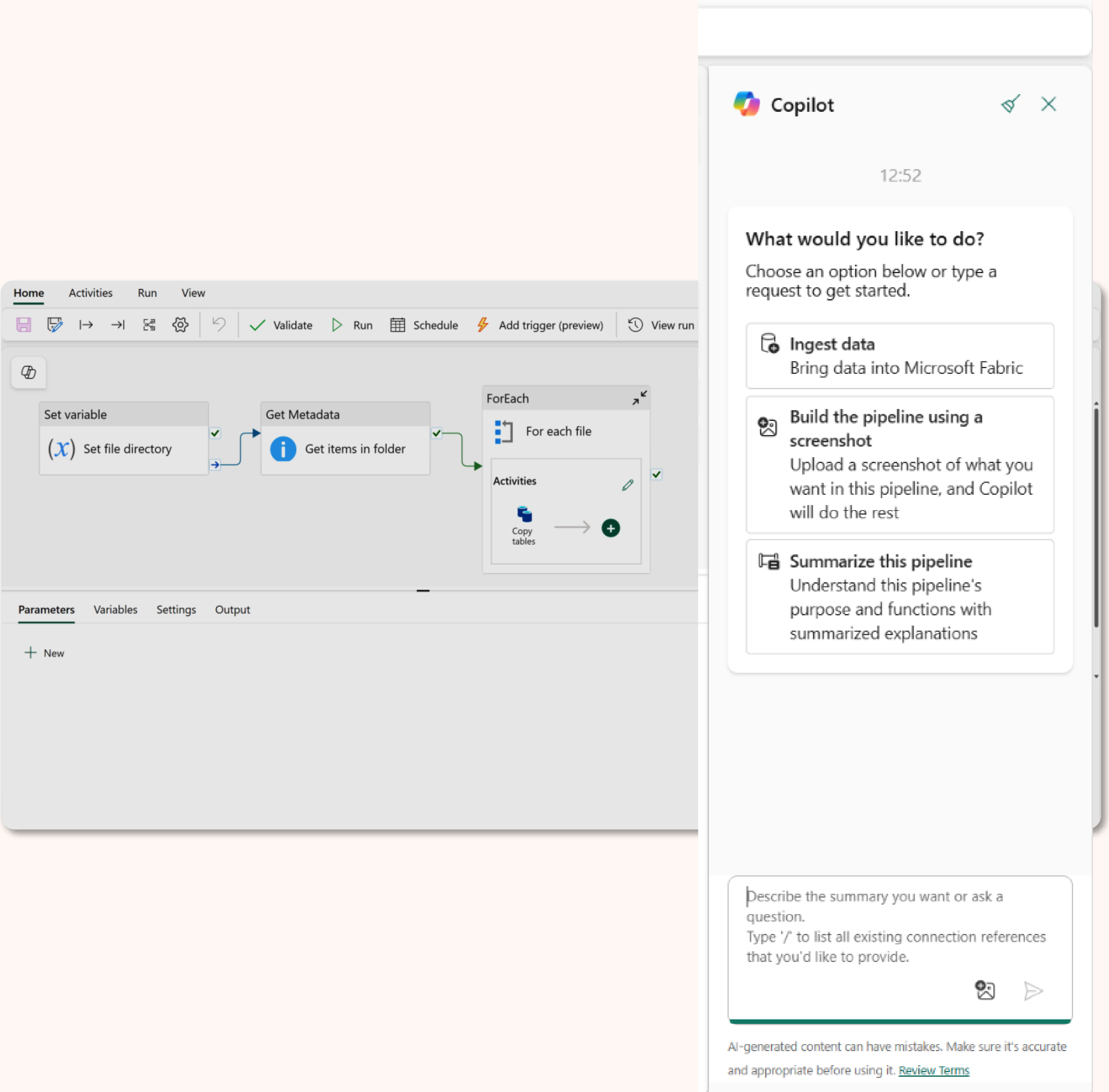
Copilot for Data Factory
in Data pipelines

AI-powered Intelligence

Ingest data into Fabric via pipeline's Copilot

Transform data by invoking a Dataflow Gen2 into your pipeline with Copilot

Receive a **summary** of your pipeline in natural language



Copy activity
in Data pipelines

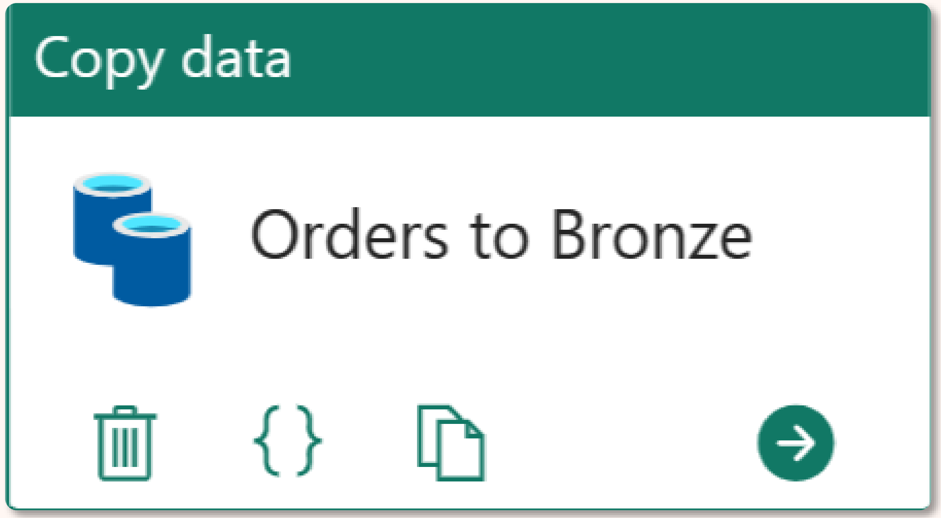
Copy activity is the core of data pipelines

Jumpstart with **Copy job** or **Copy assistant**

Fine grain control of file type conversion, column mapping, merging/splitting of files, and more..

Workspace source and destinations include Lakehouse (tables & files), Warehouse, SQL Database in Fabric, and KQL Database

Supports external sources and destinations



And so much more...

Capability	Description
Control flow	Conditional paths Loops
Nesting	Invoking data pipelines from other data pipelines Bi-directional communication via output variables
Parameterization	Parameters of varying data types
Expression language	Almost every field can be made dynamic Extensive built-in functions allow for massive customizations (data driven designs)
Repeatable	Schedulable

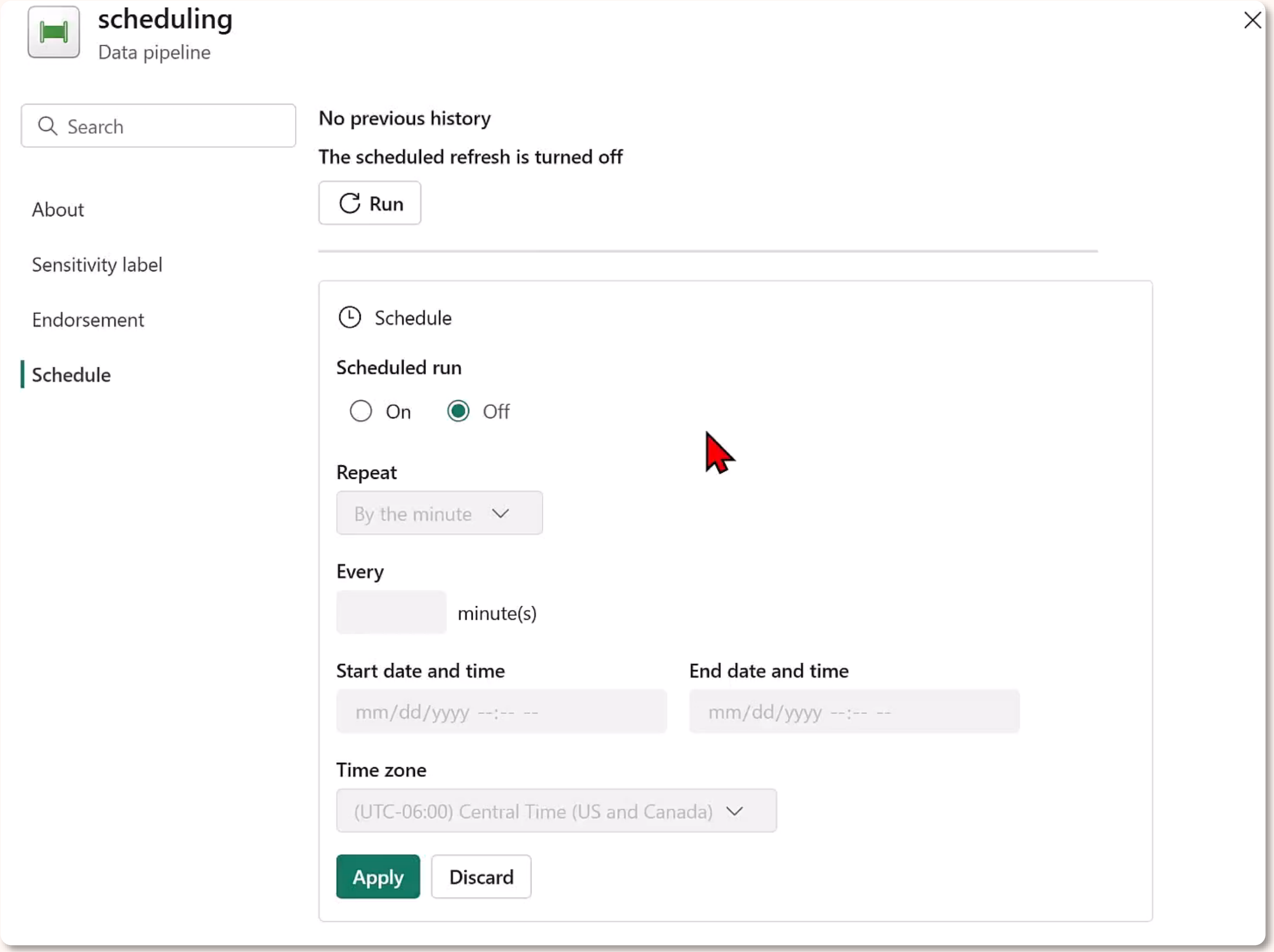
Platform scheduler
Schedule events at desired intervals

Supports **Data pipelines**, **Copy job**, **Dataflow Gen2**, and more.

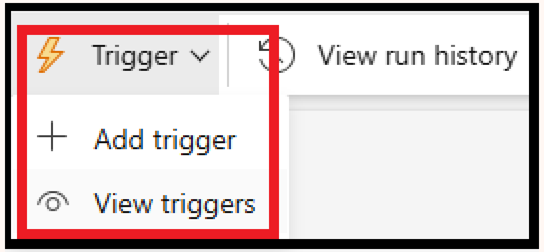
Deeply integrated with **monitoring hub**

Scheduled:

- By the minute
- Hourly
- Daily
- Weekly
- Monthly



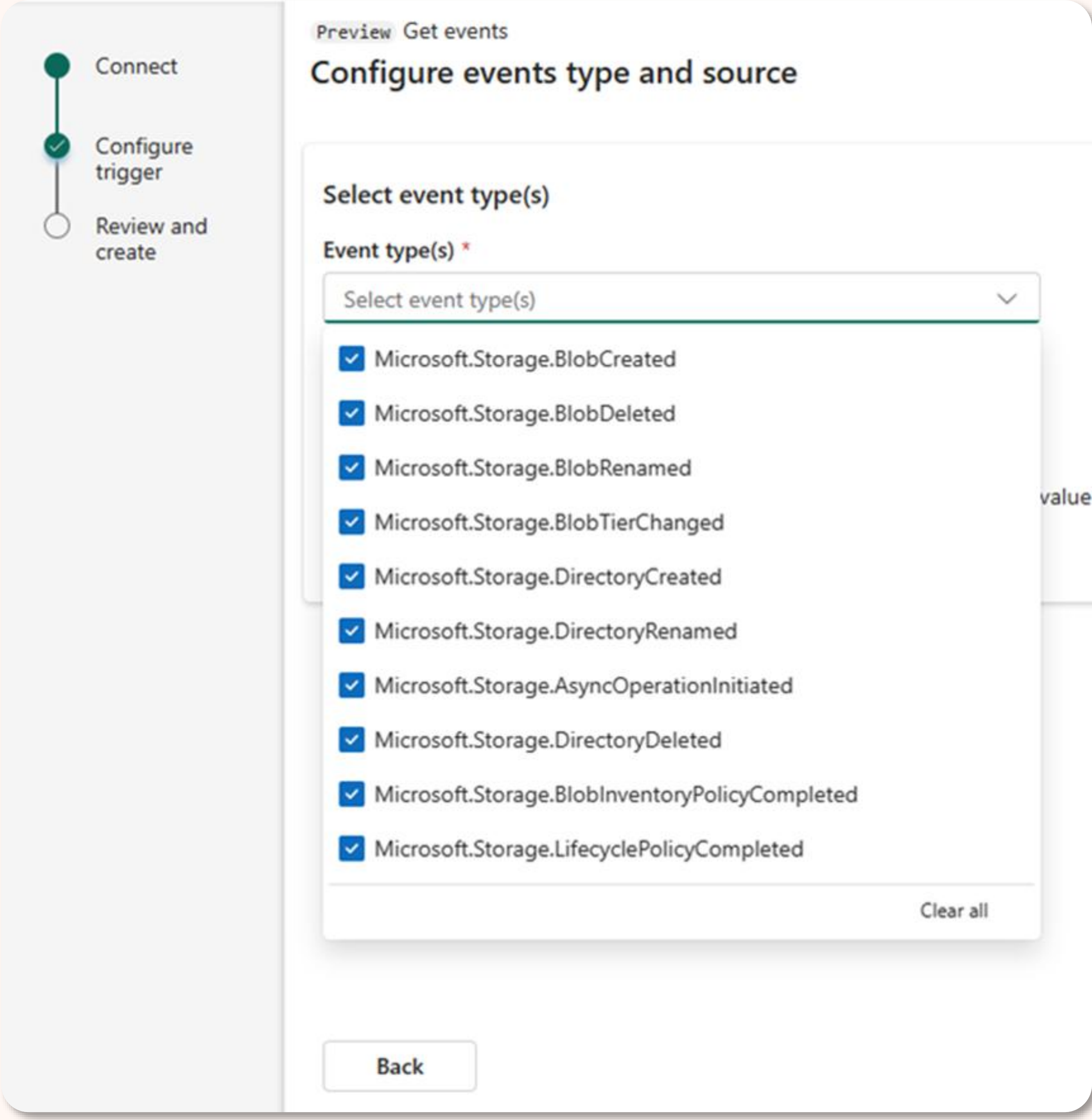
Event-driven triggers
in Data pipelines



Invoke a Data pipeline upon a file event using **Eventstreams** and **Reflex triggers**

Extensive event types and filtering capabilities

Ability to link multiple pipelines to a single event

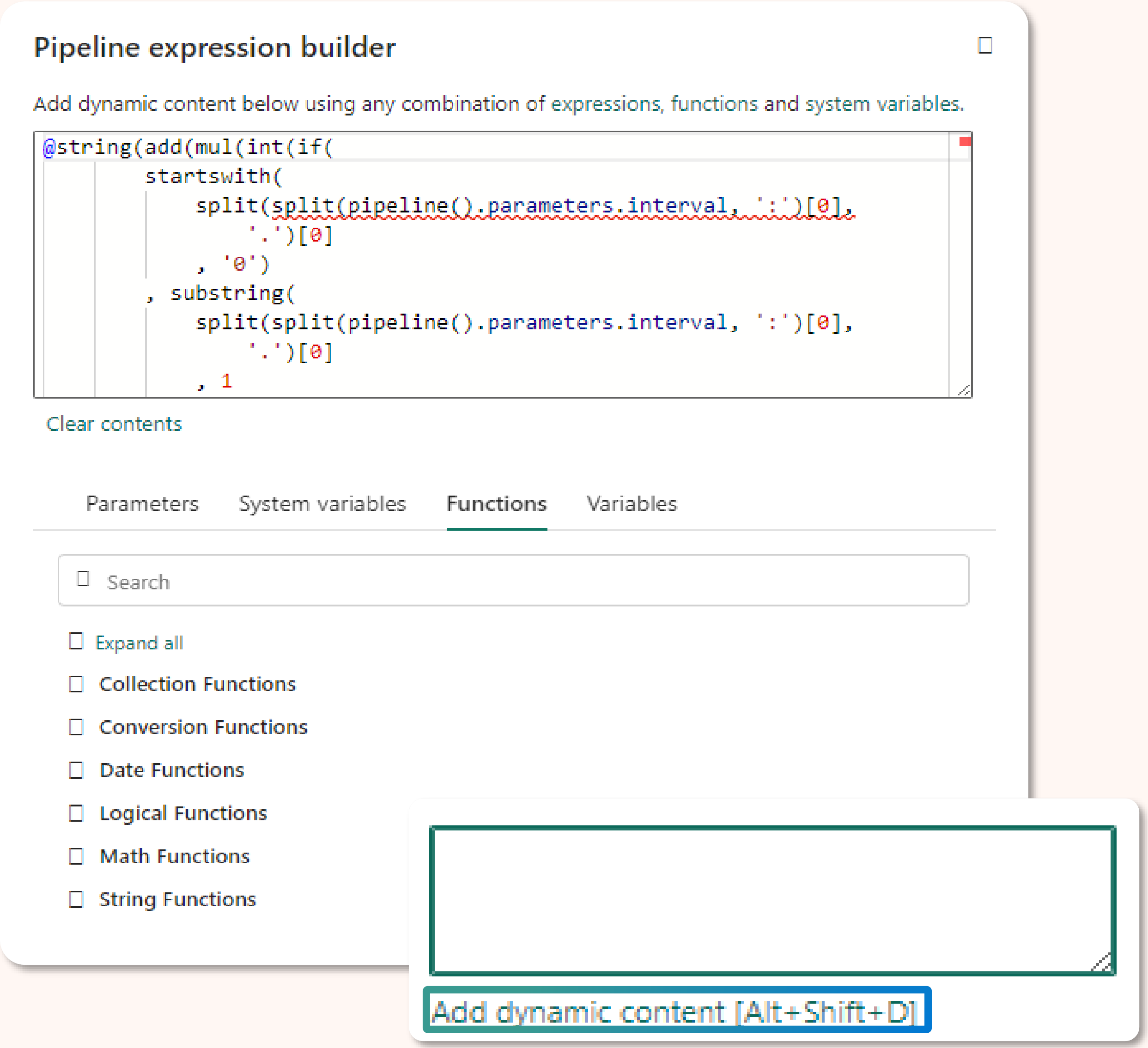


Pipeline expression builder
in Fabric Data Factory

Create and manage dynamic data-driven content

Wide range of functions and operators that can be used to manipulate and transform data

Greater **flexibility** and **reusability**, and to perform advanced data transformations without the need for custom code

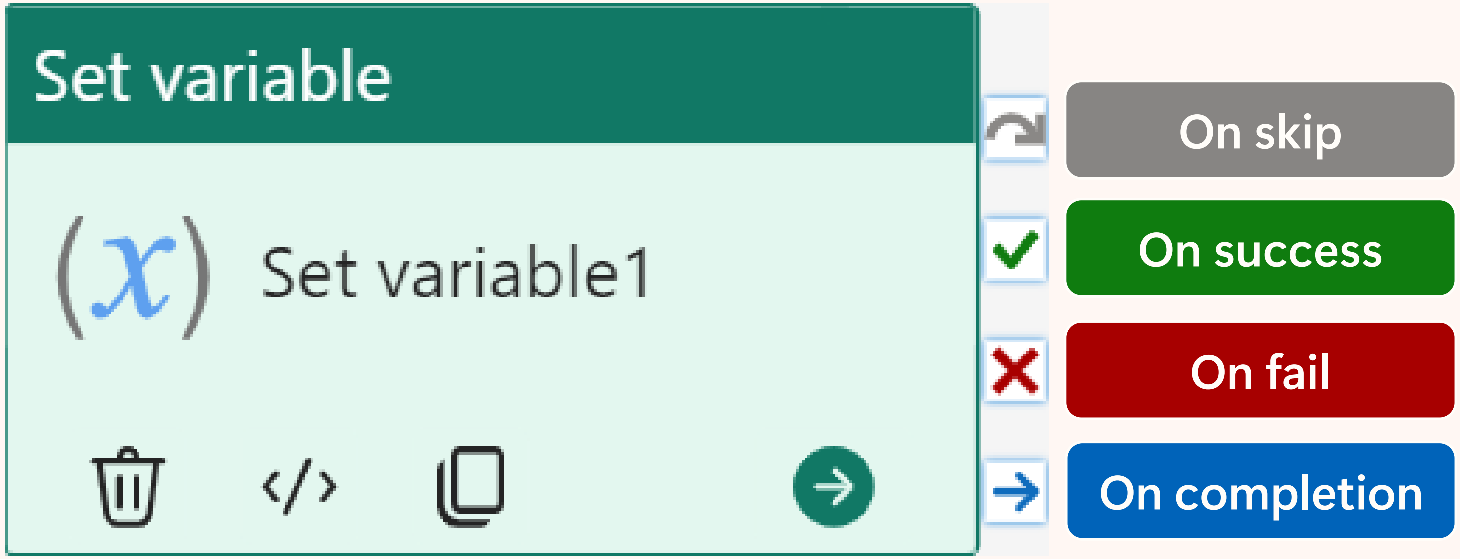


Conditional paths
in Data pipelines

Enable you to build robust pipelines with **error handling and branching logic**

There are four types:

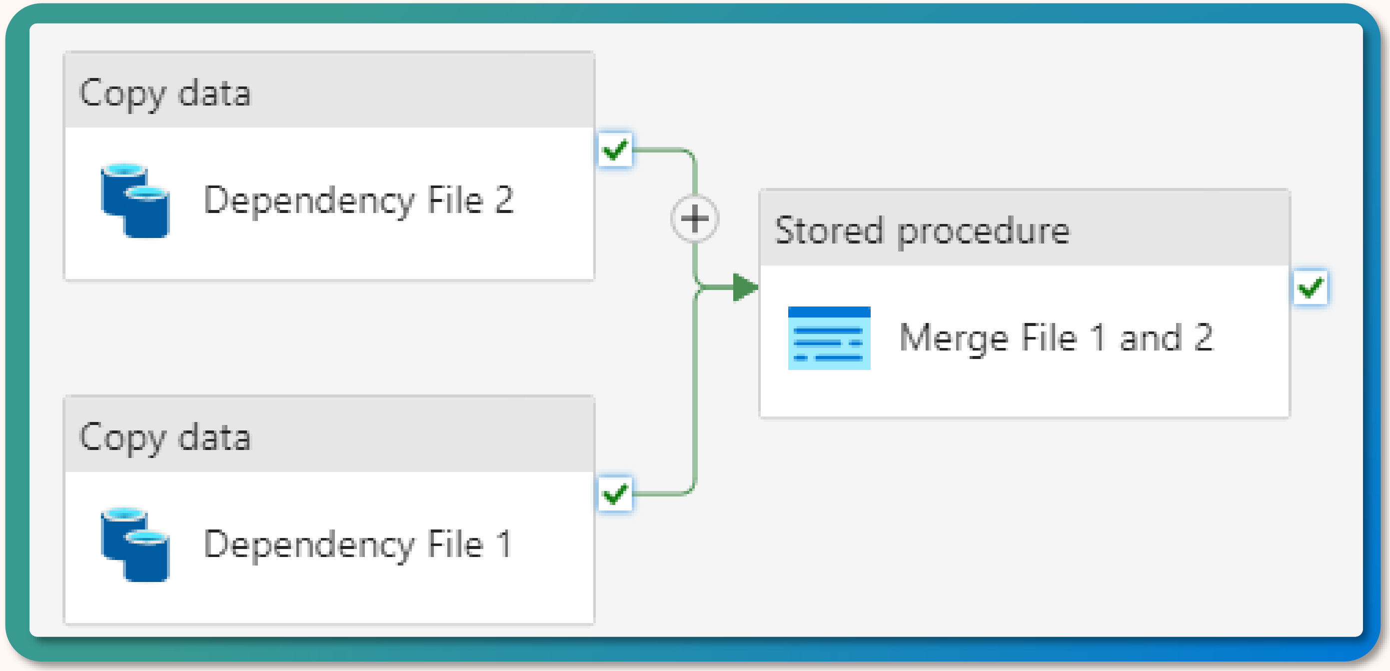
- **On Skip**
- **On success**
- **On fail**
- **On completion**



Conditional paths
in Data pipelines

Define different execution paths based on the outcome of a previous activity

- **Blocking dependencies**
- Non-blocking dependencies
- Error handling

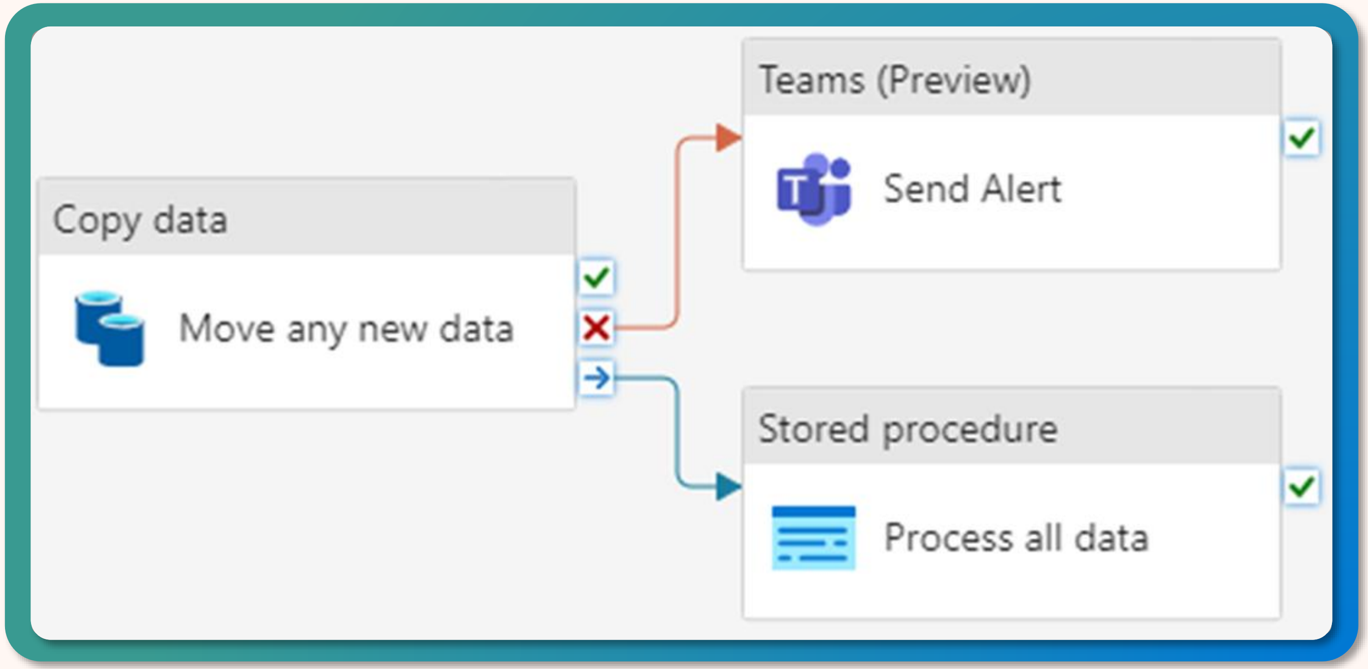


Both activities must be successful before proceeding.

Conditional paths
in Data pipelines

Define different execution paths based on the outcome of a previous activity

- Blocking dependencies
- **Non-blocking dependencies**
- Error handling

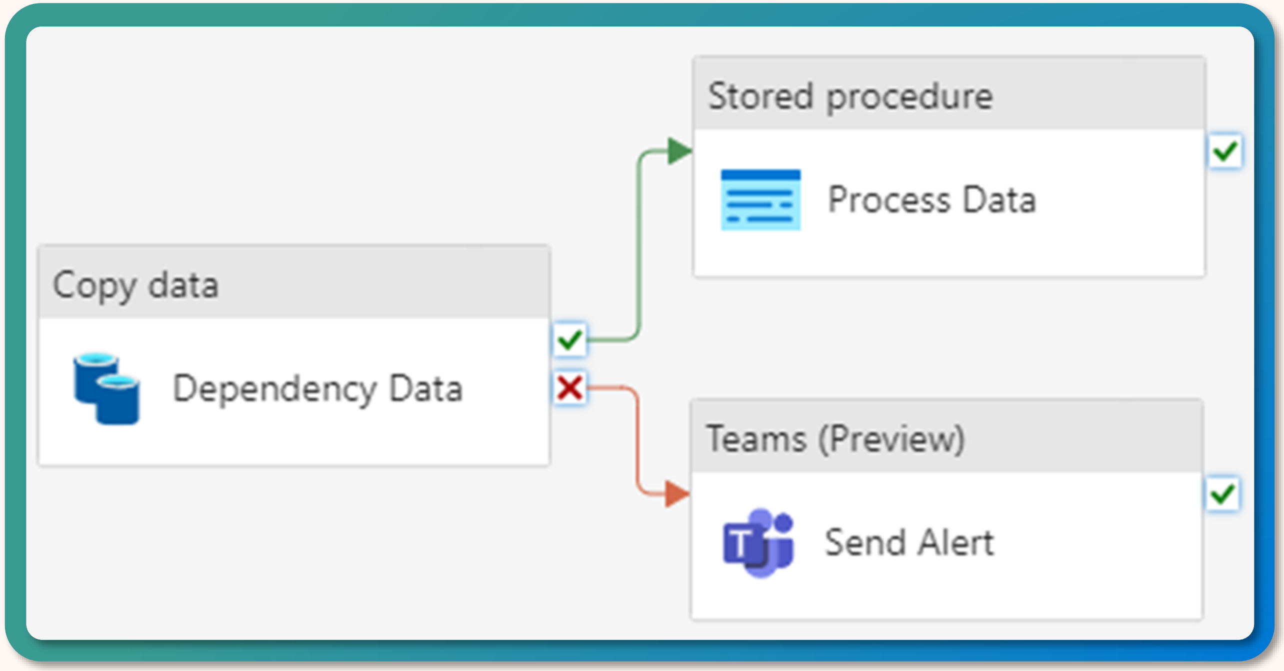


Regardless of the success or failure of an activity, the orchestrator can proceed to the next activity.

Conditional paths
in Data pipelines

Define different execution paths based on the outcome of a previous activity

- Blocking dependencies
- Non-blocking dependencies
- **Error handling**



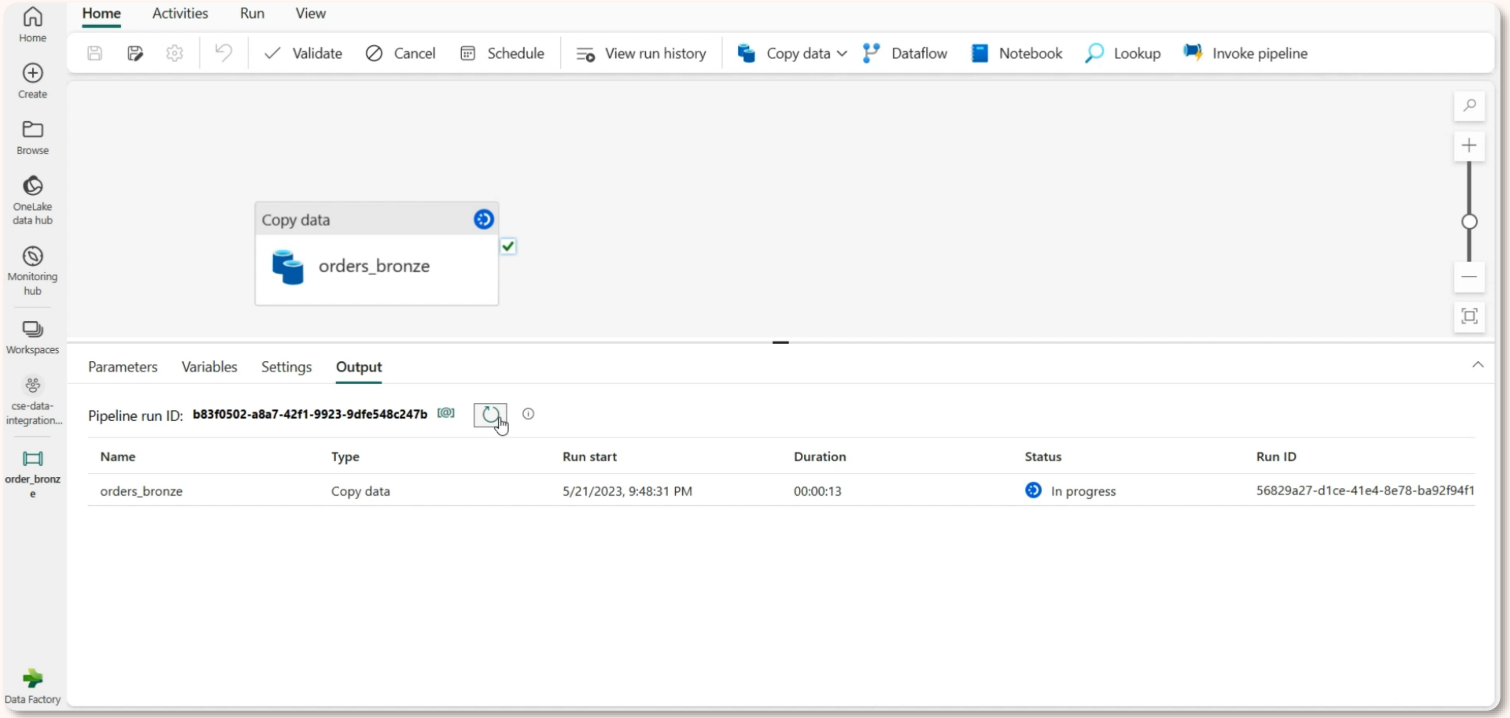
Creating a conditional path for both success and failure events.

Troubleshooting tips

Leverage the **Output window** to find and review errors

Determine if it is an **activity error** or an **error from the source/destination**

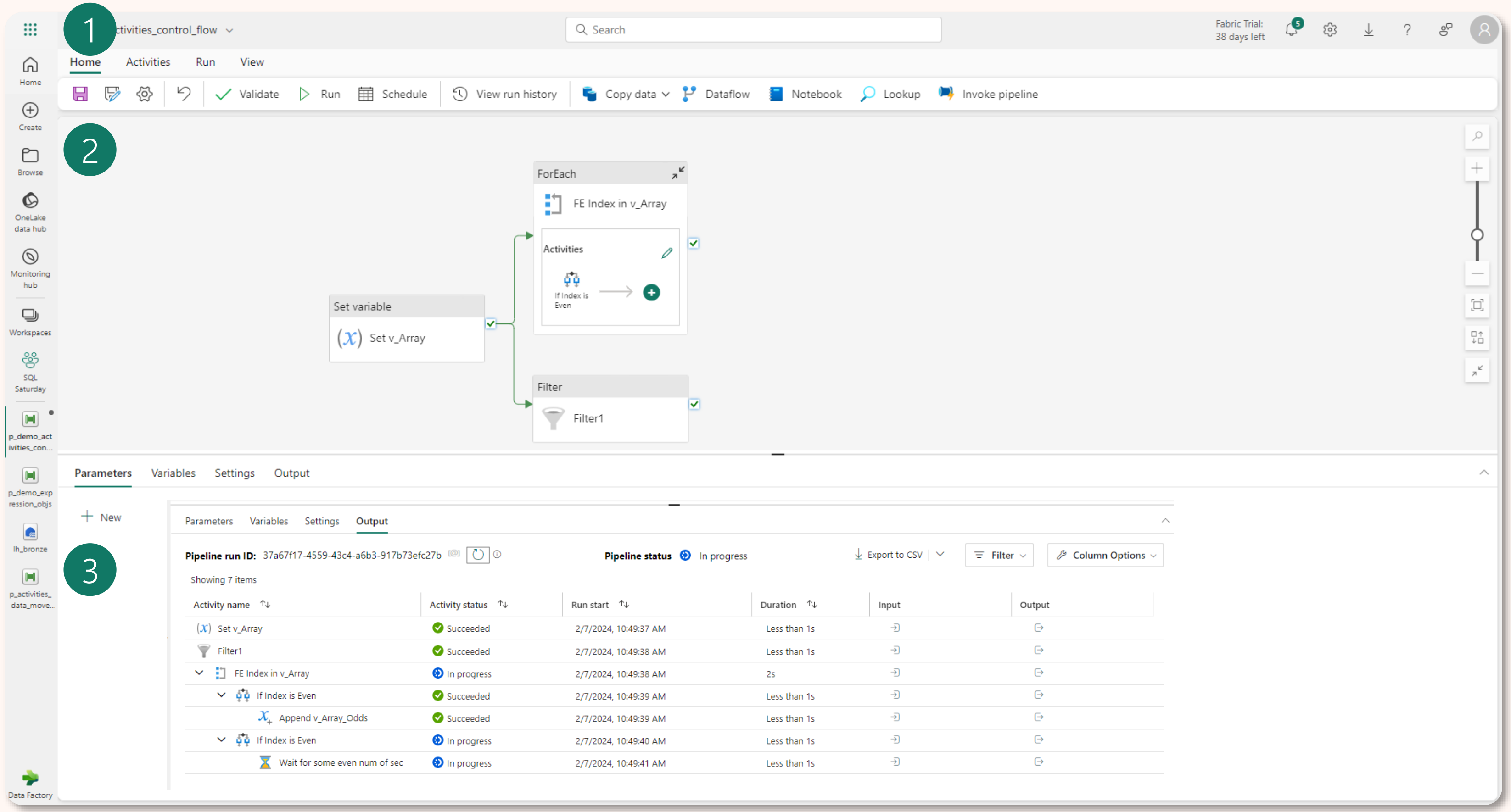
Set a **retry attempt** for activities



Data pipelines
Exploring data pipelines

Navigation

1. Command ribbon
2. Authoring canvas
3. Properties/output



Dynamic expression driven orchestration

Build enterprise data products with a multi-layered approach

Varied Complexity: Architecture designs can range widely in complexity due to the vast array of design options.

Reusability: Design with reusability in mind, incorporating parameters and components that can be reused, such as invoking other pipelines, running dataflows, notebooks or data functions.

Dynamic Designs: Aims for designs that are fully dynamic and driven by expressions, utilizing metadata parameterization.

Low

Static schedule

Data pipeline with no parameters or dynamic content

Medium

Event driven or dynamic schedule

Data pipelines with parameters, light dynamic content and nested data pipelines

High

Event driven, dynamic schedules or tumbling window

Meta-data driven data pipelines with parameters and most fields leveraging dynamic expressions

Low design time
Limited scale

Higher design time
Infinite scale

Migrate from ADF/Synapse Pipelines to Fabric Data pipelines

Feature	Fabric Data Factory	Azure Data Factory	Azure Synapse Pipelines
Office 365 Outlook Activity (Email activity)	Yes	*No	*No
MS Teams Activity	Yes	*No	*No
Refresh a Dataflow Gen2	Yes	*No	*No
Refresh a Power BI semantic model	Yes	*No	*No
Change Data Capture	Yes	Yes	No
Disable Activity	Yes	Yes	Yes
Managed Airflow	Yes	Yes (Preview)	No
Validation activity	No	Yes	Yes

* Supported by external services / REST APIs

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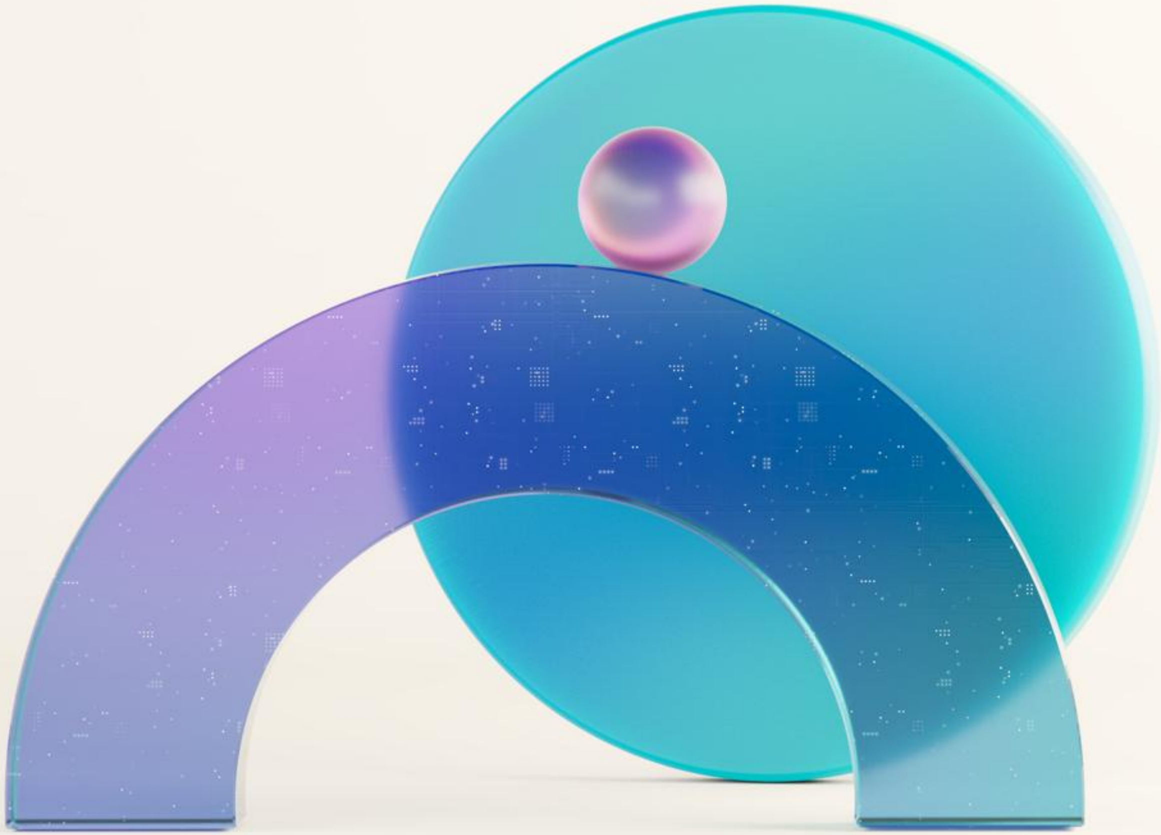


Lab Demo: Orchestrating Data Movement

- Create a dynamic **Data pipeline** with **pipeline expression builder**
- Navigate **Copy activity** settings
- **Enrich raw data** with **variables**, **loops**, and **conditional paths**
- Utilize **AI-powered Copilot** to summarize Data pipelines
- Move data between **medallion layers** in the **task flow**

Reminder of how to access lab instructions

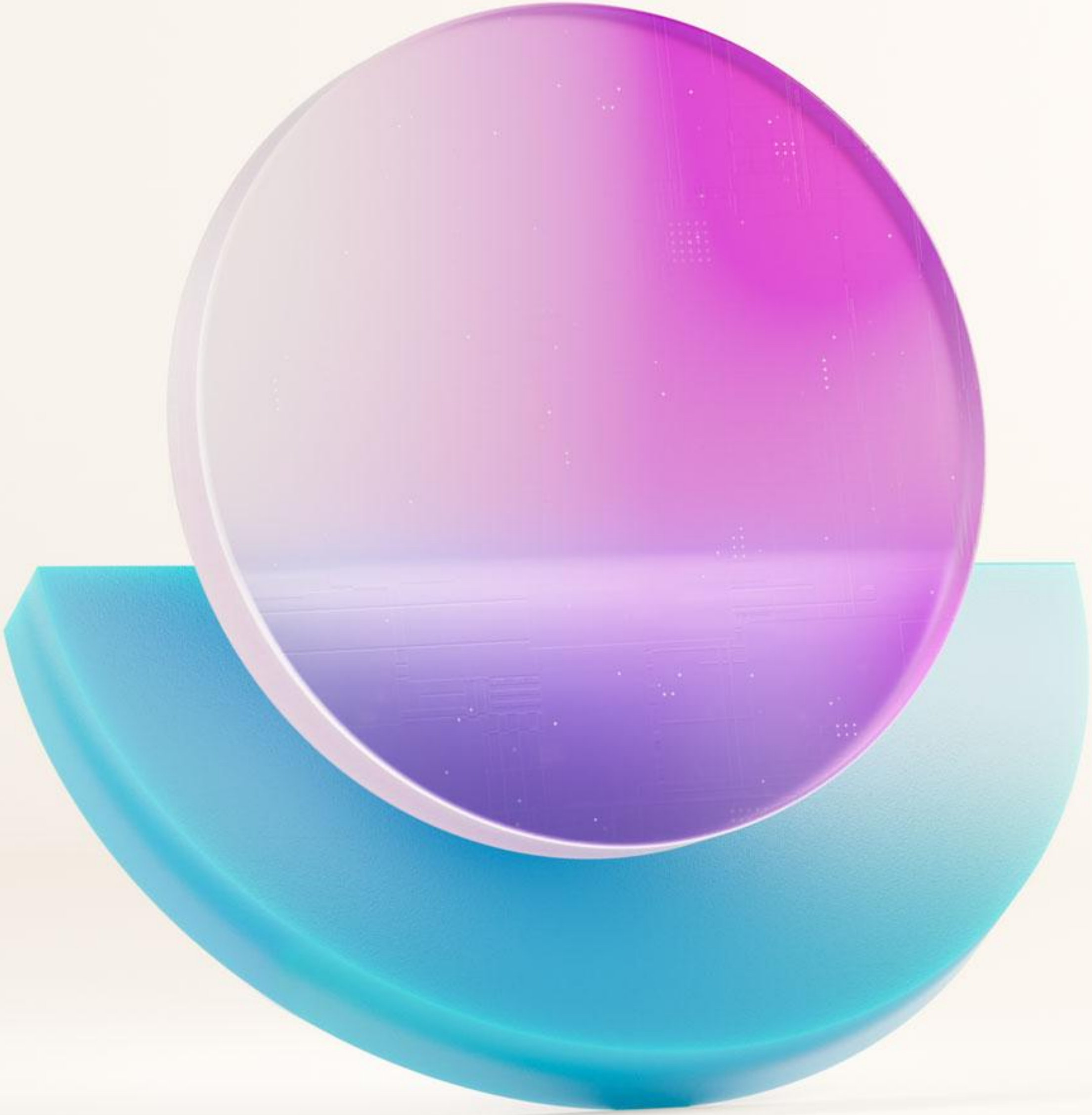
All materials are available at:

https://aka.ms/dfiad

Lab Instructions:

Orchestrating data movement

Microsoft Fabric
Community Conference



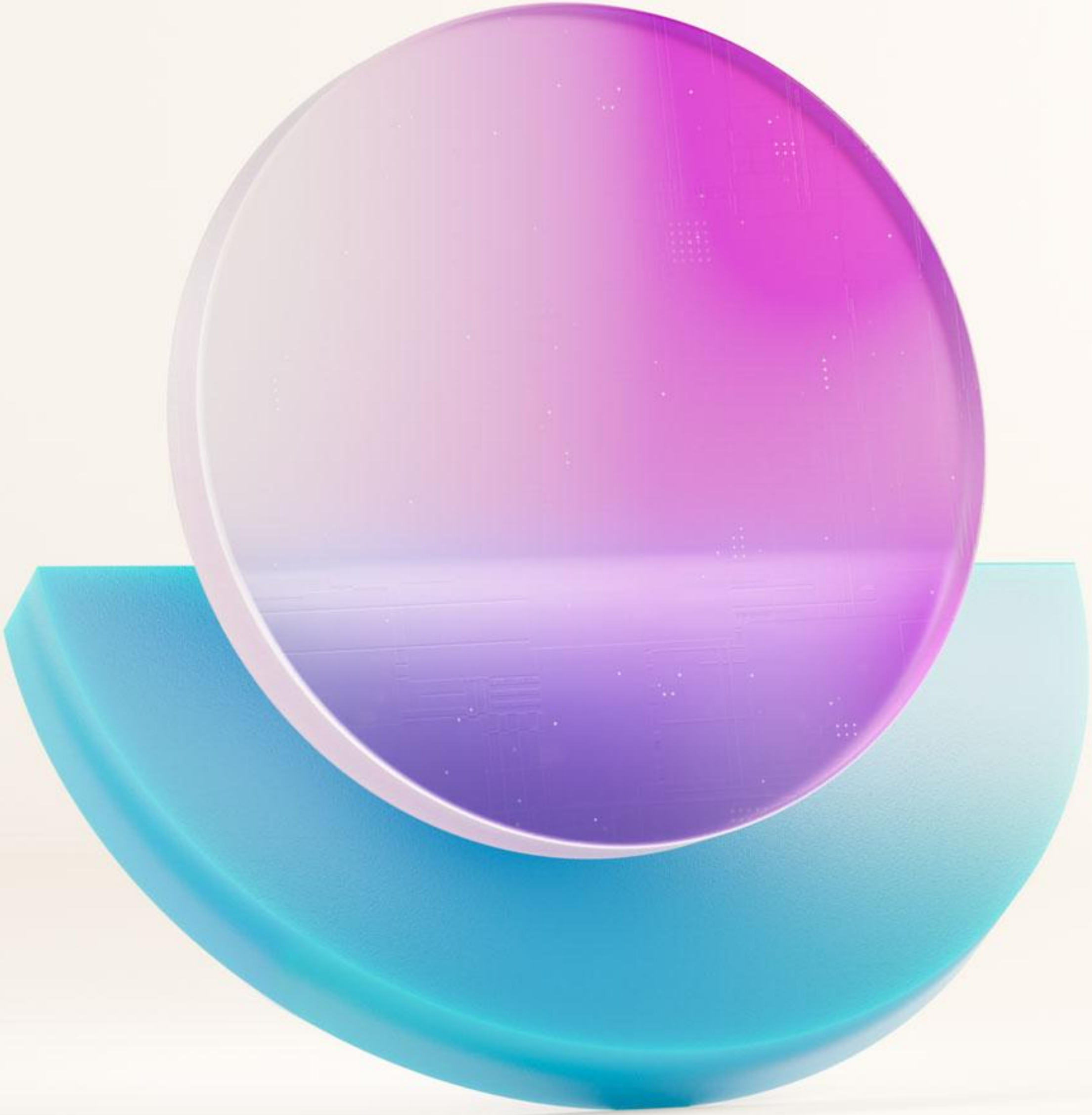
Questions?

Let’s take a short break…

Be back @ 2:00 PM

Influence our product roadmap
and ensure Fabric meets your real-life needs
https://aka.ms/fabricideas

Microsoft Fabric
Community Conference



Ingesting and transforming data with Dataflow Gen2

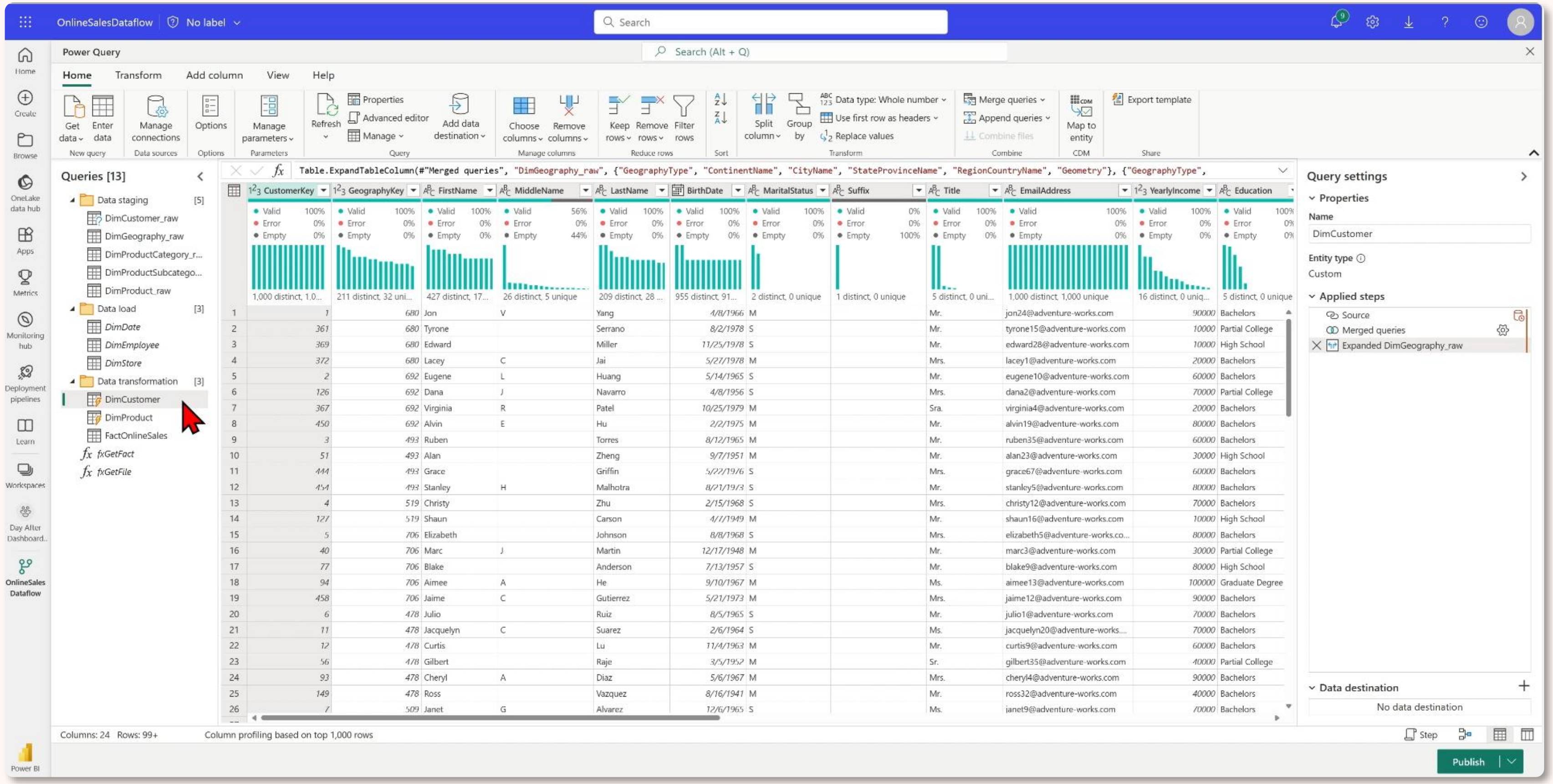
Next generation of data preparation with Dataflows Gen2
in Fabric Data Factory

Easy to use, no-code ETL & ELT

Includes **smart AI**-based data prep

More than **300+ transformations**

Output data destinations
Write output of dataflows to Azure SQL database, Data warehouse, Lakehouse and more

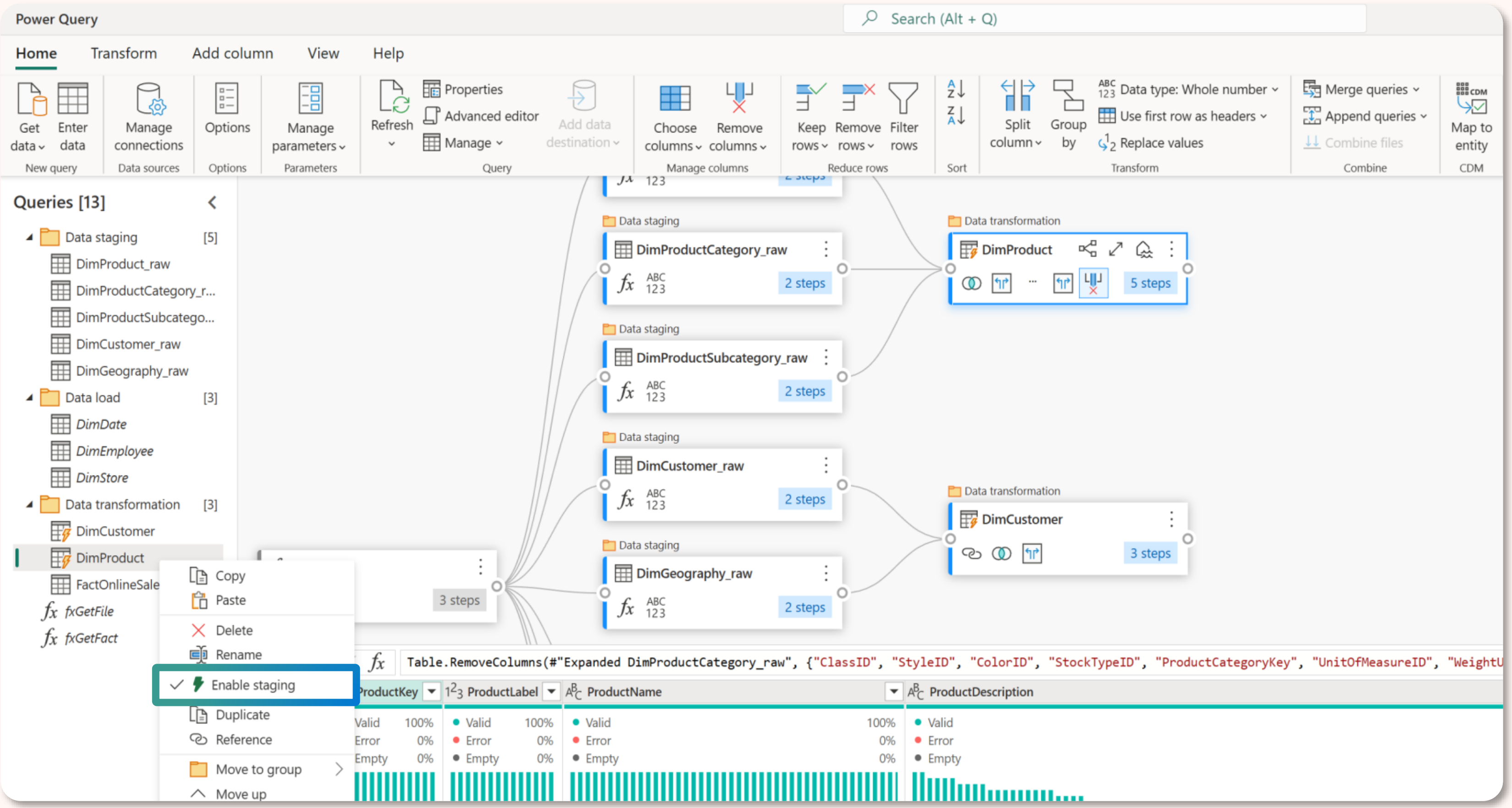


Optimized performance using Fabric compute and staging in Dataflows Gen2

Highly scalable using **Fabric compute**

A **seamless experience** - yielding fast, easy and powerful results

Abstracts away the complexities of traditional ETL and ELT

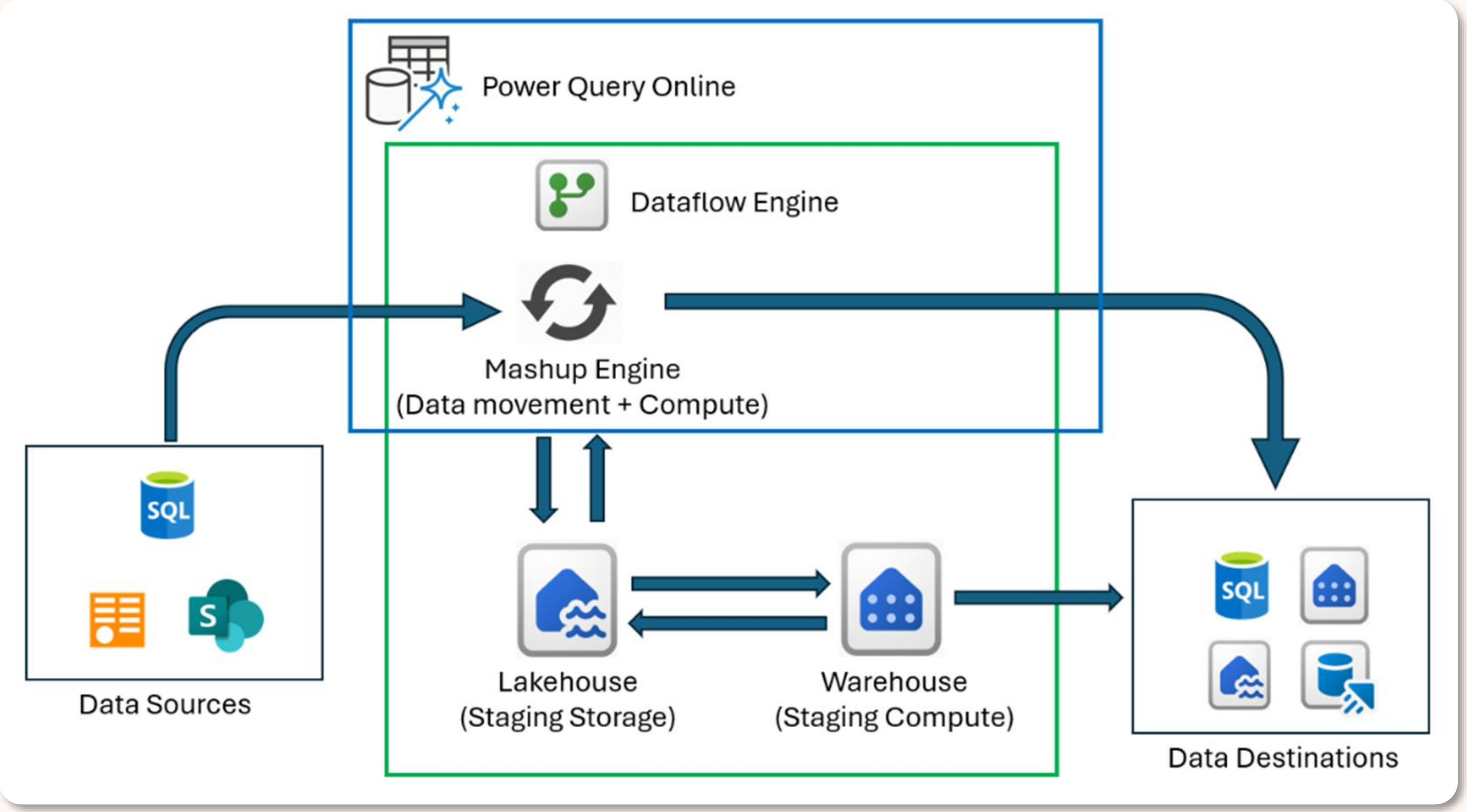


Optimized performance using Fabric compute and staging in Dataflows Gen2

Connect to your data and **copy** it into Fabric using the ***Enable staging** option
(*On by default)

Create a **reference** query in a new query

Apply **transformation steps** to the **computed** table for complex ETL operations such as join, distinct, filter and group by – leveraging Fabric compute



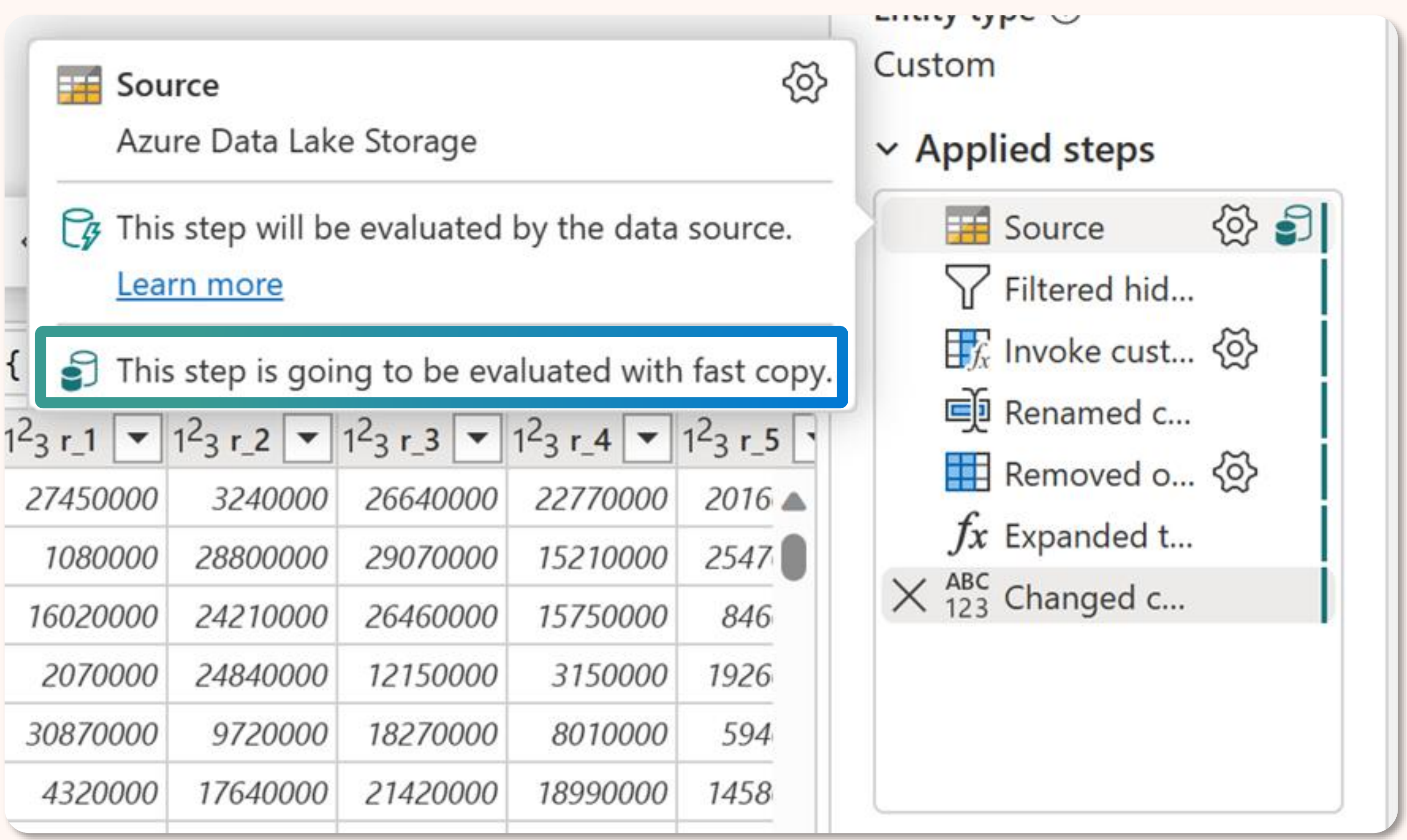
High scale data movement with Fast Copy
in Dataflows Gen2

General Availability

Ingest terabytes of data **effortlessly** with dataflows, powered by the scalable backend of the Copy activity

A limited set of transformations are supported:

- Combine files
- Select or remove columns
- Change data types
- Rename a column



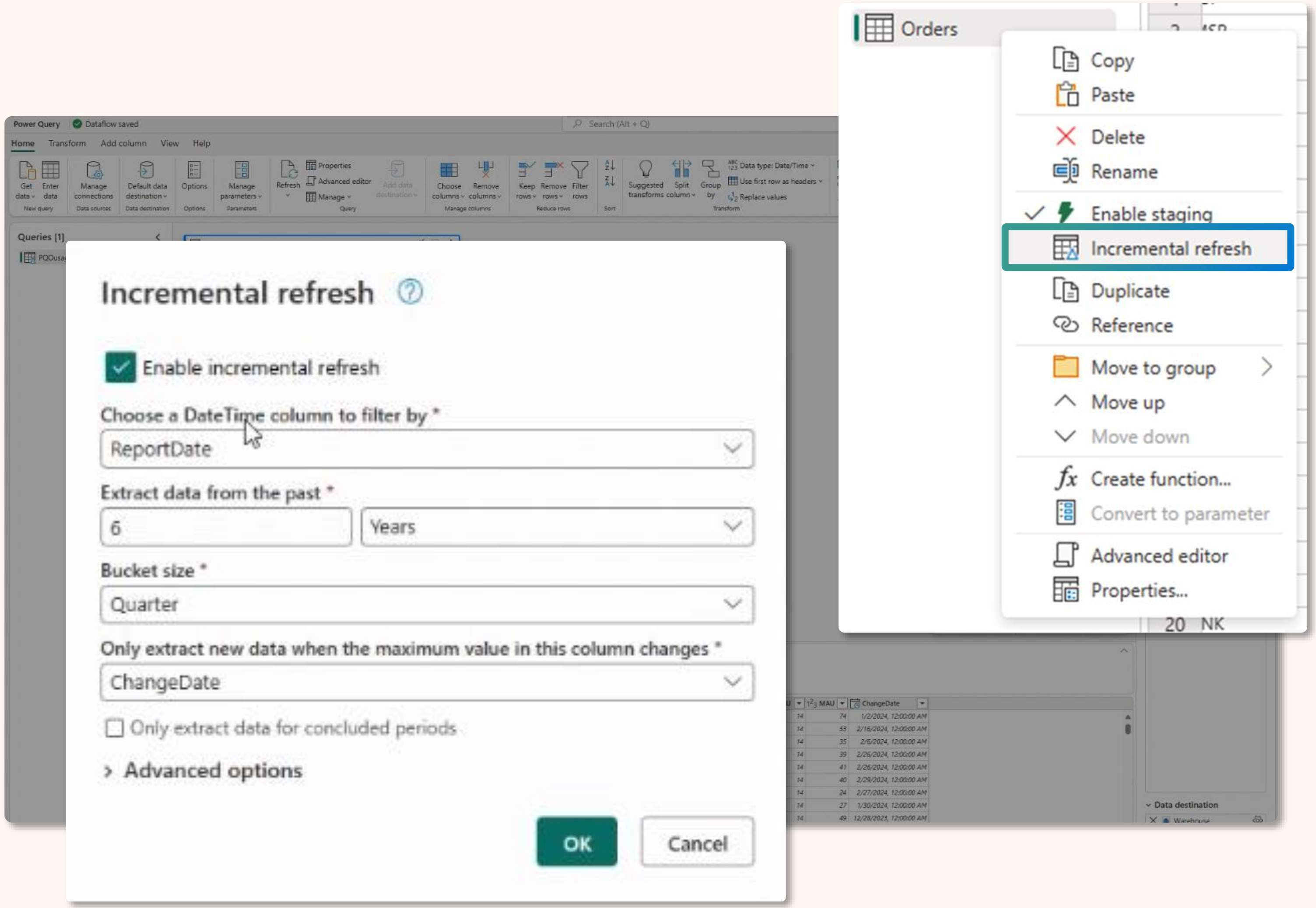
Optimized performance using incremental refresh

In Dataflows Gen2

General Availability

Process only changed data since the last refresh to **save time and resources**

- Evaluate changes by comparing the maximum **DateTime** value with the previous refresh.
- Retrieve and load data for changed buckets in parallel, loaded to staging
- Replaces the destination data with the new data, affecting only updated buckets.



Data destinations
in Dataflows Gen2

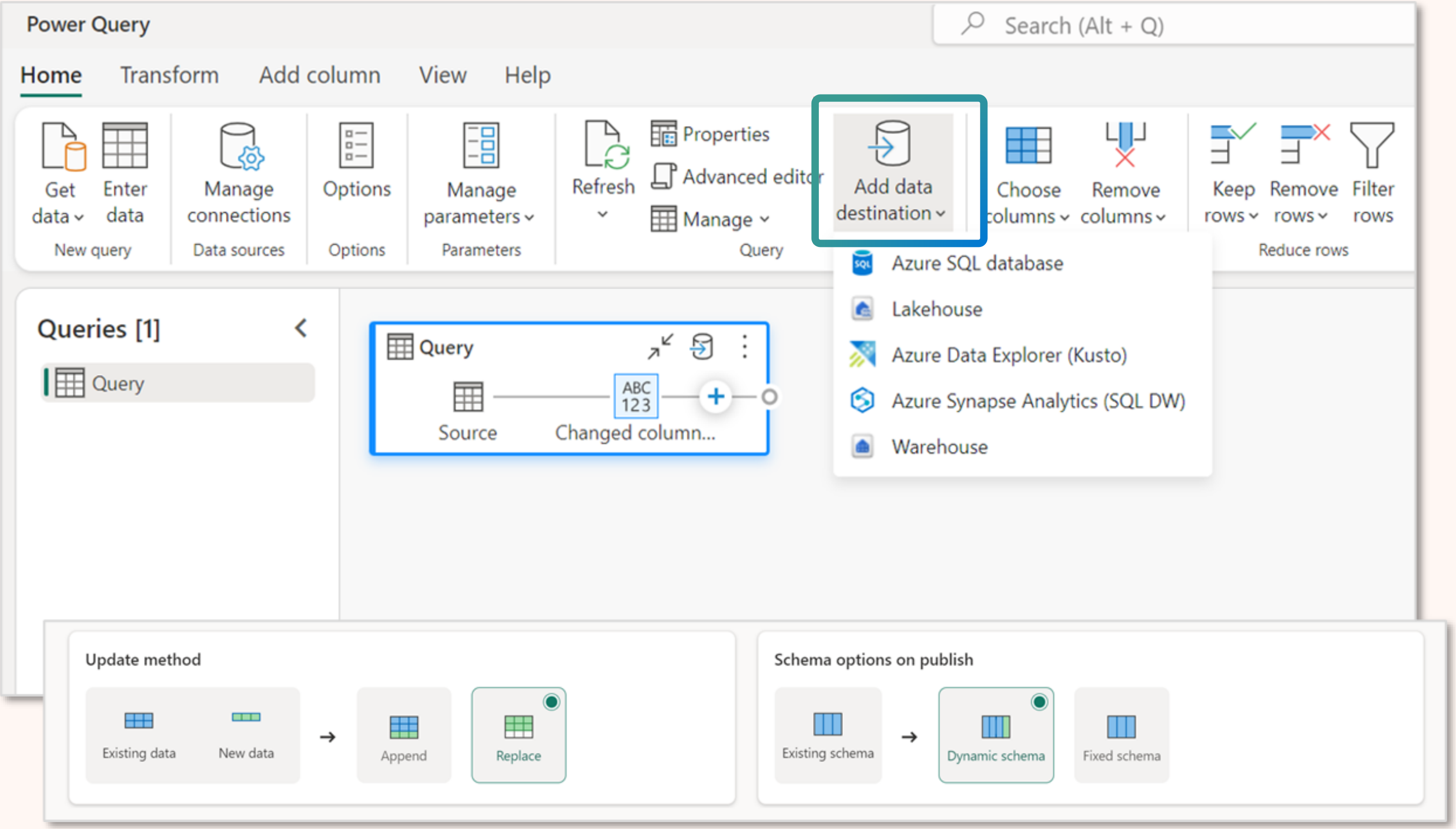
Land your data into a destination for further analysis and reporting

Supported destinations:

- Lakehouse
- Warehouse
- Eventhouse
- SQL database in Fabric
- Azure SQL database
- Azure Synapse Analytics
- Azure Data Explorer

Update methods:

- Replace
- Append

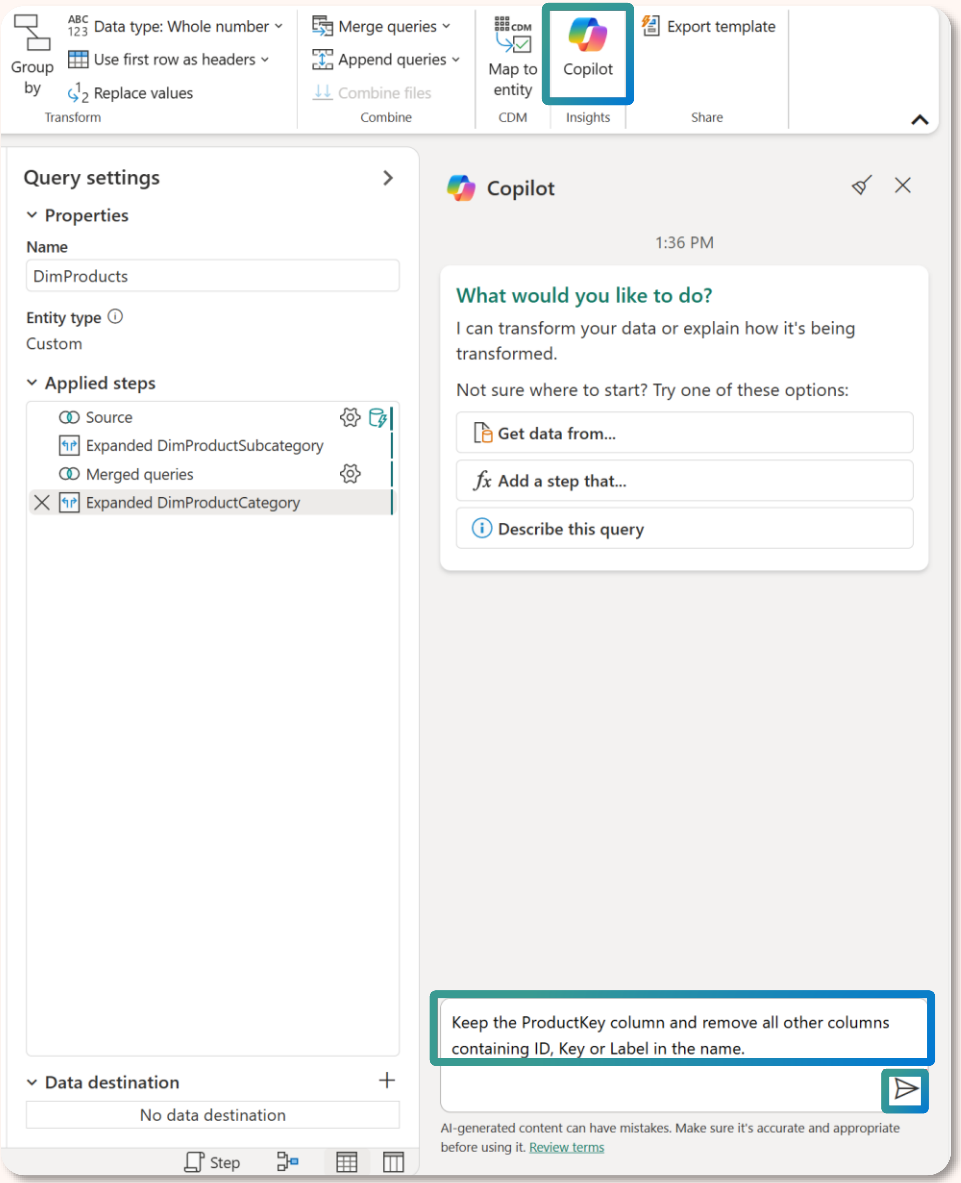


AI-powered Dataflows Gen2 with Copilot

Utilize the **get data** experience to ingest data into your dataflows

Transform data by using natural language to add a step to your query

Receive a **summary** of your queries in your Dataflows Gen2

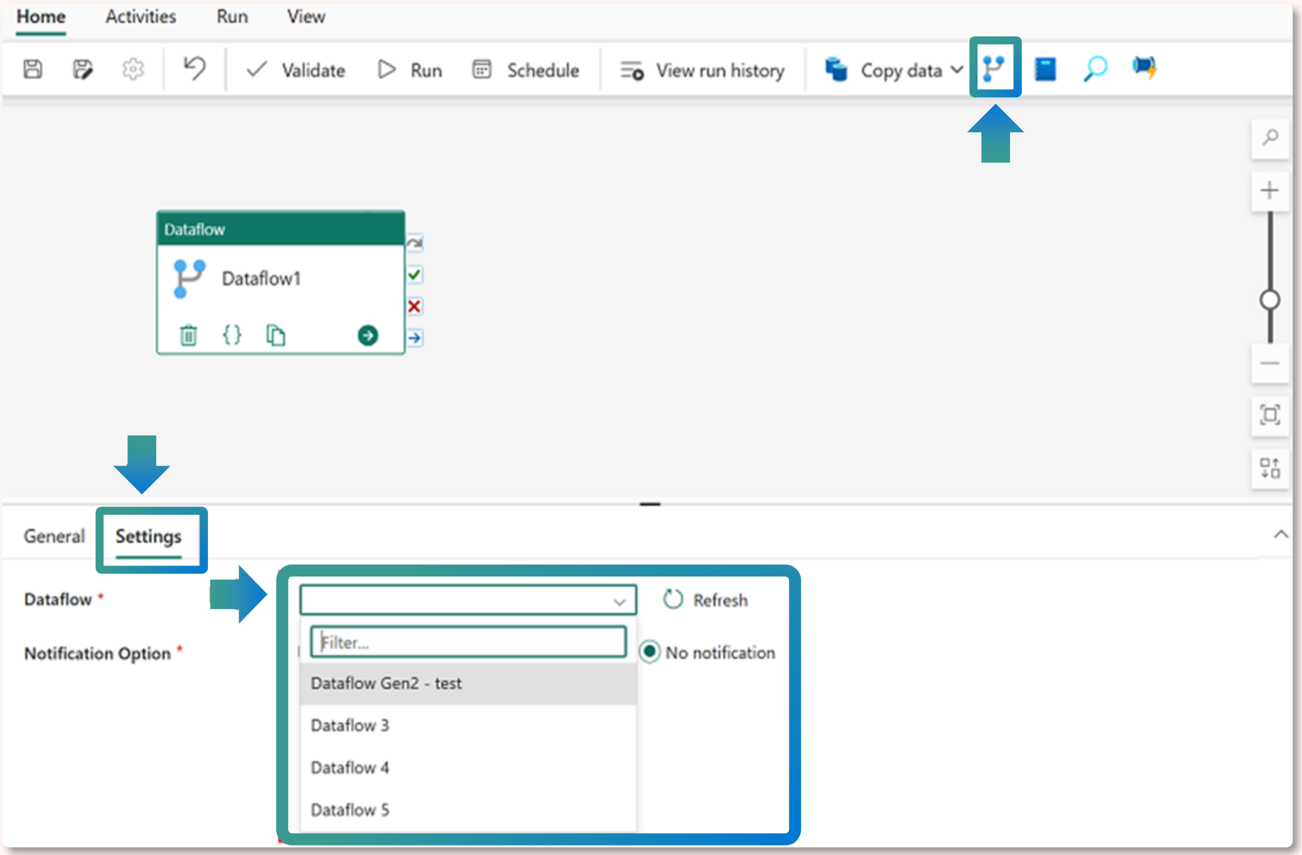


Dataflows Gen2
in Fabric Data Factory

Orchestrating Dataflows Gen2 with Data pipelines

Create a dataflow for **data ingestion and transformation**, and landing into a Lakehouse using dataflows

Then **incorporate the dataflow into a pipeline** to orchestrate additional activities



Check your knowledge | Dataflows Gen2

A single dataflow has a limit of how many queries/tables?

25	50	100

Dataflows Gen2 scale recommendations

Update method separation

Separate **dimension** tables and **fact** tables into separate dataflows based on update method

Staging separation

Separate tables with **staging** and **without staging** into separate dataflows

Fast copy separation

Separate tables **with fast copy** support and **without fast copy** support into separate dataflows

Query folding separation

Separate tables based on the data source and if they **support query folding** or **do not support query folding**

Item distinction

Use copy job, mirroring or data pipelines to **ingest data** and dataflows to **transform**

Migrate from Power BI dataflow to Fabric dataflow gen2

Feature	Power BI Dataflow	Dataflow Gen2
Incremental refresh	Yes	Yes
Accessible file outputs	*No	Yes
Fast copy	No	Yes
Data destination output	No	Yes
Premium capacity required	No	Yes
AI Insights	Yes	*No
AutoML	Deprecated	Deprecated
Attach Common Data Model (CDM) folder	Yes	No
Linked Tables	Yes	No (use Shortcuts)
On-premises data gateway	Yes	Yes
Vnet data gateway	No	Yes

* Supported by external services / REST APIs

Upgrade Pathways for Power BI dataflows customers to Fabric

New Dataflow

Copy-Paste queries from Dataflow Gen1 to Dataflow Gen2

1

General Availability

Power Query Templates

Export template from an existing dataflow, create new dataflow from template

2

General Availability

Save as Dataflow Gen2

Save a copy of an existing dataflow as a Dataflow Gen2

3

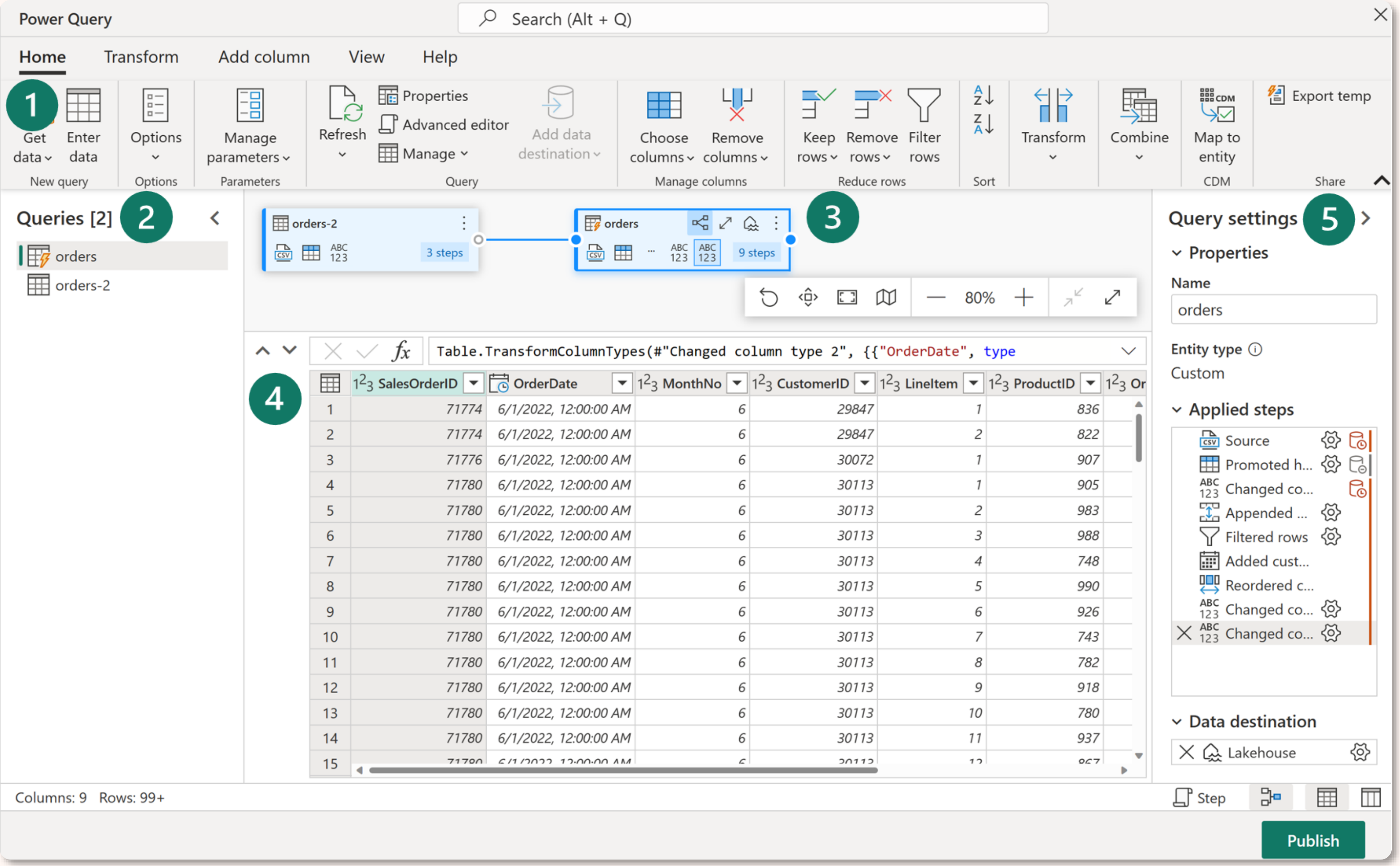
General Availability

Dataflows Gen2

Exploring Dataflows Gen2

Navigation

1. Command ribbon
2. Queries pane
3. Visual diagram
4. Data preview grid
5. Query settings



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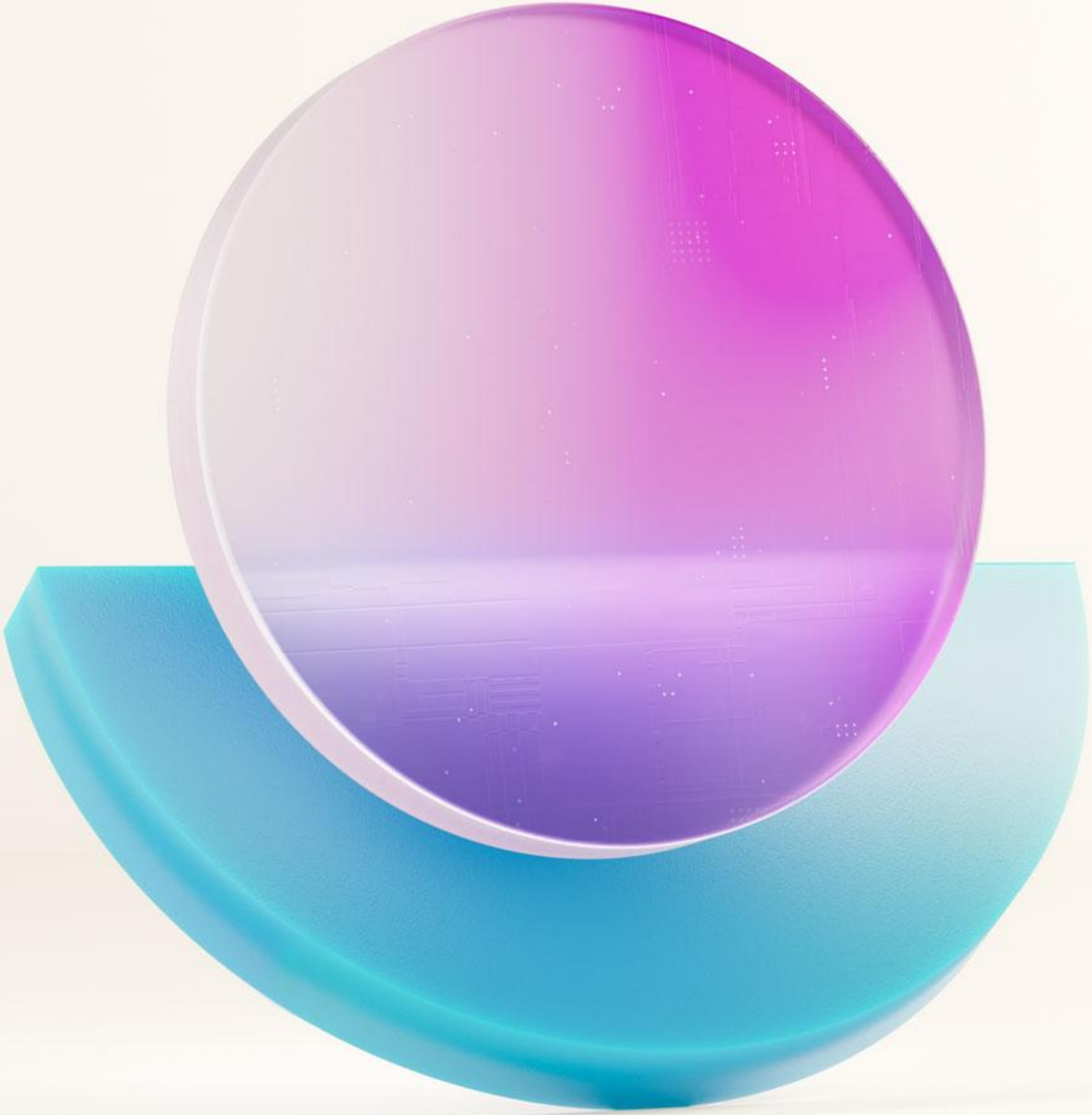


Copying data with a Copy job

Copy job
in Fabric Data Factory

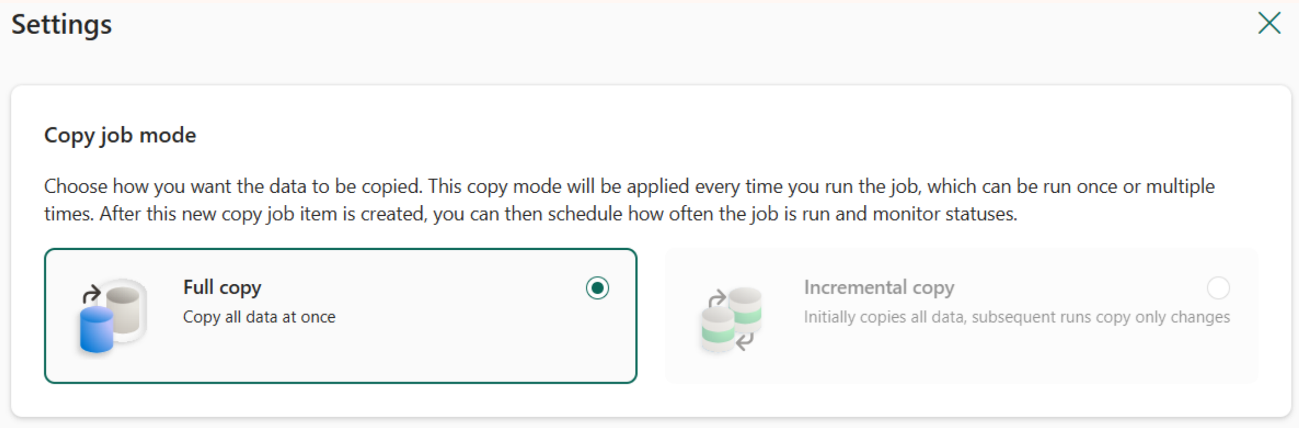
Generally Available

Ingest data from any source to any destination with Copy job

Choose your data delivery style

Full copy mode copies all data from the source to the destination **at once**

Incremental copy mode copies all data in the initial job run, and subsequent job runs only copies **changes since the last run**



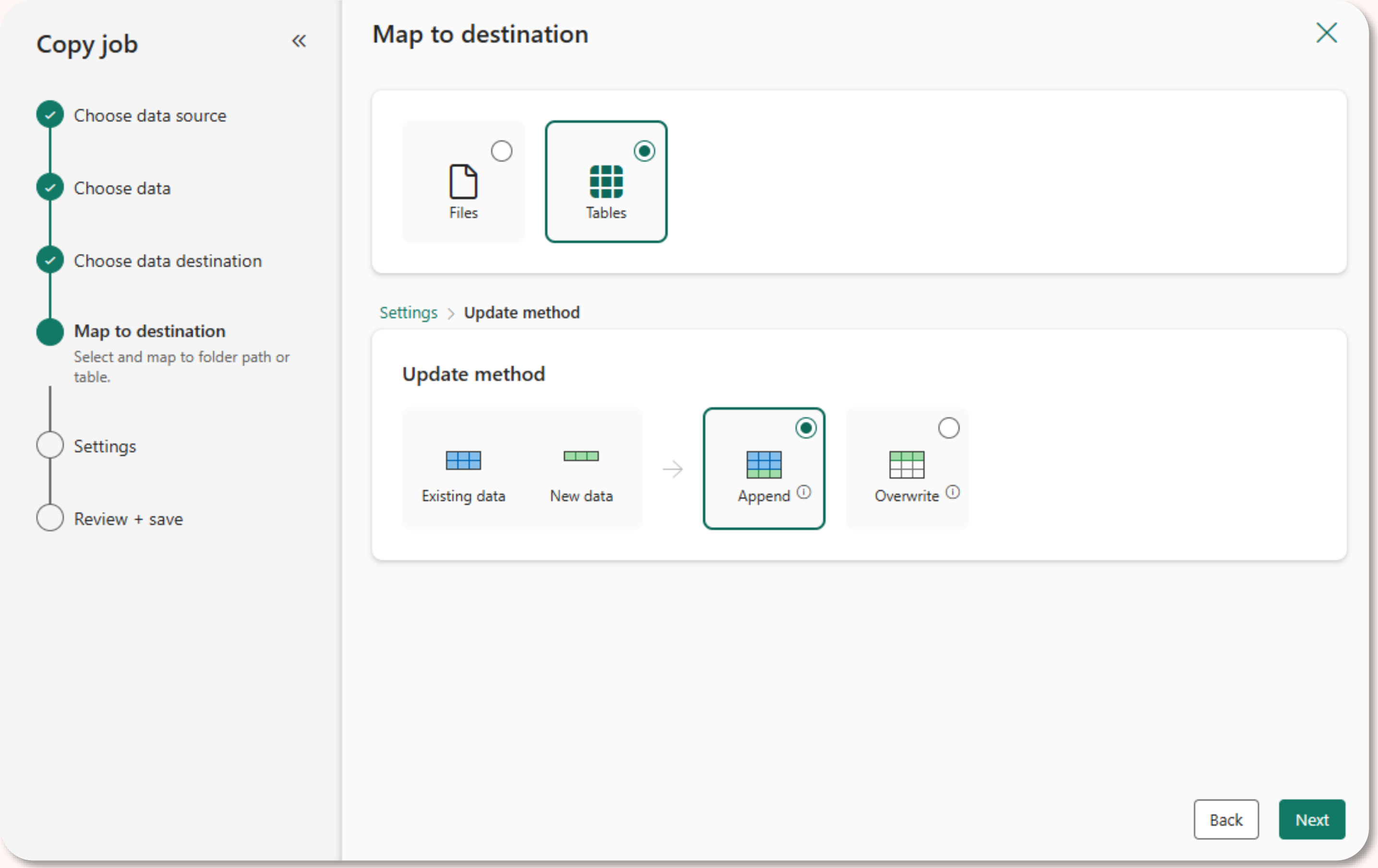
Copy job
in Fabric Data Factory

Generally Available

Personalize your data update method

By default, Copy job **appends** data to your destination

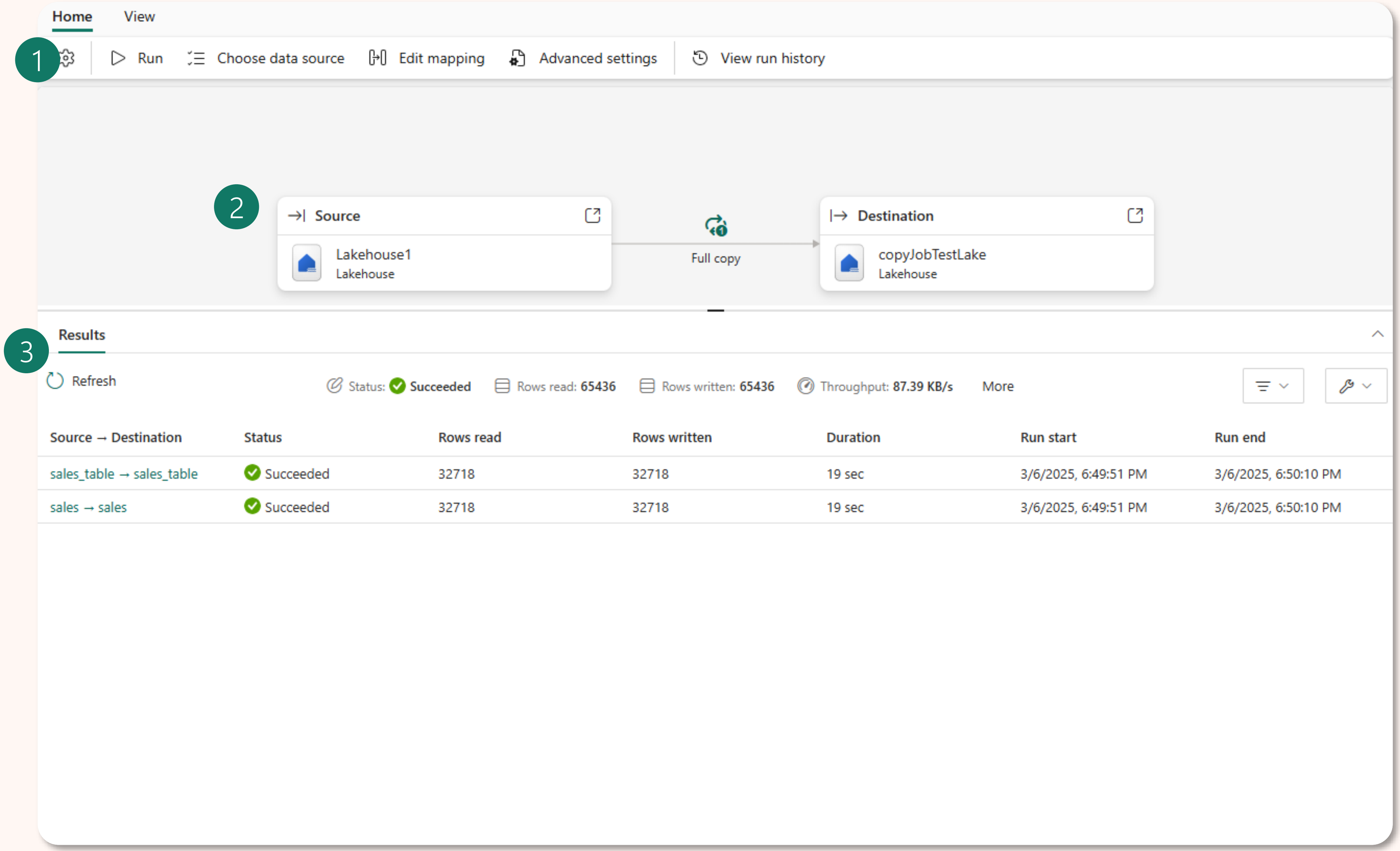
But you can also write behavior to **overwrite**



Copy job
Exploring Copy job

Navigation

1. Command ribbon
2. Visual diagram
3. Copy job monitoring



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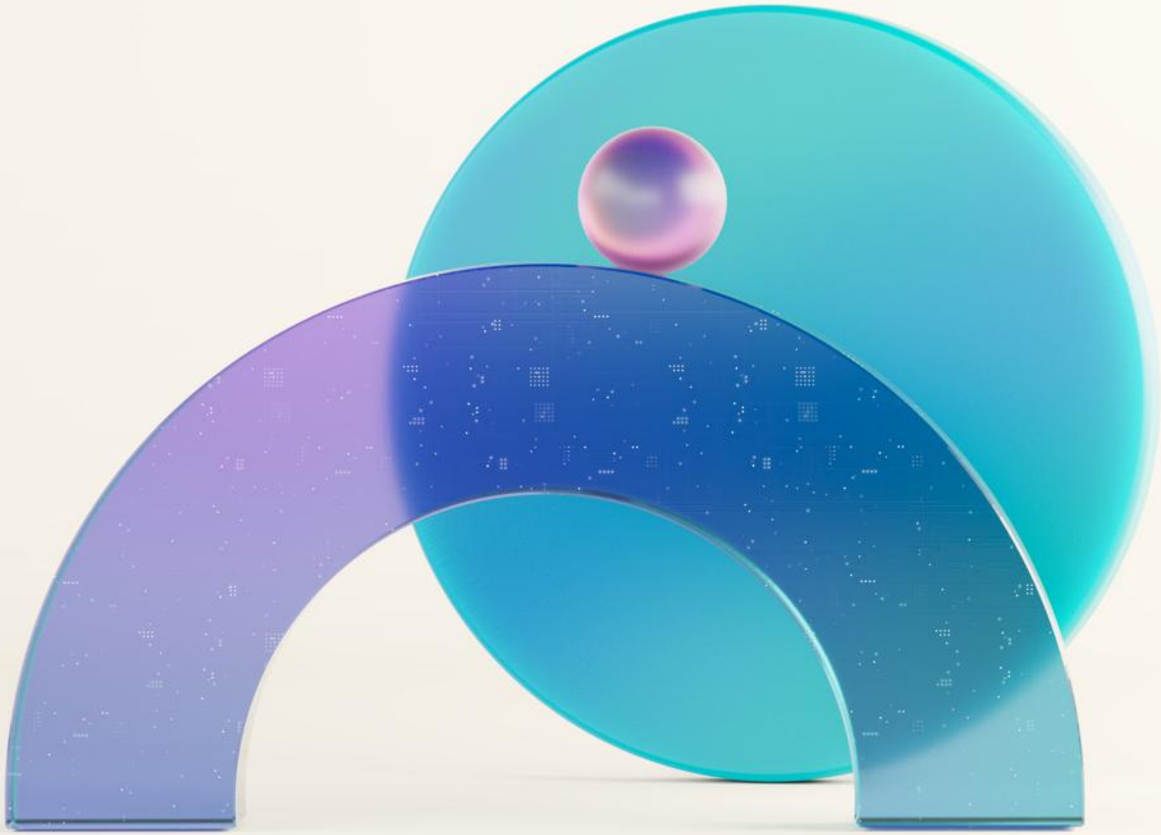

Lab Demo: Transforming and copying data

- Create a **Dataflow Gen2** and prepare data using **Power Query Online**
- Utilize **AI-powered Copilot** to transform data
- **Output data** to a **Warehouse** destination
- Copy data into a Warehouse with a **Copy Job** item

Reminder of how to access lab instructions

All materials are available at:

https://aka.ms/dfiad

Lab Instructions:

Transforming and copying data

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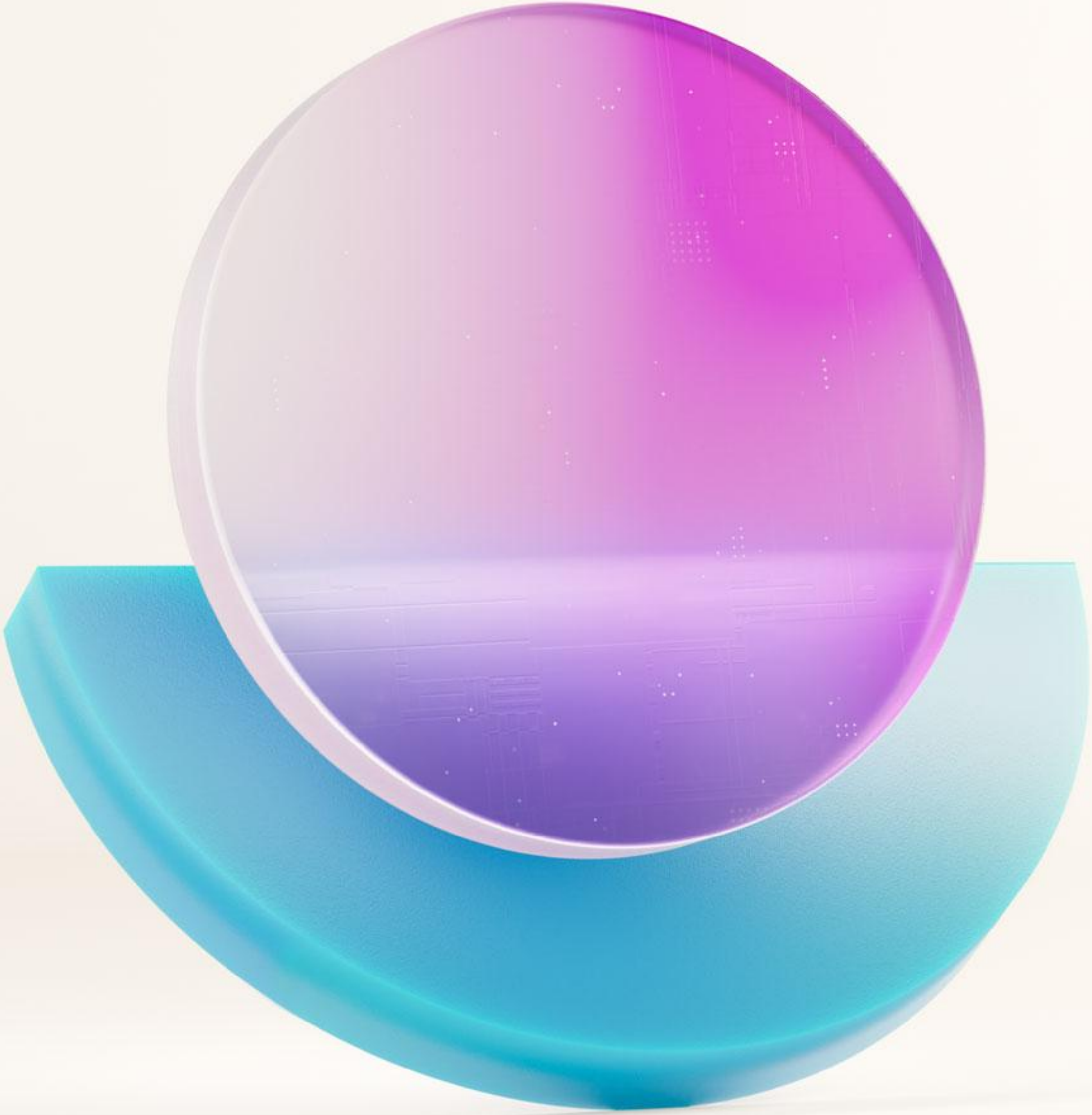


Questions?

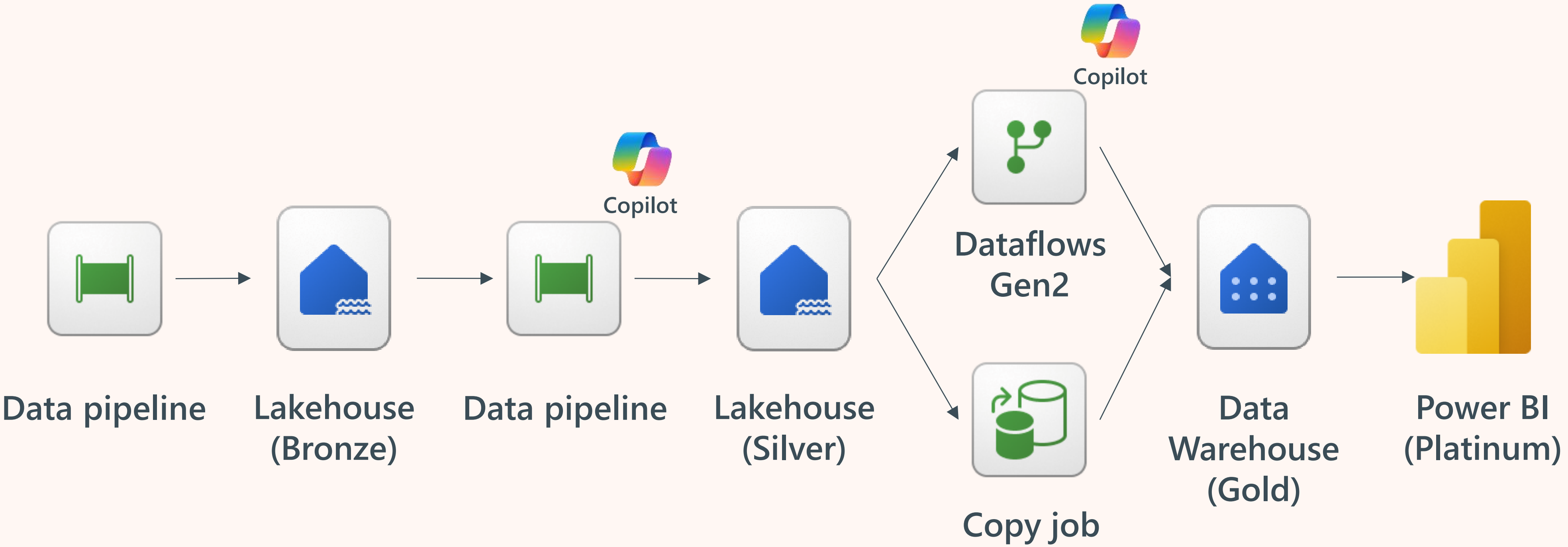
Microsoft

Let’s take a short break…

Be back @ 3:45 PM

Influence our product roadmap
and ensure Fabric meets your real-life needs
https://aka.ms/fabricideas

A quick recap of what you did today!



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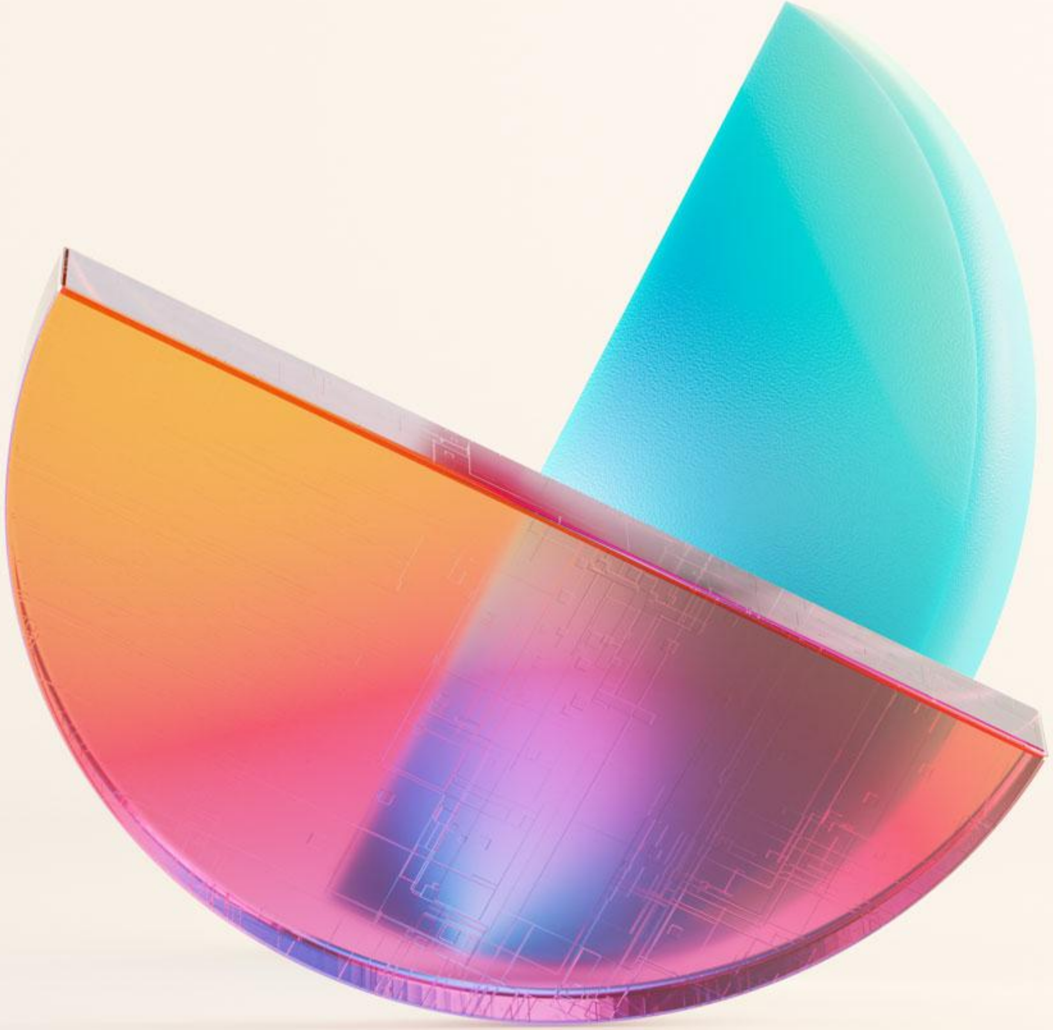

One last thing...

Thank you!

Generally Available

- Mirroring support for on-prem sources & sources behind firewall (Azure SQL MI, Snowflake, Azure SQL DB)
- Mirroring for Azure SQL MI
- Pipeline variable libraries
- Save Dataflow Gen1 as Dataflow Gen2 (CI/CD)
- Dataflow Gen2 Public APIs w/ SPN support
- Copy job & Email & Teams pipeline activities
- Dataflow Gen2 – Performance enhancements across Run time and Design Time
- Azure Key Vault integration in Connections
- Easily add SPN connections, Notebooks, and Pipelines to Airflow DAGs

- Open Mirroring Support for Delimited Text
- Copy job – more copy patterns including Merge, Reset, Truncate
- Datra Pipelines are now "Pipelines"
- Fast Copy supports Share point files & SQL in Fabric
- Invoke Pipeline & Dataflow Activity w/param support
- Multiple schedules per pipeline
- Pipeline and Copy Job integration with VNet Data Gateway
- Snowflake Key-Pair authentication support in Connections
- Workspace Identity authentication support in Connections

Public Preview

- Mirroring for Oracle/Google Big Query
- Mirroring Support for Data Agent / Workspace PL
- Dataflow Gen2 ALM: Fabric Variables
- Copy job – Native CDC with SQL Server, SQL MI, SQL DB, Fabric Lakehouse
- Pipeline expression evaluator
- Copy job – Variable library for CICD
- Dataflow Gen2 – SharePoint (XLSX), Snowflake & Lakehouse Files output destinations
- Pipeline Copy Job Activity
- Copilot in Modern Get Data
- Workspace Private Link support for Data Factory

Microsoft Fabric Conference Vienna 2025


Fabric Data Factory:
What's New and Roadmap

Tues, 10:30 – 11:30


Performance Tuning Tips for Fabric Data Factory

Tues, 14:15 – 15:15


Bringing data to OneLake: a deep dive into shortcuts and mirroring

Wed, 10:30 – 11:30


Fabric Data Factory–Maximize Business Impact with Best-In-Class Connectivity to SAP, Oracle and more

Thurs, 12:00 – 13:00


Unlock the power of Pipelines and Data Movement with Fabric Data Factory

Thurs, 14:15 – 15:15


Unlock the power of Dataflow Gen2 with Fabric Data Factory

Thurs, 15:30 – 16:30

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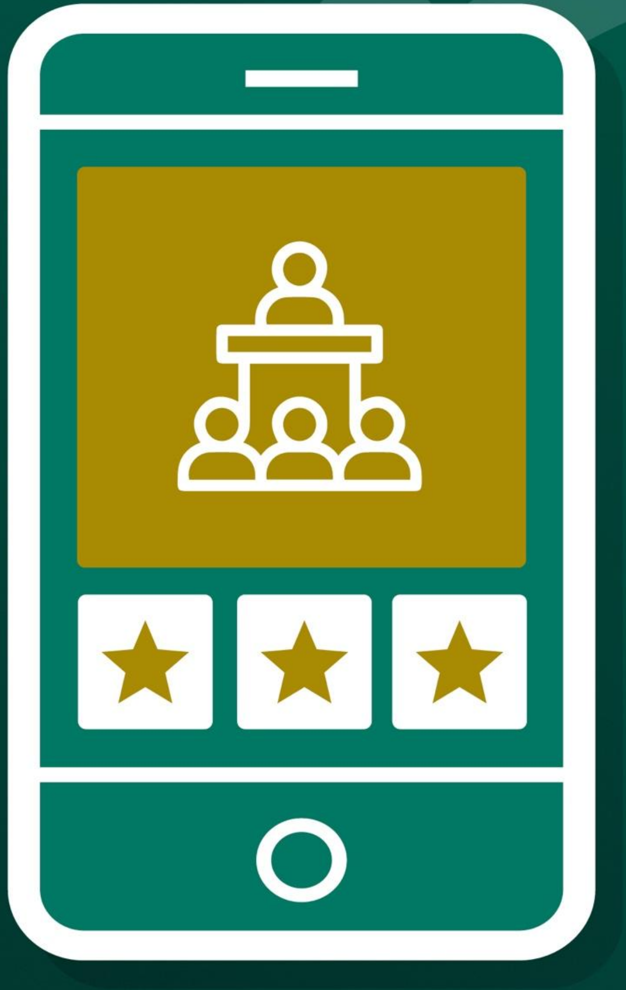

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